FIN5005 Fall 2026

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1 Course Overview

1.1 Course Overview

What Business Analytics (BA) is about?

The aim of this course is to provide students advanced knowledge, skills and competencies they can use to make data driven decisions for organizations.

Data-driven decision making

- **Describe the data**: what happened

  Using introductory statistics to identify patterns, trends, and insights embedded in historical data. E.g., summary statistics.

- **Predictive**: what will happen

  Using statistical models on historical data and forecast future outcomes. E.g., regression, time series.

- Prescriptive: what should we do

  Using optimization, simulation, and decision analysis techniques to suggest the best course of action based on data, predictions, and constraints. E.g., determine the optimal price for a product to maximize profit.

- Autonomous: automated decision-making using Machine Learning (ML) and Artificial Intelligence (AI) techniques.

Outline of models and applications we will cover in this course:1

Category	**Topics**	**Applications in Business**
Descriptive	Data Visualization, Descriptive Analysis	Asset return → **Finance** California school expenditure and academic performance → **Education**
Predictive	Linear regression	Asset return prediction (Factor models) → **Finance**Test Score Determinants → **Sociology**

Category	Topics	Applications in Business
	Hypothesis testing	Test Score determinants → **Sociology**
	Classification, logistic regression	Boston housing value determinants → **Finance**

[1] This is a preliminary plan. Topics and applications are subject to change as the course moves forward.

Software: R programming

R is an open-source programming language widely used for statistical computing and data analysis. It provides a rich ecosystem of packages and libraries for data manipulation, visualization, and modeling.

Why open-source stands out in the competition?

AI is the game changer here. AI significantly improves coding efficiency and productivity. You don't need to memorize every function or syntax anymore. The current role for humans is to communicate your needs to AI, and AI will generate the code for you.

- Open-source software integrates cutting-edge AI tools, faster and more efficiently than paid software.
- By contrast, paid software is often slower to implement new features due to development cycles and licensing restrictions.

Demonstration in VS Code.

By typing `# create a function taking the n-th power of a number`, AI will generate the code for you.

Here is how to use GitHub Copilot in RStudio: RStudio User Guide: Tools, GitHub Copilot.

1.2 Course Materials

The course materials will be self-contained.

- **Lecture notes**: lecture notes will be provided in the course website.

  You can download the lecture notes in PDF format for offline reading. Click on the icon in the top-left corner to download the lecture notes in PDF format.

- **R code**: R code will be provided in form of Lab Jupyter Notebooks.

  You may copy and paste the code into RStudio to run it.

If you want to learn in-depth, you can refer to the following textbooks and resources.

Textbooks

- Evans, J.R. (2021) *Business analytics: methods, models, and decisions*. Third edition. Harlow: Pearson.

  Statistics primer. Basic introduction to business analytics at the undergraduate level.

- James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). *An Introduction to Statistical Learning with Applications in R*. 2nd edition. Springer. Online version

  Main textbook for the course. It covers the fundamental concepts and methods of statistical learning, with practical applications in R.

- Huntsinger, R. (2025). *Business Analytics: Methods and Cases for Data-Driven Decisions.* Cambridge: Cambridge University Press. eTextbook available through Cambridge University Press.

  Advanced methods and case studies for data-driven decision-making. We might use case studies from this book in the course.

Resources:

- Distribution Tables
- Formula Sheet

1.3 Course Evaluation

Compound Assessment

❗ In case of any discrepancies between the dates listed on the course website and StudentWeb, **ALWAYS use the dates on StudentWeb as the reference.**

- **Oppgave** (mandatory home assignment): 40%

  Group work (1-3 students) is possible; including a case study with data analysis and visualization; a report will be submitted;

Date: week 42
Duration: 2 weeks.

The assignment will be released in WISEflow, and you will have **two weeks** to complete it.

More details will be provided as the course progresses.

- **Eksamen** (school exam): 60%
  Digitally in WISEflow;
  Date: Refer to Studentweb

1.4 Study Objectives

- **Knowledge**:
  - Familiarity with statistical methods and models used in business analytics.
  - Focus on descriptive analytics and predictive analytics. Prescriptive analytics will be introduced if time allows.
  - Knowledge of data visualization techniques and tools.
- **Skills**:
  - Ability to analyze and interpret data using statistical methods.
  - Proficiency in using R for data analysis and visualization.
  - Competence in applying business analytics techniques to real-world problems.
- **Competencies**:
  - Develop critical thinking skills to evaluate data-driven decisions.
  - Ability to communicate findings effectively through reports and presentations.

1.5 How to Reach Me

Instructor:	Menghan Yuan
Email:	menghan.yuan@nord.no
Office Hours:	By appointment
Office:	Hovedbygning A257

Part I

Introduction

2 Environment Setup

This guide will help you set up your coding environment for FIN5005, with both cloud-based and local options available.

- **Cloud option:** Google Colab with R Kernel (Course Labs will be run in this)
- **Local option:** RStudio Desktop (You will most likely use this for the home assignment)

At the end of the session, you will learn R basics using Google Colab .

Follow the instructions below to get started.

2.1 Cloud Solution

Cloud solution means that you don't need to install anything on your computer. You can run R code directly in your web browser. This is the easiest way to get started without worrying about installation issues.

However, convenience comes at the cost of performance and control. Below is a comparison between cloud and local solutions.

Feature	Cloud Solution	Local Solution
Compute power	1 CPU core, 1 GB RAM	Depend on your computer
Performance	Slower; up to the server	Faster; dependNon your computer
Environment control	Limited; when your session ends, your installed packages will be removed	Full control; environment is persistent across sessions;
Internet connection	Required	Not required

Feature	Cloud Solution	Local Solution
Storage	Limited; files need to be uploaded/downloaded	Files are stored directly on your computer
Best for	Learning, experimenting, and small projects	Regular coursework, larger datasets, and research projects

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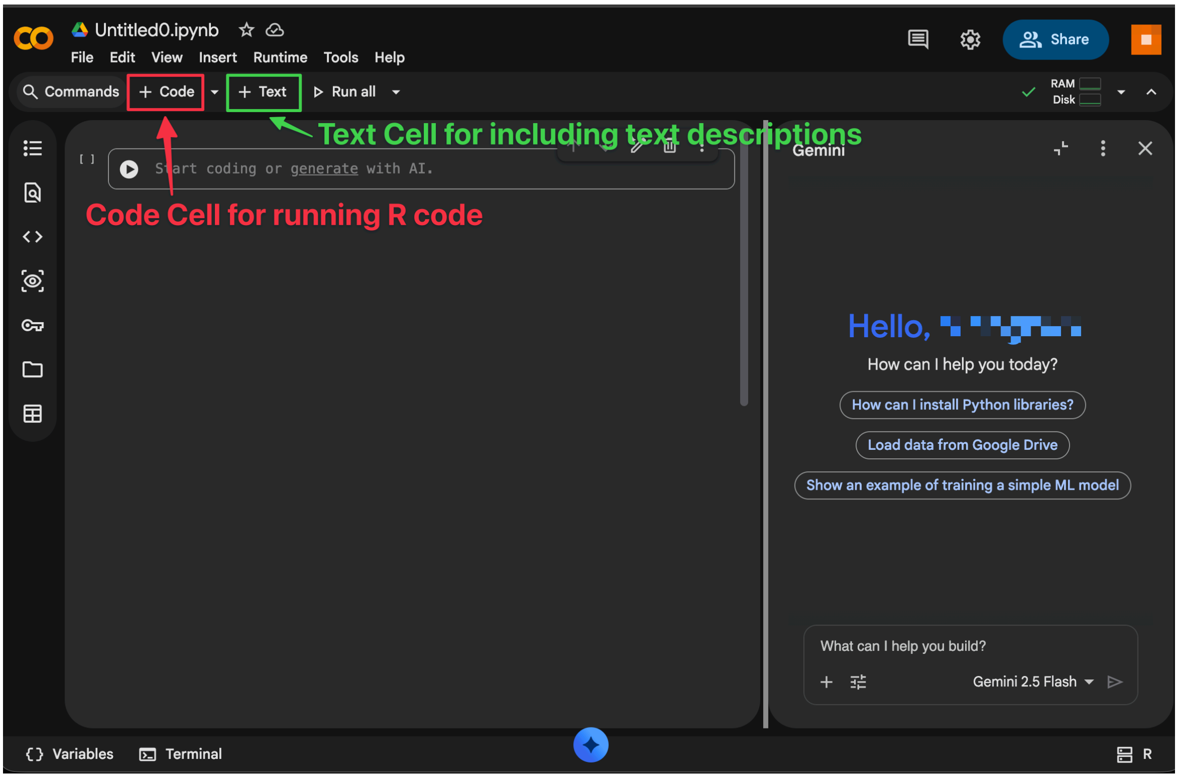
2.1.1 Google Colab

1. **Create a new R-Notebook using the following link:**

   colab.research.google.com/notebook#create=true&language=r



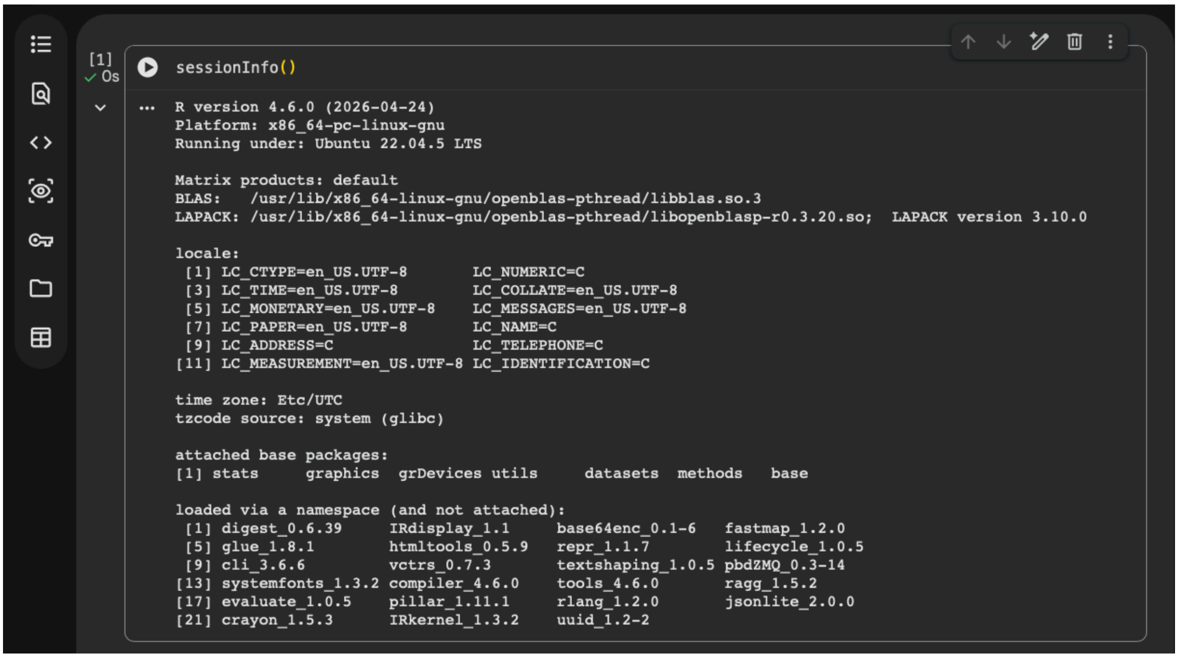
2. **Type your first R code.**

   A notebook consists of cells, which can contain either code or text. Use the toolbar to add new cells.

   For example, type the following code in a code cell and click the play button to run it:

```
sessionInfo()
```

The code output will show the R version and the packages currently loaded in your session.



Congratulations! You have successfully set up your cloud-based R environment using Google Colab. You can now start coding and exploring R without any installation hassles.

Further readings about Google Colab:

- Introduction to Google Colab Notebook
  This tutorial introduces how to upload/download data, sharing with others, editing cells, and more.

FAQ

Q: My environment is set to Python, how do I change it to R?
A: Click on the "**Runtime**" in the menu bar > Select "**Change runtime type**" > Select "**R**" from the dropdown menu > Click "**Save**".

2.2 Local Solution

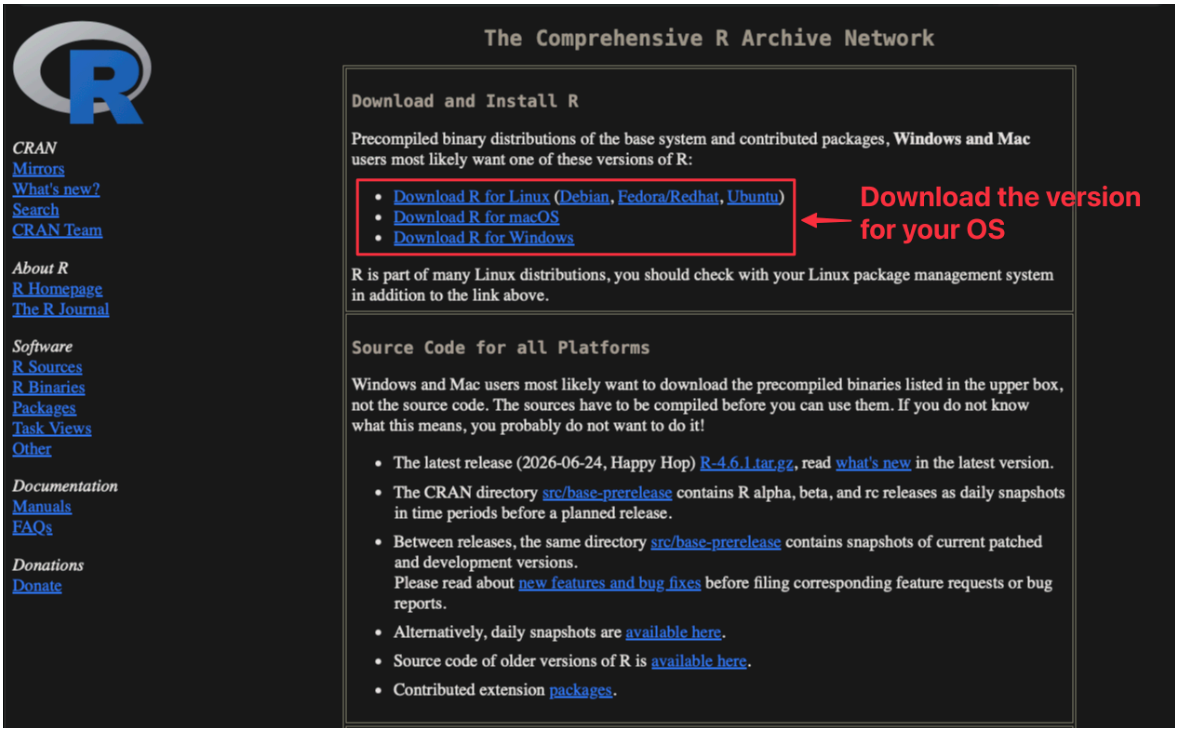
2.2.1 RStudio Desktop

RStudio Desktop is a free and open-source integrated development environment (IDE) for R. It provides a user-friendly interface for writing and executing R code, making it easier to work with R scripts, data analysis, visualization.

To install RStudio Desktop, follow these steps:

1. **Install the R programming language**

   Download the R installer from the CRAN website. Choose the appropriate version for your operating system (Windows, macOS, or Linux) and follow the installation instructions.

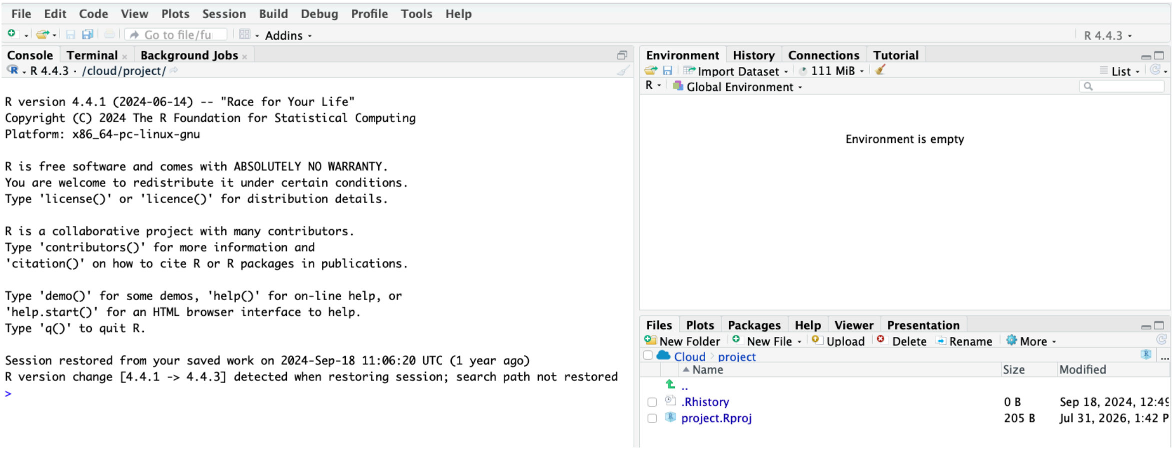
   

2. **Install RStudio IDE**

   Go to RStudio IDE Direct Download and download the installer for your operating system (Windows, macOS, or Linux).

3. **Check the installation is working**

   After installing both pieces of software, open RStudio to check the installation. It should start, and R should run in the Console window, listing the R version.



2.3 AI Coding Assistant

What AI coding assistant can do for you:

- **Code autocompletion**: generating suggestions while typing the code.
- **Code generation**: Copilot will use the context of the active document to generate suggestions for code that might be useful.
- **Answering questions**: it can also be used to ask simple questions while you are coding (e.g., "What is the definition of mean?").
- **Language support**: supports multiple programming languages, including R, Python, SQL, HTML, and JavaScript.

How to use AI coding assistant in Google Colab and RStudio:

- Google Colab has built-in support for **Gemini**.
- RStudio uses **GitHub Copilot** as the AI coding assistant.

  GitHub Copilot is a powerful AI tool that offers autocomplete-style suggestions as you code. The tool can give you suggestions based on the code you want to use or by simply inquiring about what you want the code to do.

  It is developed by GitHub in partnership with OpenAI, and it is designated to best assist developers in writing code more efficiently.

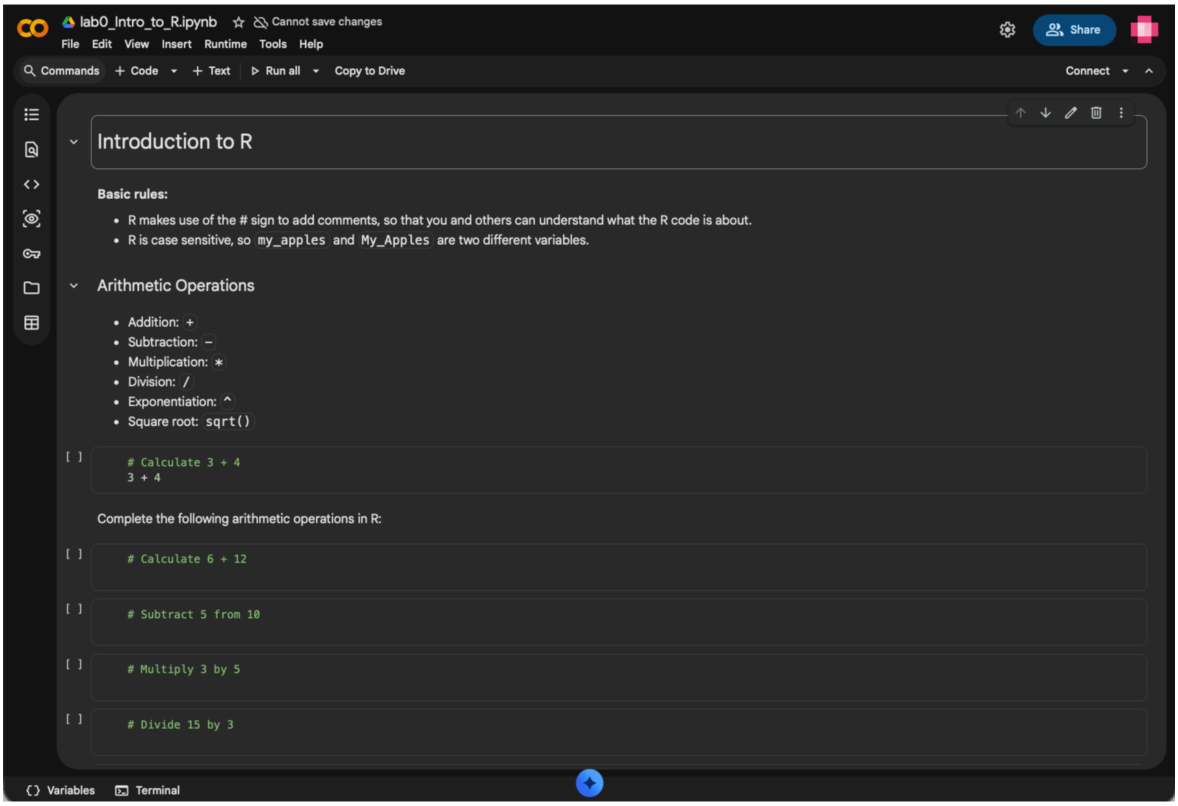
  To enable Copilot in RStudio: click on Tools → Global Options → Copilot → tick the box saying "Enable GitHub Copilot" → sign in to your GitHub account, and there you go; you are ready to start!

2.4 Exercise

Now it is your turn to get started with R programming.

Use THIS LINK to open the R Notebook in Google Colab.

The Notebook looks like the following:



Note: You only have View access to the notebook. To edit it, you need to make a copy of the notebook by clicking on "**File**" → "**Save a copy in Drive**". You can then edit the notebook in your own Google Drive.

Follow the instructions in the notebook to complete the exercises.

3 Lab 0: Introduction to R

Basic rules:

- R makes use of the # sign to add comments, so that you and others can understand what the R code is about.
- R is case sensitive, so `my_apples` and `My_Apples` are two different variables.

3.1 Arithmetic Operations

- Addition: `+`
- Subtraction: `-`
- Multiplication: `*`
- Division: `/`
- Exponentiation: `^`
- Square root: `sqrt()`

```
# Calculate 3 + 4
3 + 4
```

7

Complete the following arithmetic operations in R:

```
# Calculate 6 + 12
```

```
# Subtract 5 from 10
```

```
# Multiply 3 by 5
```

```r
# Divide 15 by 3
```

```r
# Take the square of 3
```

```r
# Take the square root of 16
```

3.2 Variable Assignments

A basic concept in (statistical) programming is called a variable.

A variable allows you to store a value (e.g. 4) or an object (e.g. a function description) in R. You can then later use this variable's name to easily access the value or the object that is stored within this variable.

```r
# Assign the value 4 to a variable called my_apples
my_apples <- 4

# Print out the value of my_apples
my_apples
```

4

```r
# Assign the value 5 to a variable called my_oranges

# Add the values of my_apples and my_oranges and assign the
↪  result to a variable called my_fruit
```

3.3 Basic Data Types

R works with numerous data types. Some of the most basic types to get started are:

- Decimal values like `4.5` are called **numerics**.
- Whole numbers like `4` are called **integers**. Integers are also numerics.
- Boolean values (`TRUE` or `FALSE`) are called **logical**.
- Text (or string) values are called **characters**.

Use `class()` function to check the data type of a variable.

```
# Assign "Nord University" to a variable called my_school

# Check class of my_school

# Check class of my_apples
```

Go to DataCamp's Introduction to R course to learn more about R programming. The course is for free if you join the DataCamp Classroom I have created for this course (Otherwise, you will see many contents are locked). The course is 4 hours long and will help you to know the basics of R programming.

Use your **Nord University email** (@nord.no or @student.nord.no) to sign up for a free account and join the DataCamp Classroom using THIS LINK. Contact me if you have any issues joining the classroom.

Part II

Basics

4 Probability and Data in Business Analytics

🎯 **Study Objectives**

- Apply probability rules to real business events.
- Distinguish clearly between population parameters and sample estimators.
- Compute and explain expectation, variance, and standard deviation in business contexts.
- Interpret higher moments: skewness and kurtosis for tail and asymmetry assessment.
- Combine random variables (e.g., portfolio or multi-channel forecast) and decompose variance with covariance terms.
- Calculate and interpret covariance and correlation; relate independence vs. zero correlation.

Probability is the foundation for making data-driven decisions in business. This section covers the essential concepts, formulas, and business examples you'll use in analytics.

4.1 Probability Axioms

Probability measures how likely an event is.

$$P(A) = \frac{\text{Number of ways A can occur}}{\text{Total possible outcomes}}$$

1. $P(A) \geq 0$ for any event A
2. $P(\text{all possible outcomes}) = 1$
3. $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$

4. If A and B are mutually exclusive, then

$$P(A \text{ and } B) = 0$$

and

$$P(A \text{ or } B) = P(A) + P(B)$$

Mutually exclusive means "cannot happen together."

Example 4.1. A retailer estimates a 30% chance of a supply chain disruption and a 50% chance of a demand spike. If these events are mutually exclusive, what is the probability of either event occurring?

Solution

Solution 4.1*.* Since the events are mutually exclusive, $P(A \text{ and } B) = 0$, then

$$P(A \text{ and } B) = P(A) + P(B) - P(A \text{ and } B) = 0.30 + 0.50 = 0.80.$$

Example 4.2. In a multi-channel marketing campaign, 35% of customers click an email offer (Event A) and 25% engage with a social media ad (Event B). Data shows 10% of customers do both. What is the probability a randomly selected customer engages with at least one of the two channels (email OR social)?

Solution

Solution 4.2*.*

$P(A) + P(B) - P(A \text{ and } B) = 0.35 + 0.25 - 0.10 = 0.50$

4.2 Sample vs. Population: Estimating the Big Picture

In business, we use samples to estimate population characteristics.

Sample mean:

$$\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

Example 4.3. A telecom company surveys a randomly selected sample of 500 customers about service quality. The sample mean satisfaction score is 4.2. How can this estimate help guide company-wide improvements?

Solution

Solution 4.3.

- The sample mean (4.2) is based on 500 surveyed customers (the sample).
- The population is all customers of the telecom company.
- To guide company-wide improvements, we need to know if the sample is representative of the population.
- There is uncertainty in the estimate; larger samples make the sample mean closer to the true population mean.
- The sample mean is useful, but its reliability depends on sample representativeness and size.

Example 4.4. Why is it important to distinguish between sample and population in business analytics?

Solution

Solution 4.4.

Because we use samples to estimate population parameters, and understanding the difference helps us make better decisions and avoid bias.

4.3 Moments

4.3.1 Expectation, Variance, and Standard Deviation

- **Expectation (population):** Average level of the outcome (true but unknown in practice).

$$\mathbb{E}[X] = \begin{cases} \sum_i x_i \, P(X = x_i) & \text{(discrete)} \\ \int_{-\infty}^{\infty} x \, f(x) \, dx & \text{(continuous)} \end{cases}$$

  Sample estimator: Replace the distribution by observed data:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^{n} X_i.$$

  The sample mean $\bar{X}$ estimates the population mean $\mathbb{E}[X]$.

- **Variance (population):** How much values fluctuate around the true mean.

$$\mathrm{Var}(X) = E[(X - E[X])^2]$$

The variance can also be expressed as:

$$\begin{aligned}\mathrm{Var}(X) &= \mathbb{E}\left[\left(X - \mathbb{E}[X]\right)^2\right] \\ &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2.\end{aligned}$$

This second form is often faster to compute when you already have (or can easily get) $\mathbb{E}[X^2]$ (e.g., from grouped data or a probability model). In finance, we frequently estimate variance from simulated or historical returns by computing the average of squared returns (giving $\mathbb{E}[X^2]$) and subtracting the square of the average return.

Sample estimators:

  - **Biased** (population style) estimator:

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2$$

  - **Unbiased** estimator:

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$$

    The latter (s^2) subtracts 1 from n in the denominator, which is known as a degrees of freedom correction. This version has some desirable properties but we will not discuss these for now. Suffice to say that both versions are usually fine in practice.

Example 4.5. An operations planner models next-hour demand arrivals for a micro-warehouse (number of urgent orders) as $X \in \{0, 1, 2\}$ with probabilities 0.2, 0.5, 0.3. Compute the variability (variance) of order arrivals using the shortcut formula $\mathbb{E}[X^2] - (\mathbb{E}[X])^2$ to inform staffing.

Solution

Solution 4.5.

$$\begin{aligned}\mathbb{E}[X] &= 0(0.2) + 1(0.5) + 2(0.3) = 1.1 \\ \mathbb{E}[X^2] &= 0^2(0.2) + 1^2(0.5) + 2^2(0.3) = 0 + 0.5 + 1.2 = 1.7 \\ \mathrm{Var}(X) &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = 1.7 - 1.1^2 = 1.7 - 1.21 = 0.49\end{aligned}$$

Example 4.6. A SaaS company tracks daily change in Daily Active Users (DAU) (in %) over 5 days: 0.4, 0.6, −0.2, 0.5, 0.7. Verify that the direct variance definition and the shortcut formula match; interpret what the variance implies for short-term engagement volatility.

Solution

Solution 4.6. Daily % changes: 0.4, 0.6, −0.2, 0.5, 0.7. Let units be **percentage points** (pp).

$$\mathbb{E}[X] = \frac{0.4 + 0.6 - 0.2 + 0.5 + 0.7}{5} = 0.4 \text{ pp}$$

$$\mathbb{E}[X^2] = \frac{0.16 + 0.36 + 0.04 + 0.25 + 0.49}{5} = \frac{1.30}{5} = 0.26 \text{ (pp)}^2$$

Variance (percentage-point squared):

$$\text{Var}(X) = 0.26 - (0.4)^2 = 0.26 - 0.16 = 0.10 \text{ (pp)}^2$$

Standard deviation = $\sqrt{0.10} \approx 0.316$ pp (about 0.32 percentage points).
Direct (long) method matches (0.10).
Interpretation: Moderate short-term volatility compared to the mean change of 0.4 pp. This suggests some fluctuations in user engagement, but not extreme.

- **Standard Deviation (population):** Square root of variance, showing average deviation from the true mean.

$$\text{sd}(X) = \sqrt{\text{Var}(X)}$$

  Sample estimator: $s = \sqrt{s^2}$ (using the unbiased s^2 above).

Example 4.7. A hedge fund analyzes daily returns of two portfolios. Portfolio A has higher mean but also higher variance. How should risk-adjusted performance be compared?

Solution

Solution 4.7.

Compare risk-adjusted metrics (e.g., Sharpe ratio = (mean - rf)/sd). Higher mean with disproportionate variance may yield lower Sharpe.

Example 4.8. A business has daily profits of \$200, \$250, \$180, \$220, and \$210. Calculate the mean and variance.

Solution

Solution 4.8. **Mean:** $\bar{x} = \dfrac{200 + 250 + 180 + 220 + 210}{5} = \dfrac{1060}{5} = 212.$

Variance calculations:

1. **Sum of squared deviations**

$$\begin{aligned}(200-212)^2 + (250-212)^2 + (180-212)^2 + (220-212)^2 + (210-212)^2 \\ = 144 + 1444 + 1024 + 64 + 4 = 2680\end{aligned}$$

2. **Population variance** (divide by $n = 5$)

$$\sigma^2 = \frac{2680}{5} = 536$$

3. **Sample variance** (unbiased, divide by $n - 1 = 4$)

$$s^2 = \frac{2680}{4} = 670$$

Population variance uses n when treating these 5 observations as the entire population; sample variance uses $n - 1$ to unbiasedly estimate $\mathrm{Var}(X)$ of a larger process.

4.3.2 Sums of Random Variables: Expectation & Variance

Population linearity of expectation:

$$E[X + Y] = E[X] + E[Y]$$

Sample counterpart: $\overline{X + Y} = \bar{X} + \bar{Y}$ (sample means add component-wise).

If X and Y are independent, the population variance of their sum is the sum of their variances:

$$\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$$

If not independent, add twice the covariance:

$$\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\mathrm{Cov}(X, Y)$$

Sample counterpart (unbiased style):

$$s_{X+Y}^2 = s_X^2 + s_Y^2 + 2s_{XY},$$

where $s_{XY} = \dfrac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X})(Y_i - \bar{Y})$.

Intuitive Example: In portfolio management, the total risk (variance) of a portfolio depends on the variances of individual assets and how they move together (covariance).

Example 4.9. Two independent projects have profit variances of 100 and 150. What is the variance of total profit?

Solution

Solution 4.9.

$100 + 150 = 250$.

4.3.2.1 Variance of Linear Transformations & Portfolios

The variance operator is the expectation of a quadratic function, so it is not linear. Instead, for constants a, b:

$$\mathrm{Var}(a + bX) = b^2 \,\mathrm{Var}(X).$$

More generally, for constants $a_i \in \mathbb{R}, i = 1, \dots, n$:

$$\mathrm{Var}\left(\sum_{i=1}^{n} a_i X_i\right) = \sum_{i=1}^{n} a_i^2 \,\mathrm{Var}(X_i) + \sum_{i=1}^{n}\sum_{j=1}^{n} a_i a_j \,\mathrm{Cov}(X_i, X_j).$$

If the X_i are pairwise independent (or uncorrelated), the double sum of covariance terms drops out for $i \neq j$, leaving only the first sum.

Example 4.10. A portfolio allocates 40% to Asset A and 60% to Asset B. $\mathrm{Var}(A) = 0.04$, $\mathrm{Var}(B) = 0.09$, and $\mathrm{Cov}(A, B) = 0.01$. Compute portfolio variance using the general formula.

Solution

Solution 4.10.

$\mathrm{Var}(P) = 0.4^2(0.04) + 0.6^2(0.09) + 2(0.4)(0.6)(0.01) = 0.0064 + 0.0324 + 0.0048 = 0.0436$.

Example 4.11. A production planner combines forecasts: Final forecast $F = 2 + 0.7X_1 + 0.3X_2$. If $\mathrm{Var}(X_1) = 25$, $\mathrm{Var}(X_2) = 9$, and $\mathrm{Cov}(X_1, X_2) = 6$, find $\mathrm{Var}(F)$.

Solution

Solution 4.11.

Constant 2 adds no variance. $\text{Var}(F) = 0.7^2(25) + 0.3^2(9) + 2(0.7)(0.3)(6) = 12.25 + 0.81 + 2.52 = 15.58$.

4.3.3 Skewness

Skewness measures asymmetry in a distribution.

- **Skewness = 0:** Symmetric (e.g., normal)
- **Skewness < 0:** Longer left tail (large losses more likely than large gains)
- **Skewness > 0:** Longer right tail (occasional big gains)

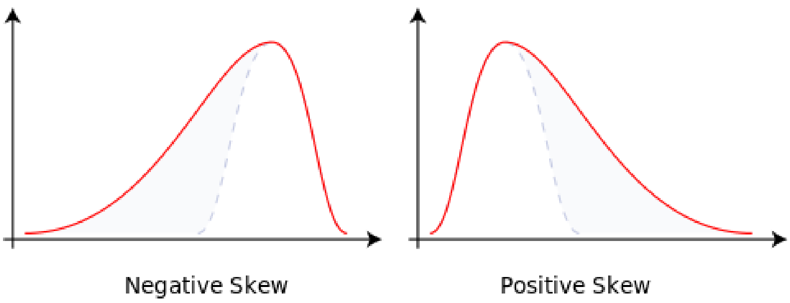


Figure 4.1 Diagram of Skewness.

Business reading: Positive skew in quarterly sales means you usually hit average numbers but sometimes land a very big contract. Negative skew in operational losses could mean rare but severe downside events that need contingency planning.

Let $\mu_3 = E[(X - \mathbb{E}[X])^3]$ denote the third central moment. Standardized (population) skewness:

$$\gamma_1 = \frac{\mu_3}{\text{sd}(X)^3}.$$

Sample (bias-adjusted) skewness estimator:

$$\hat{\gamma}_1 = \frac{n}{(n-1)(n-2)} \sum_{i=1}^{n} \left(\frac{X_i - \bar{X}}{s} \right)^3.$$

> **Tip**
>
> Note for skewness and kurtosis, it is NOT expected to hand calculate the values, the focus is on

- the interpretation of given results.
- be able to recognize skewness and kurtosis in data visualizations (e.g., histograms, box plots).

4.3.4 Kurtosis

Kurtosis measures tail weight (propensity for outliers) relative to a normal distribution.

- **Kurtosis ≈ 3:** Normal tail thickness
- **Kurtosis > 3 (leptokurtic):** Heavy / fat tails, more extreme outcomes
- **Kurtosis < 3 (platykurtic):** Light / thin tails, fewer extremes

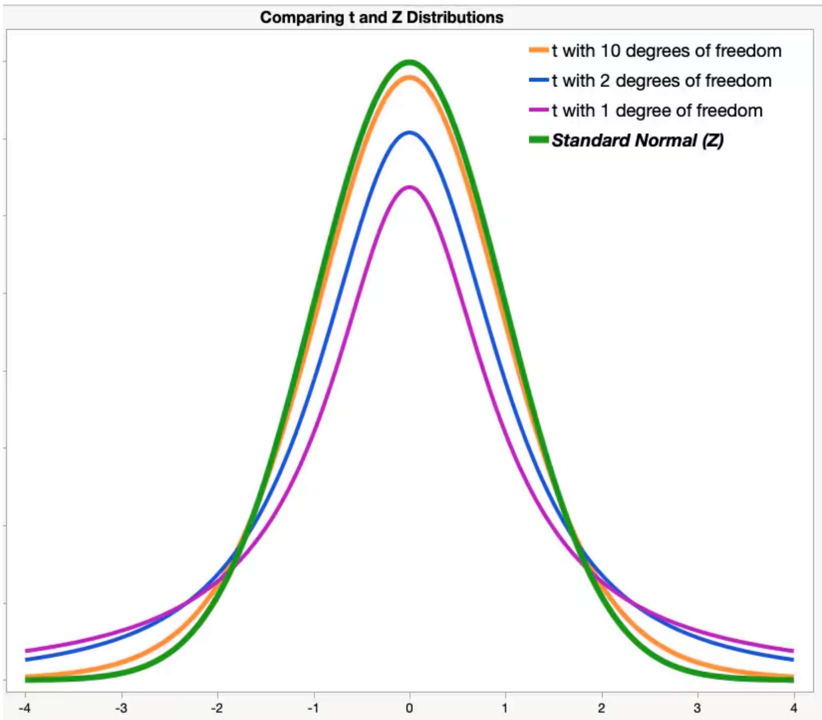


Figure 4.2 Examples of heavy-tailed distributions.

t-distribution has *higher* kurtosis than normal distributions.

- Meaning that t-distribution has a higher probability of obtaining values that are far from the mean than a normal distribution.
- It is less peaked in the center and higher in the tails than normal distribution.

- As the degree of freedom increases, t-distribution approximates to normal distribution, kurtosis decreases and approximates to 3.

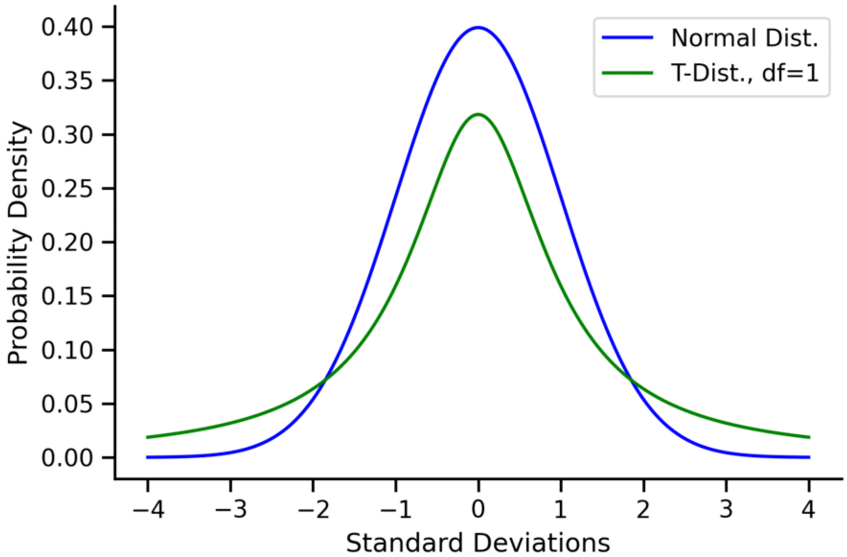


Figure 4.3 The comparison between the t-distribution and the normal distribution at degrees of freedom ranging from 1 to 50. Figure source: from T.J. Kyner.

For the t-distribution:

- As the degrees of freedom (df) increase, the t-distribution approaches the normal distribution.
- A common rule of thumb is that for $df > 30$, one can pretty safely use the normal distribution in place of a t-distribution unless you are interested in the extreme tails.

Let $\mu_4 = E[(X - \mathbb{E}[X])^4]$ be the fourth central moment. Standardized (population) kurtosis:

$$\kappa = \frac{\mu_4}{\mathrm{sd}(X)^4}, \qquad \text{Excess kurtosis} = \kappa - 3.$$

Sample (Fisher) excess kurtosis estimator:

$$\hat{g}_2 = \frac{n(n+1)}{(n-1)(n-2)(n-3)} \sum_{i=1}^{n} \left(\frac{X_i - \bar{X}}{s} \right)^4 - \frac{3(n-1)^2}{(n-2)(n-3)}.$$

High excess kurtosis signals greater tail risk; low or negative excess indicates fewer extreme outcomes.

Interpretation link: High positive skewness without high kurtosis means upside spikes with limited extra tail risk; high kurtosis (regardless of skew sign) means fatter tails and a need to focus on outlier management.

Example 4.12. A company's quarterly profits have excess kurtosis of 4. What does this mean for risk management?

Solution

Solution 4.12. Excess kurtosis is defined relative to a normal distribution (which has kurtosis 3).

- Excess kurtosis = Kurtosis − 3
- So an excess kurtosis of 4 means the total kurtosis is **7**, much higher than a normal distribution.

This implies the distribution of quarterly profits has **fat tails** — extreme profit or loss outcomes are more likely than under a normal distribution. Recommend to focus risk management on outliers.

4.4 Covariance, Correlation & Independence

Population covariance measures the direction of comovement between two variables:

- **Positive covariance:** Both variables increase or decrease together
- **Negative covariance:** One increases while the other decreases
- **Magnitude:** Does not indicate strength of relationship

Population formulas:

$$\mathrm{Cov}(X,Y) = E[(X - E[X])(Y - E[Y])]$$

$$\mathrm{Cov}(X,Y) = E[XY] - E[X]E[Y]$$

Sample covariance (unbiased):

$$s_{XY} = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y}).$$

Population correlation standardizes covariance to a scale from -1 to 1:

$$\mathrm{Corr}(X,Y) = \frac{\mathrm{Cov}(X,Y)}{\mathrm{sd}(X)\cdot\mathrm{sd}(Y)}$$

Sample correlation:

$$r_{XY} = \frac{s_{XY}}{s_X s_Y}.$$

- **Correlation = 1:** Perfect positive relationship
- **Correlation = -1:** Perfect negative relationship
- **Correlation = 0:** No linear relationship

Independence means knowing one event doesn't affect the other:

$$P(A \text{ and } B) = P(A) \cdot P(B)$$

Independence vs. Correlation

Independence means no influence between events, while correlation measures linear relationship strength.
Independence implies zero correlation, but zero correlation does not imply independence.
E.g., X and $Y = X^2$ have zero correlation but are not independent.

Business Example: Suppose a retailer finds that sales and weather are correlated. This information can be used to adjust inventory planning for seasonal effects. If two marketing campaigns are independent, the outcome of one does not affect the other.

Example 4.13. Suppose a customer visits a supermarket. Let:

- A = "customer buys a loaf of bread today" and $P(A) = 30\%$.
- B = "customer buys laundry detergent today" and $P(B) = 10\%$.

What is the probability the customer buys both bread and detergent today? What does observing a bread purchase tell you about $P(B)$?

Solution

Solution 4.13. Bread (frequent staple) and detergent (infrequent refill) decisions are unrelated, so it is reasonable to assume independence. Hence

$$P(A \text{ and } B) = 30\% \times 10\% = 3\%$$

Observing bread does not change $P(B) = 10\%$.

Example 4.14. If the correlation between two stocks is -0.8, what does this mean for portfolio diversification?

Solution

Solution 4.14.

They tend to move in opposite directions, aiding diversification and reducing portfolio variance.

Mutually Exclusive vs Independence

Mutually exclusive: Cannot happen together

$$P(A \cap B) = 0$$

Independent: Knowledge of one does not change probability of the other

$$P(A \cap B) = P(A)P(B)$$

Think: Disjoint = "never together"; Independent = "no influence".

4.5 Population vs. Sample: Summary & Estimators

When analysing data we distinguish between:

- **Population parameters (target):** Fixed (but usually unknown) numerical characteristics of the underlying process (e.g., $\mathbb{E}[X], \mathrm{Var}(X), \gamma_1, \kappa, \mathrm{Cov}(X, Y)$).
- **Sample statistics (estimators):** Functions of observed data used to approximate population parameters (e.g., $\bar{X}, s^2, s, \hat{\gamma}_1, \hat{g}_2, s_{XY}, r_{XY}$). They vary from sample to sample.

Key principles:

1. Replace integrals / probability-weighted sums with empirical averages.
2. Center around the sample mean when the population mean is unknown.
3. Use $n - 1$ (unbiased) denominators for second moments (variance, covariance) when estimating from an i.i.d. sample.
4. Always interpret sample statistics as **estimates** with sampling variability; risk management and forecasting should account for estimation error (e.g., standard deviation and confidence intervals).

4.5.1 Quick Reference Table

Concept	Population Symbol / Definition	Sample Estimator	Notes
Mean	$\mathbb{E}[X]$	$\bar{X} = \frac{1}{n}\sum X_i$	Unbiased for mean (i.i.d.)
Variance	$\mathrm{Var}(X) = E[(X - \mathbb{E}[X])^2]$	$s^2 = \frac{1}{n-1}\sum(X_i - \bar{X})^2$	s_n^2 (divide by n) is biased low
Std. Dev.	$\mathrm{sd}(X) = \sqrt{\mathrm{Var}(X)}$	$s = \sqrt{s^2}$	Plug-in
Skewness	$\gamma_1 = \mu_3/\mathrm{sd}^3$, $\mu_3 = E[(X - \mathbb{E}[X])^3]$	$\hat{\gamma}_1 = \frac{n}{(n-1)(n-2)}\sum(\frac{X_i - \bar{X}}{s})^3$	Measures asymmetry
Kurtosis (excess)	$\kappa - 3$, $\kappa = \mu_4/\mathrm{sd}^4$	$\hat{g}_2$ (Fisher)	Tail heaviness vs. normal
Covariance	$\mathrm{Cov}(X, Y) = E[(X - \mathbb{E}X)(Y - \mathbb{E}Y)]$	$s_{XY} = \frac{1}{n-1}\sum(X_i - \bar{X})(Y_i - \bar{Y})$	Sign = direction
Correlation	$\rho = \dfrac{\mathrm{Cov}(X, Y)}{\mathrm{sd}(X)\mathrm{sd}(Y)}$	$r_{XY} = \dfrac{s_{XY}}{s_X s_Y}$	Scale-free (-1 to 1)

> **Interpretation tip**
>
> If a sample statistic looks extreme (e.g., very high skewness or kurtosis), examine sample size and outliers; small samples amplify noise in higher-moment estimates.

5 Lab 1: Probability & Descriptive Statistics

5.1 Instructions

Open in Google Colab: Link

Alternatively, copy and paste the code below into RStuido Desktop or RStudio Cloud.

Finish Quiz for Lab 1 on Canvas.

Focus: explore core descriptive statistics and dependence concepts using simple synthetic data.

You will:

- Examine independence & correlation
- Compare skewed and roughly normal distributions
- Compute mean, variance, sd, skewness, kurtosis
- Compare biased vs unbiased variance estimators
- Simulate a high-correlation ($\rho = 0.90$) two-asset return setting
- Study sampling distributions of asset and portfolio means

5.2 Datasets

Synthetic generators used:

- Right-skewed transaction values (log-normal)and approx. normal reference sample
- Two correlated asset return series ($\rho = 0.90$) for sampling exercises

Run each cell sequentially. Read the comments carefully.

```
# -------- Setup: packages --------
# Unified required package list
pkgs <- c("tidyverse", "data.table", "ggsci", "moments",
 ↪  "knitr", "kableExtra", "IRdisplay")
missing <- setdiff(pkgs, rownames(installed.packages()))
if (length(missing) > 0) install.packages(missing)

# Load all packages (silently)
invisible(lapply(pkgs, function(pkg) {
  suppressWarnings(suppressPackageStartupMessages(library(pkg,
   ↪  character.only = TRUE)))
}))

# Set default options for figures in Jupyter Notebook
options(repr.plot.width = 12, repr.plot.height = 4)  # wider
 ↪  default figures

message("\nSetup complete (packages loaded: ", paste(pkgs,
 ↪  collapse = ", "), ").")
```

```
Setup complete (packages loaded: tidyverse, data.table, ggsci, moments, knitr
```

5.3 Independence vs. Zero Correlation

Independence means no influence between events, while correlation measures linear relationship strength.

Independence implies zero correlation, but zero correlation does not imply independence.

E.g., X and $Y = X^2$ have zero correlation but are not independent.

```
# ---- Zero Correlation but NOT Independence: Y = X^2 ----
# Idea: X ~ N(0,1); define Y = X^2.  Then cor(X,Y) ~ 0 (symmetry
 ↪  cancels linear relation),
# yet Y is a deterministic function of X so they are NOT
 ↪  independent.

set.seed(2025)
n <- 5000
X <- rnorm(n)
Y <- X^2
r_xy <- cor(X, Y)
cat(sprintf("Sample correlation cor(X, Y=X^2) = %.4f (near
 ↪  0)\n", r_xy))
```

```
Sample correlation cor(X, Y=X^2) = 0.0198 (near 0)
```

We got $\rho_{X,Y} = 0.0198$. It seems to be close to zero, but we need a statistical test to confirm this formally.

→ Test for significance of the correlation coefficient

```r
cor.test(X, Y)
```

```

	Pearson's product-moment correlation

data:  X and Y
t = 1.4027, df = 4998, p-value = 0.1608
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.007887062  0.047529726
sample estimates:
       cor 
0.01983657 
```

💡 **Q: Based on the output of `cor.test`, what is the p-value for the correlation test? What can you conclude about the correlation between the two variables?** Are X and Y independent? Write your answer in the cell below.

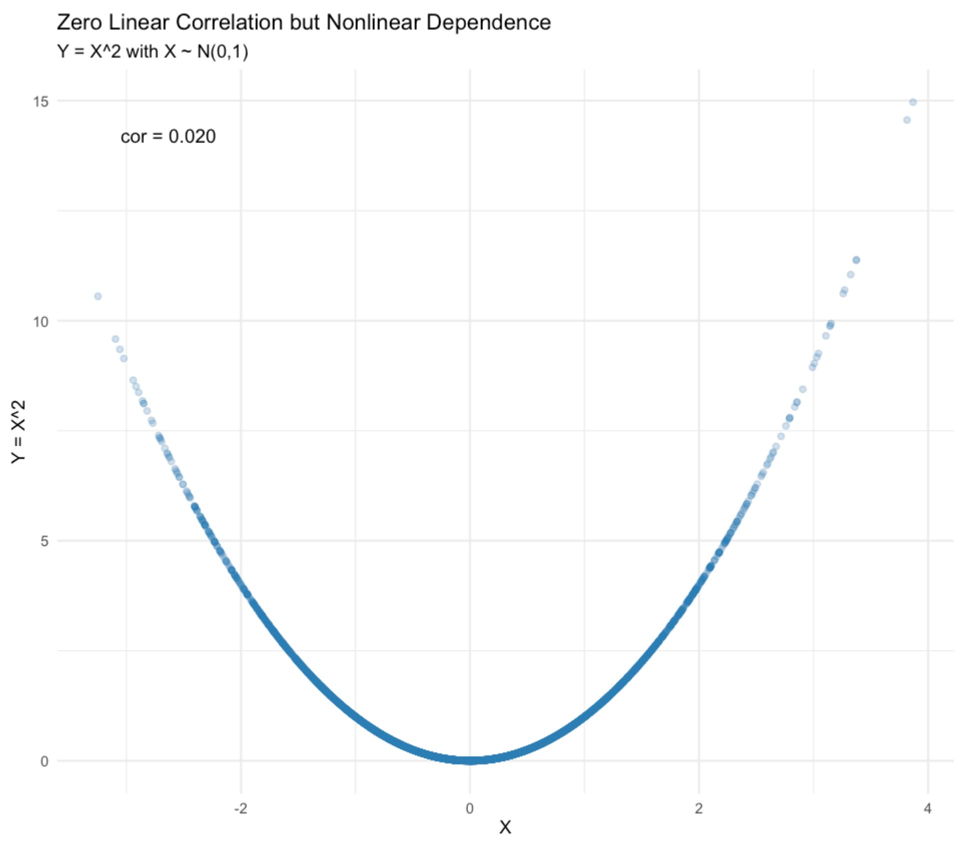
A: [Type your answer here]

Distinguish the two expressions:

- correlation coefficient equals zero ❌
- correlation coefficient (statistically) indifferent from zero ✅

Now let's visualize $Y = X^2$ by plotting the scatter plot.

```r
library(ggplot2)
options(repr.plot.width = 8, repr.plot.height = 7)
scatter_df <- data.frame(X = X, Y = Y)
p1 <- ggplot(scatter_df, aes(X, Y)) +
  geom_point(alpha = 0.25, color = '#2c7fb8') +
  annotate('text', x = min(X)+0.2, y = max(Y)*0.95,
           label = paste0('cor = ', sprintf('%.3f', r_xy)),
           hjust = 0, size = 4) +
  labs(title = 'Zero Linear Correlation but Nonlinear
   ↪  Dependence',
       subtitle = 'Y = X^2 with X ~ N(0,1)',
       x = 'X', y = 'Y = X^2') +
  theme_minimal()
print(p1)
```



The following figure shows the scatter plot of two assets with various correlation coefficients.

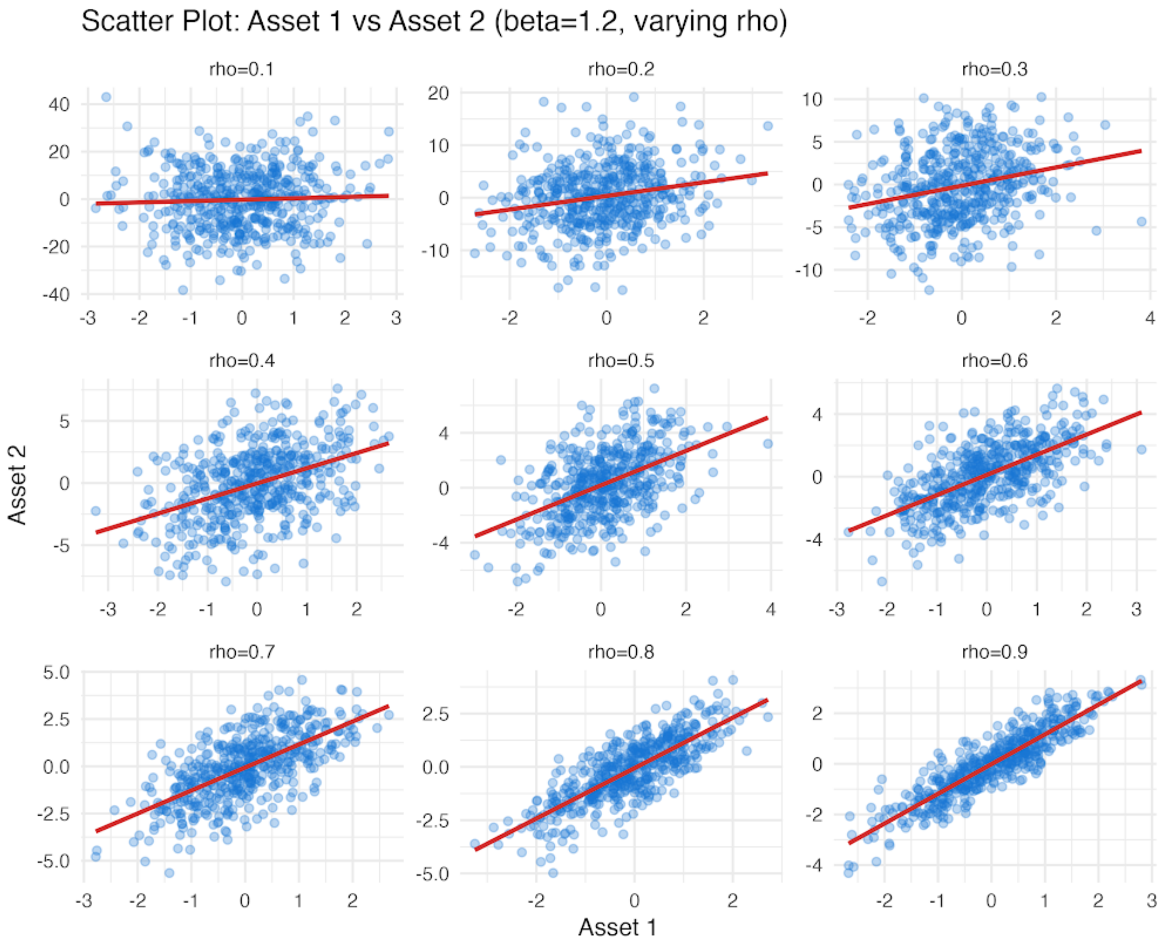


Figure 5.1 correlation_varying_rho

5.4 Descriptive Statistics on Asset Returns

```
# ---- Load Asset Returns Data ----
asset_df <-
 ↪  read_csv("https://raw.githubusercontent.com/my1396/FIN5005-Fall2025/refs/
print("Preview: first 6 rows of asset returns data frame:")
head(asset_df) %>% round(2)
```

```
Rows: 10000 Columns: 3
── Column specification ────────────────────────────────────────────
Delimiter: ","
dbl (3): Asset_A, Asset_B, Asset_C

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
[1] "Preview: first 6 rows of asset returns data frame:"
```

A tibble: 6 × 3

Asset_A <dbl>	Asset_B <dbl>	Asset_C <dbl>
17.89	4.31	9.94
10.30	4.17	10.55
20.87	14.16	6.61
18.21	22.33	9.58
17.49	2.23	10.23
29.87	15.72	8.97

We define a summary statistics function to compute the statistics of interest.

```r
quick_summary <- function(x) {
  # Function to compute basic descriptive statistics
  data.frame(
    n = length(x),
    mean = mean(x),
    sd = sd(x),
    var = var(x),
    skewness = moments::skewness(x),
    kurtosis = moments::kurtosis(x),
    row.names = NULL
  )
}
```

The summary statistic table is generated as follows:

```r
results <- lapply(asset_df, quick_summary) %>%
    do.call(rbind, .) %>%
    kable(format = "html", digits = 2, caption = "Descriptive
     ↪  Statistics of Asset Returns")
display_html(as.character(results))
```

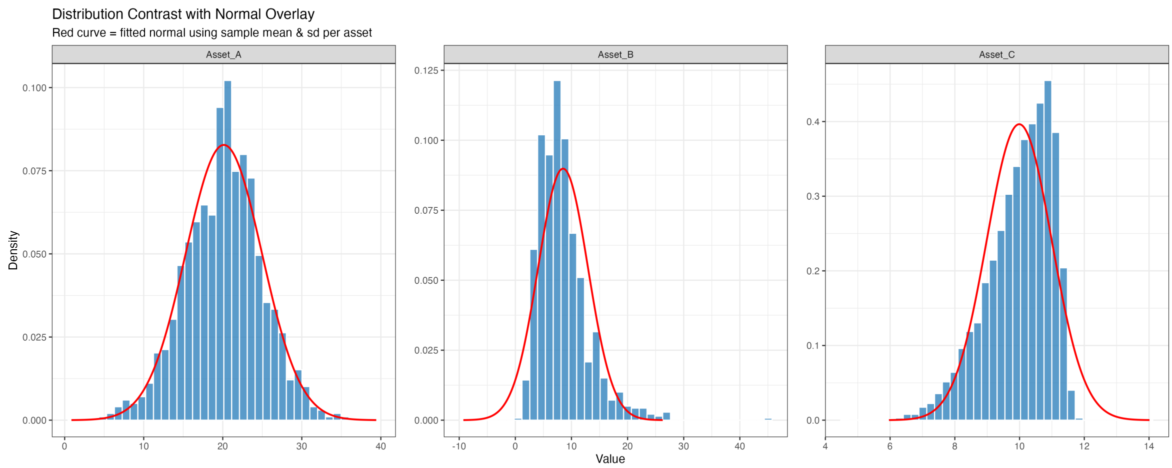
Table 5.2 Descriptive Statistics of Asset Returns

	n	mean	sd	var	skewness	kurtosis
Asset_A	10000	20.14	4.82	23.24	-0.10	3.15

	n	mean	sd	var	skewness	kurtosis
Asset_B	10000	8.50	4.44	19.71	1.71	9.19
Asset_C	10000	9.99	1.01	1.01	-0.85	3.50

Plot the histograms of the asset returns to visualize their distributions.

```
# ---- Visualization: Histograms with Normal Density Overlay ----
# Convert to long form
combined <- asset_df %>% pivot_longer(cols = everything(),
 ↪  names_to = "type", values_to = "value")

# Compute per-type mean & sd and create normal curve points
norm_params <- combined %>% group_by(type) %>% summarise(mu =
 ↪  mean(value), sigma = sd(value), .groups = 'drop')

norm_curve <- norm_params %>% group_by(type) %>% do({
  mu <- .$mu; sigma <- .$sigma
  x <- seq(mu - 4*sigma, mu + 4*sigma, length.out = 400)
  tibble(value = x, density = dnorm(x, mu, sigma))
})

p <- ggplot(combined, aes(value)) +
  geom_histogram(aes(y = after_stat(density)), bins = 40, fill =
   ↪  "#3182bd", alpha = 0.8, color = "white") +
  geom_line(data = norm_curve, aes(value, density), color =
   ↪  "red", linewidth = 0.8) +
  facet_wrap(~ type, scales = "free", nrow = 1) +
  labs(title = "Distribution Contrast with Normal Overlay",
       subtitle = "Red curve = fitted normal using sample mean &
        ↪  sd per asset",
       x = "Value", y = "Density") +
  theme_bw(base_size = 16) +
  theme(panel.spacing.x = unit(1.2, "lines"))

# Save the plot to a file
f_name <- "images/asset_returns_histograms.png"
if (!dir.exists("images")) dir.create("images")
ggsave(f_name, p, width = 15, height = 6)
```



💡 **Q: Answer the following questions based on the summary statistics and the distribution plots:**

- Which asset has the highest mean return?
- Which asset is the most risky? and which one has the most stable returns?
- Which asset shows the positive skewness and what does it mean?
- If an investor prefers assets with low risk and return distributions with tails close to normal, which asset is most appropriate?
- Based on mean and standard deviation, which asset has the best risk–return trade-off (highest mean per unit of risk)?

A: [Type your answer here]

5.5 Large Sample Gives More Accurate Estimates

We have a variable $X \sim N(0, 5^2)$.

Normal distribution notation: $X \sim N(\mu, \sigma^2)$ means that X is normally distributed with mean μ and variance σ^2.

The second parameter is the variance, **NOT** the standard deviation.

Now we draw random samples of size 5, 10, 10, ... until 5000 from this distribution and compute the sample mean and standard deviation for each sample.

For each sample, we compute the biased and unbiased sample variance

- The biased sample variance is computed as

$$\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2.$$

- The unbiased sample variance is computed as

$$\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

```
# ---- Simulation: Sample vs Population variance ----
set.seed(2025)  # for reproducibility
true_var <- 25  # sd^2 with sd=5, true population variance
sample_sizes <- c(5, 10, 20, 50, 100, 250, 500, 1000, 5000) #
 ↪  varying sample sizes from 5 to 500
results <- lapply(sample_sizes, function(n){
  x <- rnorm(n, mean = 0, sd = 5)
  data.frame("sample_size" = n, "var_biased" = mean((x -
   ↪  mean(x))^2), "var_unbiased" = var(x))
}) %>% dplyr::bind_rows()
cat('Variance estimators: var_n (biased, divide by n) vs var_n1
 ↪  (unbiased, divide by n-1).\n')
results
```

```
Variance estimators: var_n (biased, divide by n) vs var_n1 (unbiased, divide by n
```

A data.frame: 9 × 3

sample_size <dbl>	var_biased <dbl>	var_unbiased <dbl>
5	4.262707	5.328384
10	14.832929	16.481032
20	33.210505	34.958426
50	25.332410	25.849398
100	24.957059	25.209150
250	22.936725	23.028840
500	25.862610	25.914439

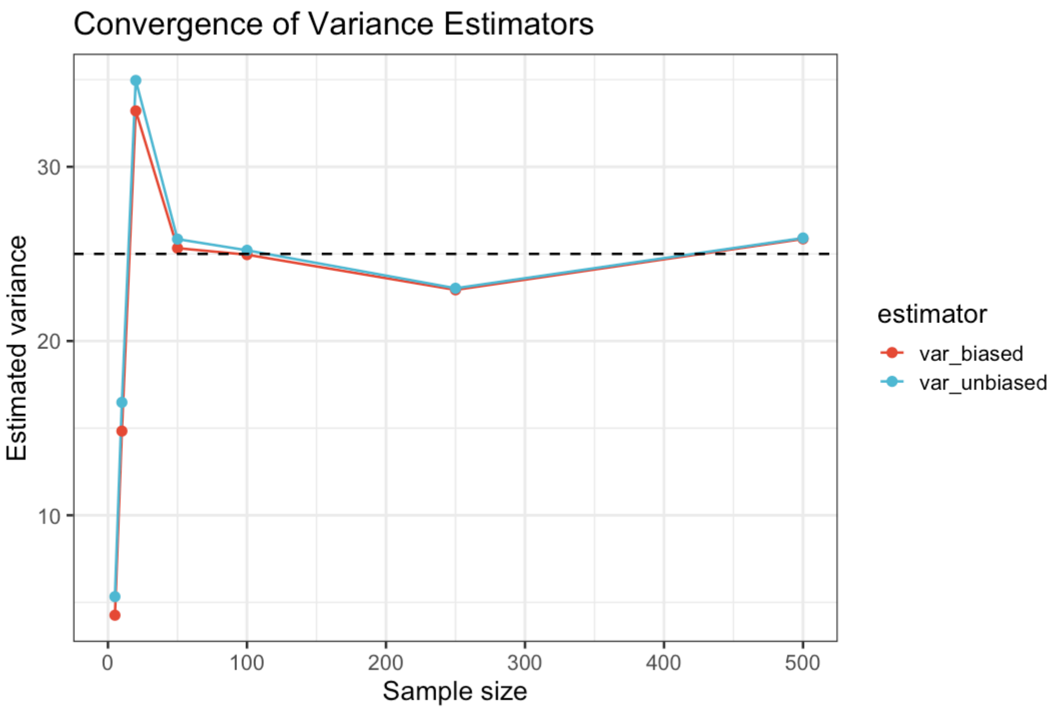
sample_size <dbl>	var_biased <dbl>	var_unbiased <dbl>
1000	25.574182	25.599782
5000	25.301658	25.306719

💡 **Q: Based on the variance estimates for different sample sizes:**

- **As the sample size increases, how do the sample variance estimates compare to the true variance 25?**
- **Does the sample variance converge to the true variance as the sample size increases?**
- **What is the difference between the biased and unbiased sample variance estimators?**

A: [Type your answer here]

```r
# ---- Plot: Convergence of variance estimators ----

results_long <- results %>%
  tidyr::pivot_longer(var_biased:var_unbiased, names_to =
  ↪  "estimator", values_to = "value") %>%
  dplyr::mutate(estimator = dplyr::recode(estimator, var_n =
  ↪  "Divide by n", var_n1 = "Divide by n-1"))

p <- ggplot(results_long, aes(sample_size, value, color =
  ↪  estimator)) +
  geom_line() +
  geom_point() +
  scale_color_npg() +
  geom_hline(yintercept = true_var, linetype = "dashed") +
  xlim(c(0,500)) +
  labs(title = "Convergence of Variance Estimators", y =
  ↪  "Estimated variance", x = "Sample size") +
  theme_bw(base_size = 16)
p
```



5.6 Correlated Asset Returns

We will simulate repeated samples of two (correlated) asset return series and compare the sampling distributions of their sample means and the distribution of the portfolio mean (equal weights).

```
# ---- Load Correlated Asset Returns Data ----
X_demo <-
 ↪  read_csv("https://raw.githubusercontent.com/my1396/FIN5005-Fall2025/refs/
colnames(X_demo) <- c("Asset_A", "Asset_B")  # Rename columns
 ↪  for clarity
X_demo %>%
    as.data.table() %>%
    .[, lapply(.SD, function(x) if (is.numeric(x)) round(x, 4)
 ↪  else x)] %>%
    print(digits = 4)
```

```
Rows: 2000 Columns: 2
── Column specification ────────────────────────────────────────
Delimiter: ","
dbl (2): V1, V2

 Use `spec()` to retrieve the full column specification for this data.
 Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
       Asset_A Asset_B
         <num>   <num>
   1:  0.0129  0.0351
   2:  0.0012  0.0094
   3:  0.0160  0.0130
   4:  0.0259  0.0550
   5:  0.0079  0.0147
  ---
1996:  0.0080  0.0177
1997:  0.0061 -0.0001
1998: -0.0299 -0.0425
1999: -0.0202 -0.0357
2000:  0.0603  0.0746
```

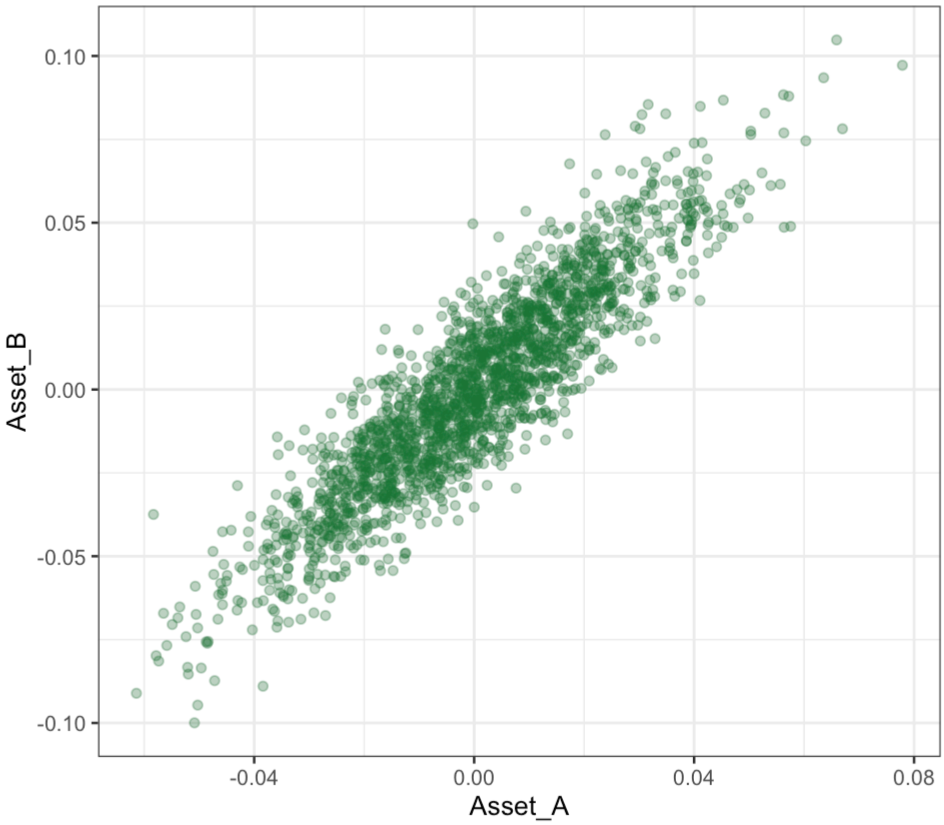
```r
emp_cor <- cor(X_demo)
emp_cor
```

A matrix: 2 × 2 of type dbl

	Asset_A	Asset_B
Asset_A	1.0000000	0.9040318
Asset_B	0.9040318	1.0000000

Plot the scatter plot of the two asset returns to visualize their relationship.

```r
options(repr.plot.width = 8, repr.plot.height = 7)
ggplot(X_demo, aes(Asset_A, Asset_B)) +
  geom_point(alpha = 0.35, color = '#1b7837') +
  theme_bw(base_size = 16)
```



💡**Q: What is the correlation between the two asset returns? How do their prices move together?**

A: [Type your answer here]

5.6.1 Exercise

Following the steps below, you will compute the mean and standard deviation of the two asset returns, construct a portfolio, and analyze its properties.

Step 1: Calculate the standard deviation of Asset A and B returns, respectively.

Step 2: Construct a equally weighted portfolio of the two assets. Calculate the portfolio returns.

$$E[r_p] = \frac{1}{2}(E[r_A] + E[r_B])$$

Step 3: Calculate the standard deviation of the portfolio returns, $\mathrm{sd}(r_p)$.

Step 4: Calculate the average of the volatility of Asset A and B (using results in Step 1). Compare it with the volatility of the portfolio returns.

5.6.1 Reflection

How does correlation between assets influence the portfolio mean's variability? Write 3–4 sentences interpreting for a diversified vs concentrated investment decision.

6 Data Visualization

🎯 **Study Objectives**

- Loads and prepares the California school dataset (CASchools) for analysis.
  - Demonstrates how to visualize data using scatter plots, line charts, histograms, box plots.
  - Explores expenditure per student across different grade spans.
  - Investigates correlations between expenditure and other variables.
- QQ plots to assess normality and interpret skewness and kurtosis in the data.
  - Given a QQ plot, you should be able to identify how does the distribution compare to normal: positively skewed, negatively skewed, has fat tails, or thin tails.

6.1 Descriptive Analysis and Data Visualization with `CASchools`

Data visualization is a powerful tool for understanding patterns, relationships, and distributions in datasets. In this session, we will explore the California Schools dataset using various visualization techniques including histograms, box plots, scatter plots, and Q-Q plots. These visualizations help us answer important questions about educational spending, student performance, and the relationships between various school characteristics.

6.1.1 Dataset Overview

The following prints the first and last 5 rows of the dataset, together with data types of each column:

```
# load packages and dataset
pkgs <- c("tidyverse", "moments", "data.table", "ggsci")
missing <- setdiff(pkgs, rownames(installed.packages()))
if (length(missing) > 0) install.packages(missing)
invisible(lapply(pkgs, function(pkg) {
    suppressPackageStartupMessages(library(pkg, character.only =
     ↪  TRUE))
}))
f_name <-
 ↪  "https://raw.githubusercontent.com/my1396/FIN5005-Fall2025/refs/heads/m
cas <- read_csv(f_name,
  col_types = cols(
    county = col_factor(), # read as factor
    grades = col_factor()
  ))
```

The following prints the first and last 5 rows of the dataset, together with data types of each column:

```
cas %>% as.data.table()
```

```
    district                           school     county grades students
       <num>                           <char>     <fctr> <fctr>    <num>
  1:   75119               Sunol Glen Unified    Alameda  KK-08      195
  2:   61499              Manzanita Elementary      Butte  KK-08      240
  3:   61549     Thermalito Union Elementary      Butte  KK-08     1550
  4:   61457 Golden Feather Union Elementary      Butte  KK-08      243
  5:   61523        Palermo Union Elementary      Butte  KK-08     1335
 ---
416:   68957         Las Lomitas Elementary  San Mateo  KK-08      984
417:   69518          Los Altos Elementary Santa Clara  KK-08     3724
418:   72611         Somis Union Elementary    Ventura  KK-08      441
419:   72744               Plumas Elementary       Yuba  KK-08      101
420:   72751            Wheatland Elementary       Yuba  KK-08     1778
    teachers calworks   lunch computer expenditure   income   english  read
       <num>    <num>   <num>    <num>       <num>    <num>     <num> <num>
  1:   10.90   0.5102  2.0408       67    6384.911 22.690001  0.000000 691.6
  2:   11.15  15.4167 47.9167      101    5099.381  9.824000  4.583333 660.5
  3:   82.90  55.0323 76.3226      169    5501.955  8.978000 30.000002 636.3
  4:   14.00  36.4754 77.0492       85    7101.831  8.978000  0.000000 651.9
  5:   71.50  33.1086 78.4270      171    5235.988  9.080333 13.857677 641.8
 ---
416:   59.73   0.1016  3.5569      195    7290.339 28.716999  5.995935 700.9
417:  208.48   1.0741  1.5038      721    5741.463 41.734108  4.726101 704.0
418:   20.15   3.5635 37.1938       45    4402.832 23.733000 24.263039 648.3
```

```
419:   5.00 11.8812 59.4059    14   4776.336  9.952000  2.970297 667.9
420:  93.40  6.9235 47.5712   313   5993.393 12.502000  5.005624 660.5
      math
     <num>
  1: 690.0
  2: 661.9
  3: 650.9
  4: 643.5
  5: 639.9
 ---
416: 707.7
417: 709.5
418: 641.7
419: 676.5
420: 651.0
```

The `cas` dataset contains information on 420 public school districts in California, specifically those that operate either kindergarten through 6th grade (K–6) or kindergarten through 8th grade (K–8) schools, with data collected for the years 1998 and 1999.

The dataset includes:

- **Student performance measures** (average reading and math scores).
- **School characteristics** (such as enrollment, number of teachers, student–teacher ratio, number of computers, expenditure per student).
- **Student demographics** (including income, percentage qualifying for reduced-price lunch, percentage of English learners, etc.).

`cas` is a data frame containing 420 observations on 14 variables. Refer to the following table for the full definition.

Variable	Description
`district`	Character. District code. Unique ID.
`school`	Character. School name.
`county`	Factor indicating county.
`grades`	Factor indicating grade span covered by the district's schools. It takes on two values: - `KK-06`: means the district's schools serve kindergarten through 6th grade.- `KK-08`: means the district's schools serve kindergarten through 6th grade.
`students`	Total enrollment.

Variable	Description
`teachers`	Number of teachers.
`calworks`	Percent qualifying for CalWorks (income assistance program).
`lunch`	Percent qualifying for reduced-price lunch.
`computer`	Number of computers.
`expenditure`	Expenditure per student.
`income`	District average income (in USD 1,000).
`english`	Percent of English learners (i.e., students for whom English is a second language).
`read`	Average reading score.
`math`	Average math score.

Data source: Stock, J. H. and Watson, M. W. (2020). Introduction to Econometrics, 4th ed. Pearson.

6.1.2 Summary Statistics

The following displays the summary for each variable in the dataset:

```
summary(cas)
```

```
    district        school                 county        grades
 Min.   :61382   Length:420         Sonoma     : 29   KK-08:359
 1st Qu.:64308   Class :character   Kern       : 27   KK-06: 61
 Median :67760   Mode  :character   Los Angeles: 27
 Mean   :67473                      Tulare     : 24
 3rd Qu.:70419                      San Diego  : 21
 Max.   :75440                      Santa Clara: 20
                                    (Other)    :272
    students          teachers          calworks           lunch
 Min.   :   81.0   Min.   :   4.85   Min.   : 0.000   Min.   :  0.00
 1st Qu.:  379.0   1st Qu.:  19.66   1st Qu.: 4.395   1st Qu.: 23.28
 Median :  950.5   Median :  48.56   Median :10.520   Median : 41.75
 Mean   : 2628.8   Mean   : 129.07   Mean   :13.246   Mean   : 44.71
 3rd Qu.: 3008.0   3rd Qu.: 146.35   3rd Qu.:18.981   3rd Qu.: 66.86
 Max.   :27176.0   Max.   :1429.00   Max.   :78.994   Max.   :100.00
```

```
    computer        expenditure        income           english
 Min.   :   0.0   Min.   :3926   Min.   : 5.335   Min.   : 0.000
 1st Qu.:  46.0   1st Qu.:4906   1st Qu.:10.639   1st Qu.: 1.941
 Median : 117.5   Median :5215   Median :13.728   Median : 8.778
 Mean   : 303.4   Mean   :5312   Mean   :15.317   Mean   :15.768
 3rd Qu.: 375.2   3rd Qu.:5601   3rd Qu.:17.629   3rd Qu.:22.970
 Max.   :3324.0   Max.   :7712   Max.   :55.328   Max.   :85.540

      read            math
 Min.   :604.5   Min.   :605.4
 1st Qu.:640.4   1st Qu.:639.4
 Median :655.8   Median :652.4
 Mean   :655.0   Mean   :653.3
 3rd Qu.:668.7   3rd Qu.:665.8
 Max.   :704.0   Max.   :709.5
```

Interpreting the Summary Statistics

This summary provides key insights into our dataset:

- **Continuous variables** (students, teachers, expenditure, income, etc.) show their distribution through quartiles, mean, and range
- **Categorical variables** (county, grades) show frequency counts
- Notice the **range of expenditure** per student: from around \$3,900 to over \$7,700 - this substantial variation will be a key focus of our analysis
- **Test scores** (reading and math) also show considerable variation, suggesting potential relationships with school resources

6.1.3 Expenditure Variation Analysis

Q: To what extent does expenditure per student vary?

Understanding the distribution of expenditure per student is crucial for education policy analysis. Let's examine detailed descriptive statistics:

```r
quick_summary <- function(x, full=FALSE) {
    # Function to compute basic descriptive statistics
    if (!full){
        # Basic summary
        data.frame(
          n = length(x),
          min = min(x),
```

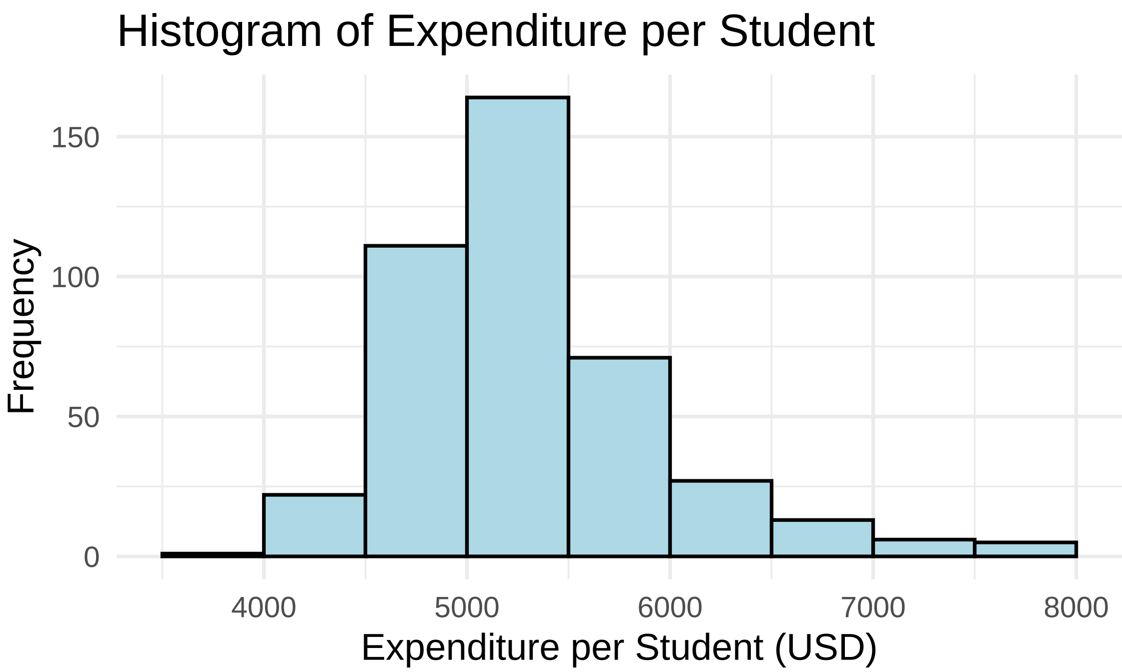
```r
          mean = mean(x),
          median = median(x),
          max = max(x),
          sd = sd(x),
          skewness = moments::skewness(x),
          kurtosis = moments::kurtosis(x),
          row.names = NULL
        )
    } else {
        # Full summary with quartiles and IQR
        data.frame(
          n = length(x),
          min = min(x),
          Q1 = quantile(x, 0.25),
          mean = mean(x),
          median = median(x),
          Q3 = quantile(x, 0.75),
          max = max(x),
          IQR = IQR(x),
          sd = sd(x),
          skewness = moments::skewness(x),
          kurtosis = moments::kurtosis(x),
          row.names = NULL
        )
    }
}
quick_summary(cas$expenditure) %>% round(2)
```

```
    n     min    mean  median     max     sd skewness kurtosis
1 420 3926.07 5312.41 5214.52 7711.51 633.94     1.07     4.88
```

Interpreting the Results:

- **Range:** Expenditure varies from about \$3,926 to \$7,712 per student - a difference of nearly \$4,000
- **Central Tendency:** The mean (\$5,312) is slightly higher than the median (\$5,196), **suggesting a slight right skew**
- **Variability:** Standard deviation of \$633 indicates moderate spread around the mean
- **Skewness (1.07):** Positive skewness confirms some districts spend considerably more than average
- **Kurtosis (4.88):** Larger than 3 indicates a distribution with fat tails relative to the normal distribution.

```
# Basic histogram
ggplot(cas, aes(x = expenditure)) +
  geom_histogram(binwidth = 500, boundary = 0, fill =
   ↪  "lightblue", color = "black") +
  labs(title = "Histogram of Expenditure per Student",
       x = "Expenditure per Student (USD)",
       y = "Frequency") +
  theme_minimal(base_size = 14)
```



What the Histogram Reveals:

- Most districts cluster around the \$5,000–\$5,500 range
- Variability is moderate
- The distribution has a long right tail, indicating positive skewness. That suggests that expenditure per student is concentrated at lower values for most districts, while a smaller number of districts spend substantially more, pulling the mean above the median
- There are **few extreme positive outliers**, indicating some districts do spend significantly more than average

6.1.4 Group Comparison: K-6 vs K-8 Schools

Q: Are there any differences in expenditure per student between K-6 and K-8 schools?

This comparison helps us understand whether grade span affects resource allocation. Let's examine the statistics for each group:

```
cas %>%
    group_by(grades) %>%
    group_map(~ {
        s <- quick_summary(.x$expenditure, full=TRUE)
        s <- cbind(
            grades = .y$grades,    # add group info
            s)
        return(s)
    }) %>%
    bind_rows()
```

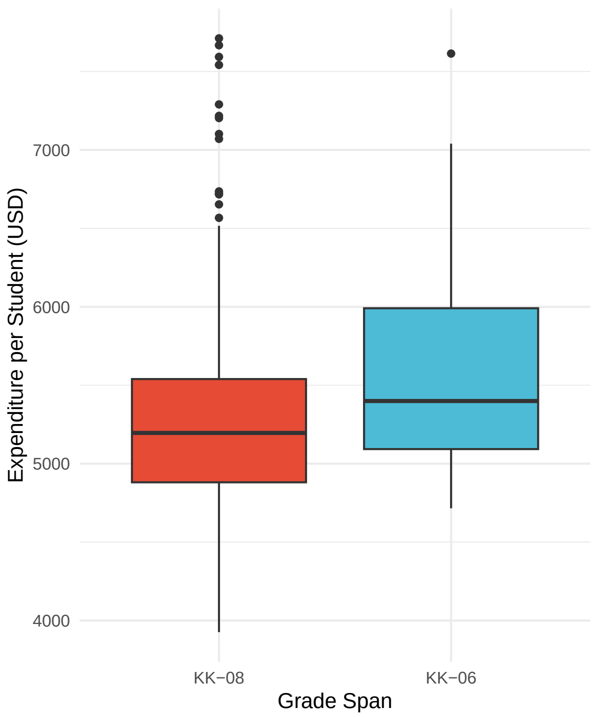
```
  grades   n     min      Q1     mean   median      Q3      max      IQR
1  KK-08 359 3926.070 4881.139 5267.365 5195.919 5539.380 7711.507 658.2412
2  KK-06  61 4715.446 5092.917 5577.493 5399.383 5990.794 7614.379 897.8770
        sd skewness kurtosis
1 618.6415 1.094514 5.284716
2 662.8033 1.019435 3.309185
```

Key Findings:

- **K–6 schools** (61 districts): Mean expenditure of \$5,577, slightly higher than K–8 schools
- **K–8 schools** (359 districts): Mean expenditure of \$5,267, with a larger sample size
- K–6 schools indicating higher variability, with a standard deviation of \$662 compared to \$618 for K-8 schools
- **Skewness** is positive for both groups, with K–8 schools showing slightly more skewness
- The difference suggests that **smaller grade span districts may have slightly higher per-pupil spending**

6.1.4.1 Box plot by grades span

```
ggplot(cas, aes(x = grades, y = expenditure, fill = grades)) +
  geom_boxplot() +
  scale_fill_npg() +
  labs(x = "Grade Span",
       y = "Expenditure per Student (USD)") +
  theme_minimal(base_size = 14) +
  theme(legend.position = "none")
```



Understanding the Box Plot:

1. **Box and middle bar (median)**

   - The **box** itself spans from the **first quartile (Q1, 25th percentile)** to the **third quartile (Q3, 75th percentile)**.
   - This range is called the **interquartile range (IQR = Q3 - Q1)**.
   - The **middle bar** inside the box represents the **median (50th percentile)**.
     - If the median is closer to the bottom of the box, the lower half of the data is more concentrated.
     - If closer to the top, the upper half is more concentrated.

2. **Whiskers**

   - The **whiskers** extend from the box to show the range of the data **excluding outliers**.

- A whisker extends to the **largest/smallest value within 1.5 × IQR from the box**:
  - Upper whisker = largest value ≤ Q3 + 1.5 × IQR
  - Lower whisker = smallest value ≥ Q1 - 1.5 × IQR

3. **Outliers**

- **Points beyond the whiskers** are plotted individually as **outliers**.
- These are values that are unusually high or low relative to the bulk of the data.

Summary of boxplot

- Box: middle 50% of the data
- Middle line: median
- Whiskers: typical range, excluding extreme values
- Outliers: extreme observations

Interpreting the Box Plot:

- **Median lines:** K–6 districts show a higher median expenditure than K-8 districts
- **Box sizes:** K–6 district show a larger interquartile ranges (IQR), indicating greater middle 50% spread
- **Outliers:** K–8 groups show more positive outliers (points beyond the whiskers), representing districts with unusually high expenditure

---

6.1.5 Correlation Analysis

Q: What predicts expenditure per student?

Understanding which factors correlate with expenditure helps identify patterns in educational resource allocation. Let's examine correlations:

Correlation coefficients in ascending order:

```r
# Get the set of numeric variables
v <- setdiff(
    names(cas),
    c("district", "school", "county", "grades")
)
corExp <- cor(cas["expenditure"], cas[setdiff(v,
 ↪  "expenditure")])
t(corExp) %>%
    as.data.frame() %>%
    rownames_to_column(var = "variable") %>%
    rename(correlation = expenditure) %>%
    arrange(correlation)
```

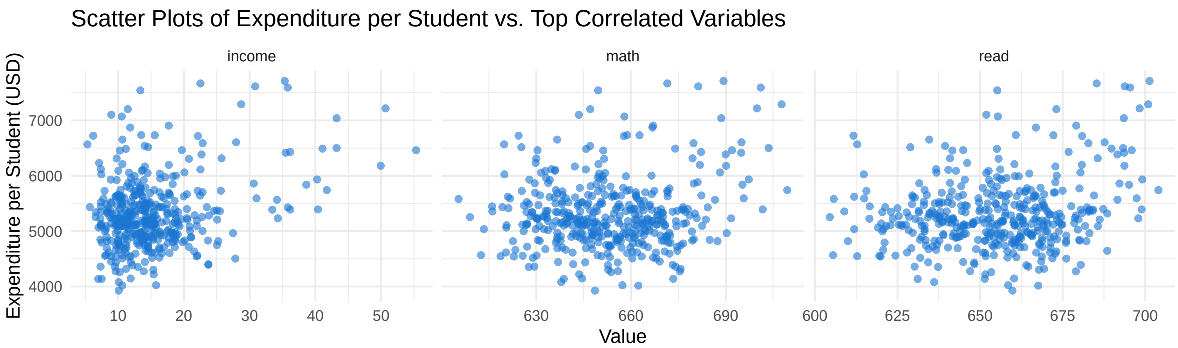
```
  variable correlation
1 students -0.11228455
2 teachers -0.09519483
3  english -0.07139604
4 computer -0.07131050
5    lunch -0.06103871
6 calworks  0.06788857
7     math  0.15498949
8     read  0.21792682
9   income  0.31448448
```

Key Correlation Insights:

- No strong correlations (e.g., > |0.7|) with expenditure
- **Positive correlations:** Income (0.31), math scores (0.22), and reading scores (0.15)
- **Moderate Negative correlations:** Students(-0.11), indicating districts with a larger amount of students tend to spend less per student
- **Interpretation:**
  - Wealthier districts tend to spend more per student
  - Higher spending correlates with better test scores
  - Districts with more students may face budget constraints leading to lower per-student expenditure

Plot scatter plots for the top three correlated variables

```r
top_vars <- c("income", "math", "read")
cas_long <- cas %>%
    select(expenditure, all_of(top_vars)) %>%
    pivot_longer(cols = all_of(top_vars), names_to = "variable",
     ↪  values_to = "value")
ggplot(cas_long, aes(x = value, y = expenditure)) +
    geom_point(alpha = 0.6, color = "#1976d2") +
    facet_wrap(~variable, scales = "free_x") +
    labs(
        title = "Scatter Plots of Expenditure per Student vs. Top
         ↪  Correlated Variables",
        x = "Value",
        y = "Expenditure per Student (USD)"
    ) +
    theme_minimal(base_size = 14)
```



What These Scatter Plots Reveal:

- **Income vs. Expenditure:** Positive relationship - wealthier districts consistently spend more per student; the trend is relatively clear with some variability;
- **Math Scores vs. Expenditure:** Positive trend suggests higher spending correlates with better math performance; the points are more scattered, indicating other factors also play a role;
- **Reading Scores vs. Expenditure:** Similar pattern to math scores, showing the relationship between resources and achievement
- **Data Distribution:** Points are fairly scattered, indicating that while correlations exist, other factors also influence these relationships

6.1.6 Academic Performance Relationship

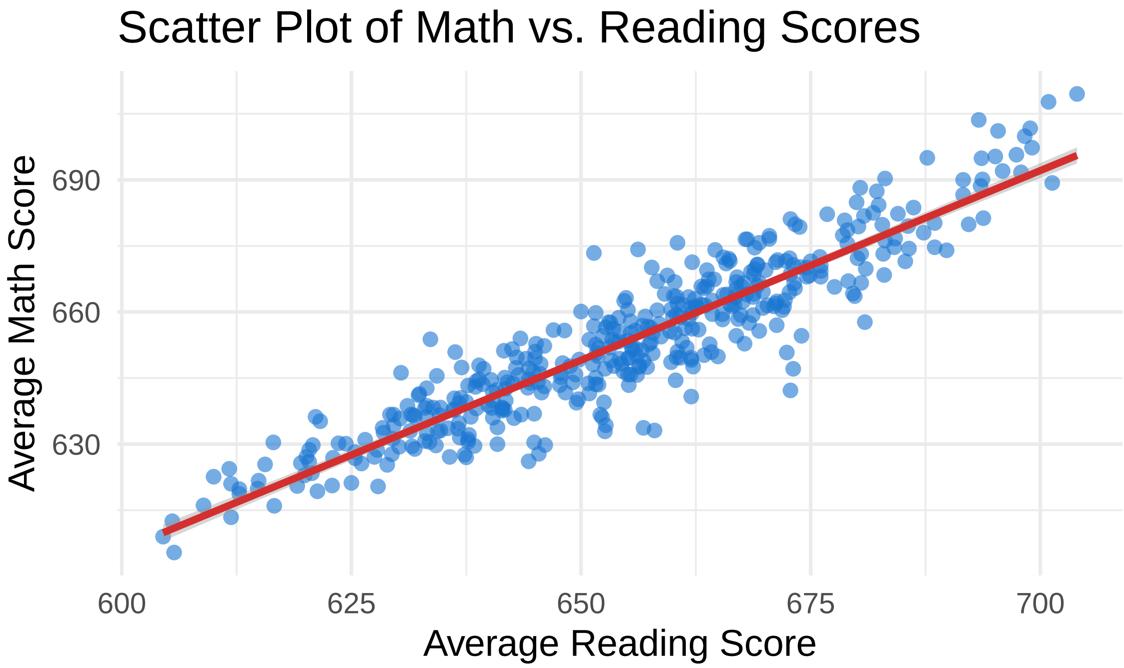
Q: what is the relationship between district level maths and reading scores?

Academic performance across different subjects often correlates, as they reflect overall educational quality and student preparation. Let's examine this relationship:

```
with(cas, cor.test(math, read))
```

```

	Pearson's product-moment correlation

data:  math and read
t = 49.005, df = 418, p-value < 2.2e-16
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.9073417 0.9359362
sample estimates:
      cor
0.9229015
```

```
ggplot(cas, aes(x = read, y = math)) +
    geom_point(alpha = 0.6, color = "#1976d2") +
    geom_smooth(method = "lm", color = "#d32f2f") +
    labs(
        title = "Scatter Plot of Math vs. Reading Scores",
        x = "Average Reading Score",
        y = "Average Math Score"
    ) +
    theme_minimal(base_size = 14)
```



Analysis of the Relationship:

- **Strong positive correlation:** Districts with higher reading scores typically have higher math scores
- **Linear trend:** The smooth line shows a clear linear relationship between the two subjects
- **Tight clustering:** Most points cluster closely around the trend line, indicating a strong relationship
- **Educational insights:**
  - This suggests that factors affecting academic performance impact multiple subjects similarly
  - Districts that excel in one area tend to excel in others

6.2 Q-Q Plot

The Q-Q plot, or quantile-quantile plot, is a graphical tool to help us assess if a set of data plausibly came from some theoretical distribution such as a normal distribution.

Understanding Q-Q Plots:

A QQ plot is a scatter plot created by plotting two sets of quantiles against one another.

- If both sets of quantiles came from the same distribution, we should see the points forming a line that's roughly straight.
- For example, if we run a statistical analysis that assumes our **residuals** are normally distributed, we can use a normal QQ plot to check that assumption.

  When the theoretical quantiles are from a normal distribution, the plot is called a **normal QQ plot**.

Why Q-Q Plots Matter:

- Many statistical tests assume normality; deviations from normality can affect the validity of statistical inferences
- Q-Q plots provide a visual method to assess these assumptions
- They help identify the type of non-normality (skewness, heavy tails, etc.)

6.2.1 QQ plot visualization tool

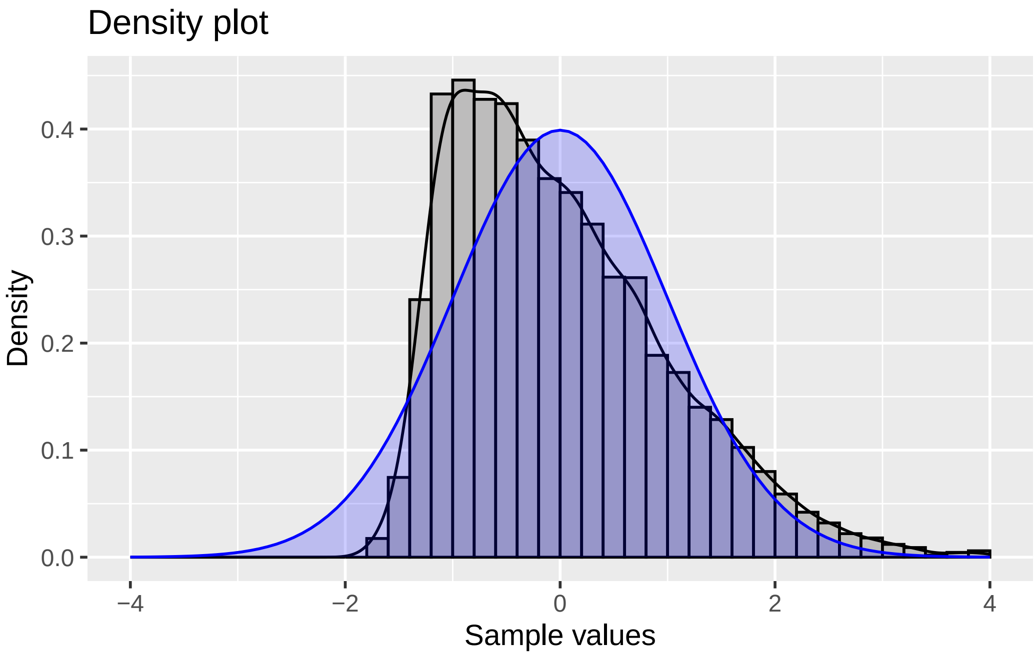
https://xiongge.shinyapps.io/QQplots/

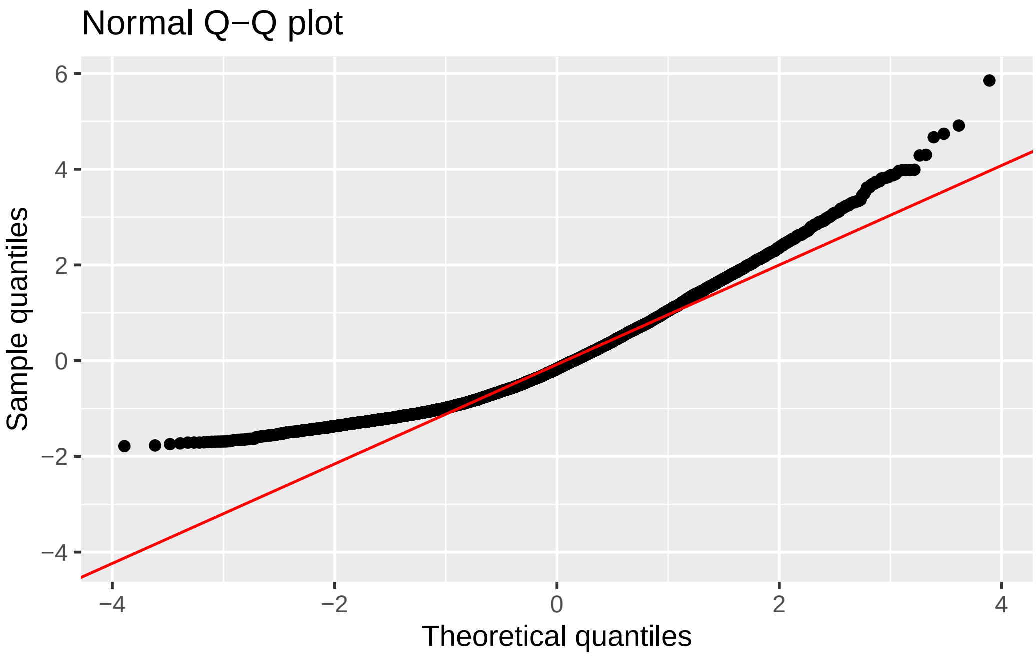
- Adjust Skewness and Kurtosis to see how the QQ plot changes.
- Note the website is not able to host many users at the same time, so if it does not load, please try again later.

6.2.2 Skewness

6.2.2.1 Positive skew

Positively skewed (mean > median) data have a J-shaped pattern in the Q-Q plot.





Summary statistics:

```
      n   min mean median  max sd skewness kurtosis
1 10000 -1.79 0.01  -0.17 5.85  1     0.86     3.57
```

💡 Note that for a positive skewed distribution, **the mean is larger than the median** because the long right tail pulls the mean above the median.

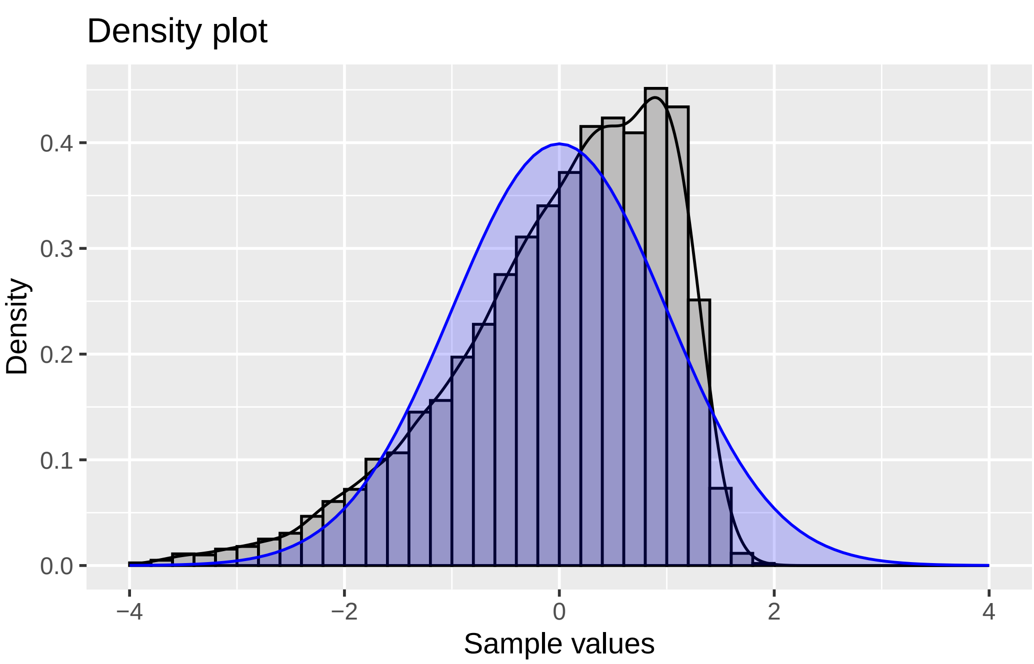
Understanding Positive Skewness:

- **Shape:** Long tail extending to the right

- **Mean vs. Median:** Mean > Median (tail pulls the mean upward)
- **Q-Q Plot Pattern:** J-shaped curve, points above the line at both ends
- **Real-world examples:** Income distribution, housing prices
- **Statistical implications:** May indicate floor effects or bounded distributions

6.2.2.2 Negative skew

Negatively skewed (mean < median) data have Q-Q plots that display an inverted J-shape.



Summary statistics:

```
      n   min mean median max   sd skewness kurtosis
1 10000 -5.23    0   0.19 1.9 1.01    -0.92     3.68
```

💡 Note that for a negatively skewed distribution, **the mean is smaller than the median** because the long left tail pulls the mean above the median.

Understanding Negative Skewness:

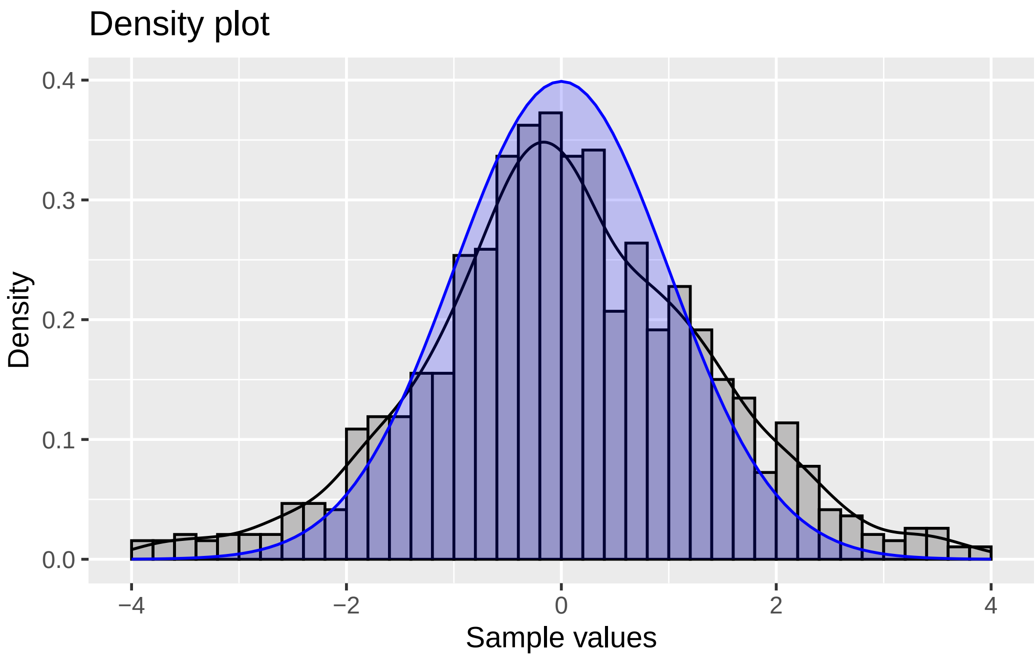
- **Shape:** Long tail extending to the left
- **Mean vs. Median:** Mean < Median (tail pulls the mean downward)
- **Q-Q Plot Pattern:** Inverted J-shaped curve, points below the line at both ends
- **Real-world examples:**
  - **Test scores (when most students perform well):** Most students score high (80-100), with fewer low scores creating a left tail
  - **Age at retirement:** Most people retire around 65-67, but some retire much earlier due to health or financial reasons
- **Statistical implications:** May indicate ceiling effects or bounded distributions

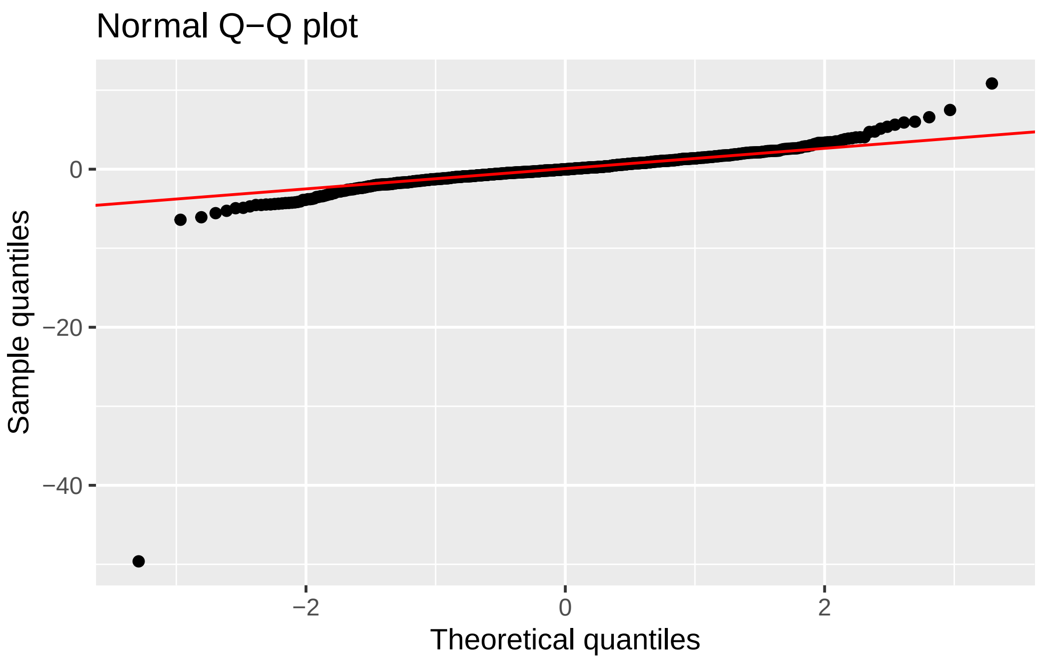
6.2.3 Kurtosis

Kurtosis measures the “tailedness” of a distribution compared to a normal distribution. Understanding kurtosis helps identify distributions with unusually many or few extreme values.

6.2.3.1 Fat tails

This plot shows a t-distribution with 3 degrees of freedom, which has heavier tails than a normal distribution.





Summary statistics:

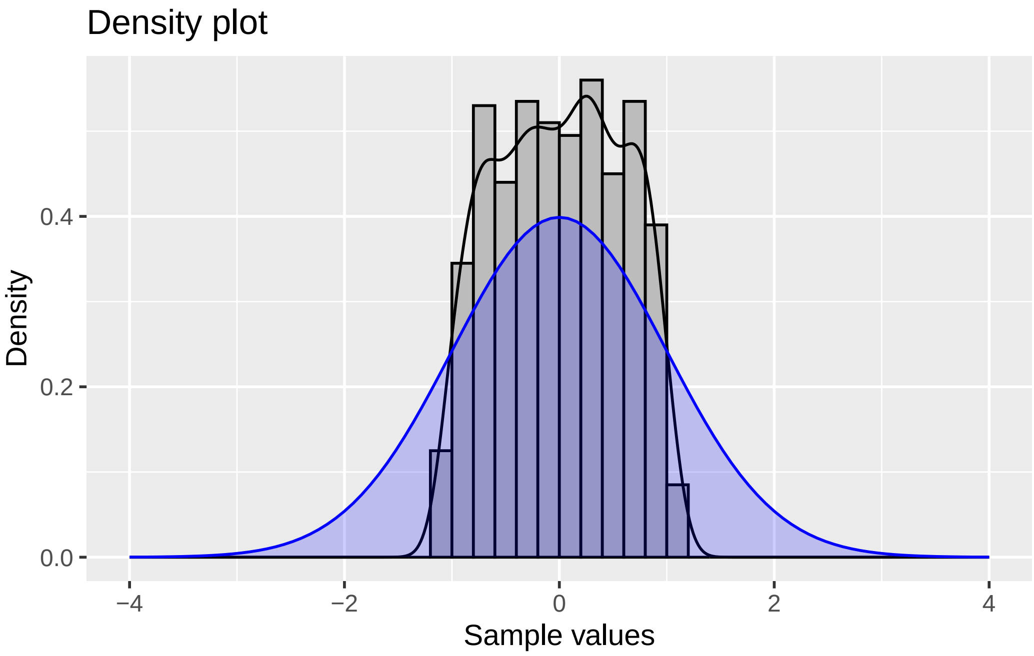
```
     n    min  mean median   max   sd skewness kurtosis
1 1000 -49.62 -0.02  -0.01 10.85 2.25   -10.61   238.79
```

Understanding Fat Tails (High Kurtosis):

- **Q-Q Plot Pattern:** Reverse S-shaped curve - points below the line at low quantiles, above at high quantiles.
- **Kurtosis > 3:** Indicates heavier tails than normal distribution
- **Practical meaning:** More extreme values (both high and low) occur than expected under normality
- **Real-world examples:**
  - **Financial returns:** Stock market crashes and booms occur more frequently than normal distribution predicts
  - **Measurement errors:** Occasionally very large errors occur due to equipment malfunction or human mistakes
  - **Response times:** Most responses are quick, but some take much longer due to system overload or user distraction
- **Statistical implications:** Higher risk of extreme events; standard confidence intervals may be too narrow

6.2.3.2 Thin tails

This example shows a distribution with lighter tails than normal.



Summary statistics:

```
     n   min mean median  max   sd skewness kurtosis
1 1000 -1.11 0.01   0.01 1.11 0.58    -0.04     1.88
```

Understanding Thin Tails (Low Kurtosis):

- **Q-Q Plot Pattern:** S-shaped curve - points above the line at low quantiles, below at high quantilees.
- **Kurtosis < 3:** Indicates lighter tails than normal distribution

- **Practical meaning:** Fewer extreme values occur than expected under normality
- **Real-world examples:** Bounded measurements, standardized test scores, some quality control data
- **Statistical implications:** Lower risk of extreme events; data more predictable than normal assumption suggests

---

Summary of Distribution Shapes

Understanding these patterns helps in:

- **Choosing appropriate statistical methods**
- **Identifying data quality issues**
- **Making informed assumptions about uncertainty**
- **Selecting proper transformations when needed**
- **Interpreting results correctly in context**

7 Simple Linear Regression

In this session, we will use the same `CASchools` dataset from the previous session and implement a simple linear regression model to explore a question of interest:

Research Question

Does expenditure per student affect student performance in elementary school education?

7.1 Data Preparation

```
# load packages and dataset
pkgs <- c("tidyverse", "moments", "data.table", "ggsci",
 ↪  "stargazer")
missing <- setdiff(pkgs, rownames(installed.packages()))
if (length(missing) > 0) install.packages(missing)
invisible(lapply(pkgs, function(pkg)
 ↪  suppressPackageStartupMessages(library(pkg, character.only =
 ↪  TRUE))))
f_name <-
 ↪  "https://raw.githubusercontent.com/my1396/FIN5005-Fall2025/refs/heads/mai
cas <- read_csv(f_name,
    col_types = cols(
        county = col_factor(), # read as factor
        grades = col_factor()
    )
)
cas %>% as.data.table()
```

```
    district                             school     county grades students
       <num>                             <char>     <fctr> <fctr>    <num>
 1:    75119                  Sunol Glen Unified    Alameda  KK-08      195
 2:    61499                Manzanita Elementary      Butte  KK-08      240
 3:    61549         Thermalito Union Elementary      Butte  KK-08     1550
 4:    61457   Golden Feather Union Elementary      Butte  KK-08      243
```

```
  5:   61523      Palermo Union Elementary     Butte  KK-08    1335
 ---
416:   68957       Las Lomitas Elementary  San Mateo  KK-08     984
417:   69518        Los Altos Elementary Santa Clara  KK-08    3724
418:   72611       Somis Union Elementary    Ventura  KK-08     441
419:   72744           Plumas Elementary       Yuba  KK-08     101
420:   72751         Wheatland Elementary       Yuba  KK-08    1778
    teachers calworks   lunch computer expenditure   income   english  read
       <num>    <num>   <num>    <num>       <num>    <num>     <num> <num>
  1:   10.90   0.5102  2.0408       67    6384.911 22.690001  0.000000 691.6
  2:   11.15  15.4167 47.9167      101    5099.381  9.824000  4.583333 660.5
  3:   82.90  55.0323 76.3226      169    5501.955  8.978000 30.000002 636.3
  4:   14.00  36.4754 77.0492       85    7101.831  8.978000  0.000000 651.9
  5:   71.50  33.1086 78.4270      171    5235.988  9.080333 13.857677 641.8
 ---
416:   59.73   0.1016  3.5569      195    7290.339 28.716999  5.995935 700.9
417:  208.48   1.0741  1.5038      721    5741.463 41.734108  4.726101 704.0
418:   20.15   3.5635 37.1938       45    4402.832 23.733000 24.263039 648.3
419:    5.00  11.8812 59.4059       14    4776.336  9.952000  2.970297 667.9
420:   93.40   6.9235 47.5712      313    5993.393 12.502000  5.005624 660.5
      math
     <num>
  1: 690.0
  2: 661.9
  3: 650.9
  4: 643.5
  5: 639.9
 ---
416: 707.7
417: 709.5
418: 641.7
419: 676.5
420: 651.0
```

Variable of interest: Test scores

- read and math are average reading and math test scores for each district
- In the last session, we show that these two variables are highly correlated (r=0.92)
- We construct an average test score as the average of the test score for reading and the score of the math test.

```
# compute TestScore and append it to CASchools
cas <- cas %>%
```

```
  mutate(TestScore = (read + math) / 2)
```

7.2 Descriptive Statistics

Refer to the previous session for detailed summary statistics and visualizations.

A quick preview

```
v <- setdiff(names(cas), c("district", "school", "county",
 ↪  "grades"))
stargazer(
  as.data.frame(cas[v]),
  type = ifelse(knitr::is_html_output(), "html", "latex"),
  digits = 2)
```

% Table created by stargazer v.5.2.3 by Marek Hlavac, Social Policy Institute. E-mail: marek.hlavac at gmail.com % Date and time: Thu, Jul 02, 2026 - 13:41:56

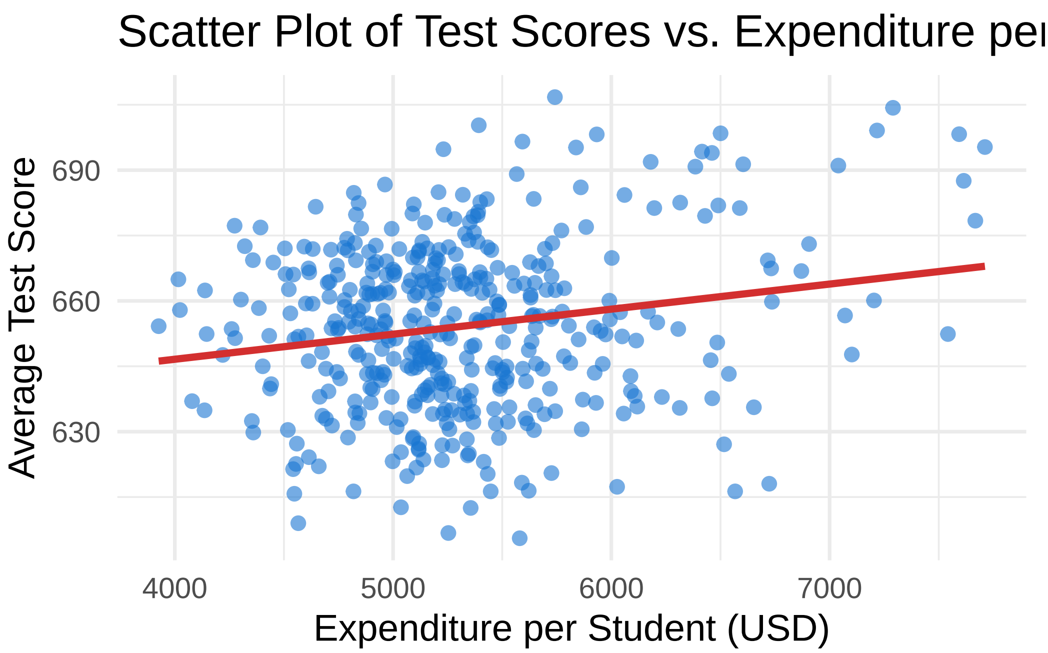
Table 7.1

Statistic	N	Mean	St. Dev.	Min	Max
students	420	2,628.79	3,913.10	81	27,176
teachers	420	129.07	187.91	4.85	1,429.00
calworks	420	13.25	11.45	0.00	78.99
lunch	420	44.71	27.12	0.00	100.00
computer	420	303.38	441.34	0	3,324
expenditure	420	5,312.41	633.94	3,926.07	7,711.51
income	420	15.32	7.23	5.34	55.33
english	420	15.77	18.29	0.00	85.54
read	420	654.97	20.11	604.50	704.00
math	420	653.34	18.75	605.40	709.50
TestScore	420	654.16	19.05	605.55	706.75

7.3 Scatter Plot

Hypothesis: Higher spending per student leads to better test scores.

```
ggplot(cas, aes(x = expenditure, y = TestScore)) +
    geom_point(alpha = 0.6, color = "#1976d2") +
    geom_smooth(method = "lm", color = "#d32f2f", se = FALSE) +
    labs(
        title = "Scatter Plot of Test Scores vs. Expenditure per
         ↪  Student",
        x = "Expenditure per Student (USD)",
        y = "Average Test Score"
    ) +
    theme_minimal(base_size = 14)
```



There seems to be a positive relationship between expenditure per student and average test scores. Also note that the points are dispersed, indicating that other factors may also influence test scores. We will further explore this using multiple regression in future sessions.

For now, we will focus on a simple linear regression model with only one independent variable: expenditure per student.

Let's calculate the correlation coefficient between expenditure and test scores.

```
with(cas, cor.test(expenditure, TestScore))
```

```
	Pearson's product-moment correlation

data:  expenditure and TestScore
t = 3.9841, df = 418, p-value = 7.989e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.09736861 0.28180139
sample estimates:
      cor
0.1912728
```

What does the correlation coefficient tell us? Is it statistically significant? Justify your answer.

Solution

- **The reported correlation coefficient** is 0.19 between expenditure and TestScore.

  This is a positive but weak correlation: higher expenditures are associated with higher test scores, but the relationship is not very strong.

- **The p-value** = 7.989e-05, which is far below common significance levels (0.05, 0.01, or even 0.001).

  This means we reject the null hypothesis that the true correlation is zero. The result is statistically significant.

7.4 Simple Linear Regression

7.4.1 Model Specification

We will fit a simple linear regression model to examine the relationship between expenditure per student and average test scores.

$$\text{TestScore}_i = \beta_0 + \beta_1 \times \text{expenditure}_i + \varepsilon_i \tag{7.1}$$

Equation 7.1 is the linear regression model with a single independent variable.

Where:

💡 Key Components of the Linear Regression Model

- TestScore_i in the Left Hand Side (LHS) is the **dependent variable** (response variable). i indexes different school districts.
- expenditure_i in the Right Hand Side (RHS) is the **independent variable** (explanatory variable or regressor).
- β_0 and β_1 are known as the parameters (coefficients) of the model.
  - β_0 is the **intercept**, representing the expected value of TestScore when expenditure is zero.
  - β_1 is the **slope coefficient**, representing the change in TestScore for a one dollar increase in expenditure.
- ε_i is the error term, capturing all other factors affecting expenditure not included in the model, e.g, teacher quality, school facilities, parental involvement, etc.

Equation 7.1 can be written more generally as:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

- β_1 is the slope coefficient, which measures the change in the dependent variable Y associated with a one-unit increase in the independent variable X.
- β_0 is the intercept, which is the expected value of Y when $X = 0$; it is the point at which the population regression line intersects the Y axis.

📌 In some econometric applications, the intercept has a meaningful economic interpretation. In other applications, the intercept has no real-world meaning; for example, when X is the expenditure per student, strictly speaking the intercept is the expected value of test scores when there is no expenditure, which might be unrealistic.

When the real-world meaning of the intercept is nonsensical, it is best to think of it simply as the coefficient that determines the level of the regression line.

7.4.2 Estimating the Coefficients of the Linear Regression Model

Based on the scatter plot and the correlation analysis, we expect a positive relationship between expenditure and test scores. We will use the Ordinary Least Squares (OLS) method to estimate the coefficients of the linear regression model.

To run this regression, we use the `lm()` function in R.

```r
# Fit the linear regression model
model <- lm(TestScore ~ expenditure, data = cas)
summary(model)
```

```

Call:
lm(formula = TestScore ~ expenditure, data = cas)

Residuals:
    Min      1Q  Median      3Q     Max 
-50.146 -14.206   0.689  13.513  50.127 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 6.236e+02  7.720e+00  80.783  < 2e-16 ***
expenditure 5.749e-03  1.443e-03   3.984 7.99e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 18.72 on 418 degrees of freedom
Multiple R-squared:  0.03659,	Adjusted R-squared:  0.03428 
F-statistic: 15.87 on 1 and 418 DF,  p-value: 7.989e-05
```

7.4.3 Interpretation of the output

7.4.3.1 A. Model Specification and Fitted Values

The regression estimates the effect of **expenditure** (per student, in USD) on **test scores** (combined reading and math).

From Theory to Practice:

We started with the theoretical model:

$$\text{TestScore}_i = \beta_0 + \beta_1 \times \text{expenditure}_i + \varepsilon_i$$

Using OLS estimation, we obtain the **fitted model**:

$$\widehat{\text{TestScore}} = 623.6 + 0.00575 \times \text{expenditure}$$

The hat symbol ($\widehat{\text{TestScore}}$) indicates these are **predicted values** based on our sample data. Notice that we no longer have the error terms (ε_i) because this equation gives us the predicted values from the regression line.

Understanding Residuals:

Since our model can't perfectly predict every observation, we need to measure how far off our predictions are from reality:

$$\text{Residual}_i = \text{TestScore}_i - \widehat{\text{TestScore}}_i$$

Residuals represent the **difference between actual and predicted values** - essentially, they capture what our model couldn't explain. In the original theoretical model, these correspond to the error terms (ε_i) that we estimated.

7.4.3.2 B. Coefficients

- **Intercept (623.6):** When expenditure = 0, the predicted test score is about **623.6**. While not meaningful in practice (since expenditure cannot realistically be zero), it serves as the baseline of the regression line.
- **Expenditure (0.00575):** For every **one-dollar increase** in expenditure per student, the average test score is predicted to increase by **0.0057 points**. Put differently: a **$1,000 increase** in expenditure is associated with a **5.75-point increase** in test scores.

7.4.3.3 C. Statistical Significance

For the slope coefficient (β_1):

- The **t-value = 3.984** and **p-value = 7.99e-05**, which is well below 0.001.
- This indicates that expenditure has a **statistically significant positive effect** on test scores.

7.4.3.4 D. Model Fit

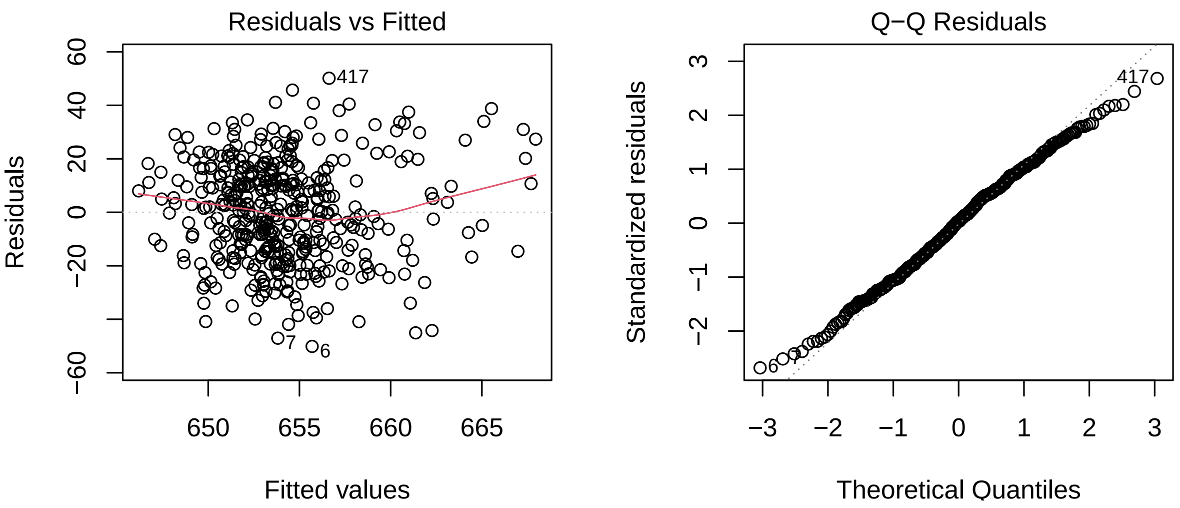
- **R-squared = 0.0366** (about 3.7%).
  - This means expenditure explains only a **small portion of the variation** in test scores.
  - Many other factors (teacher quality, socio-economic background, school resources, etc.) likely play a larger role.
- **F-statistic = 15.87** with a p-value of 7.99e-05.
  - This tests the null hypothesis that all regression coefficients are equal to zero (no effect).
  - The low p-value indicates we reject the null hypothesis, confirming that the model has some explanatory power.

7.4.3.5 E. Residuals

The investigation of residuals requires diagnostic test which checks the assumptions of linear regression (e.g., homoscedasticity, normality of errors).

We will cover them in future sessions.

Here we provide a brief overview.



- The first plot is **"Residuals vs Fitted" plot**.

  The residuals appear reasonably balanced around zero, though the spread suggests substantial unexplained variation.

  The spread of residuals seems to **increase slightly with fitted values**, indicating potential heteroskedasticity (non-constant variance of errors). This suggests that the assumption of homoskedasticity may be violated, which could affect the reliability of our coefficient estimates and standard errors.

- The second plot is the **normal Q-Q plot**, which assesses whether the residuals are normally distributed.
  - The points generally follow the reference line, suggesting that the residuals are approximately normally distributed, although there shows thin tails.
  - This indicates that there are few extreme residuals. The model predictions don't have large outliers.

From the summary output:

- The **residual standard error = 18.72**, which is the average size of prediction errors (in test score points).

✅ **Summary Interpretation:**

The regression analysis shows a **statistically significant positive relationship** between school expenditure and test scores. On average, an additional \$1,000 in per-student expenditure is associated with an increase of about **5.75 points** in test scores. However, the **explanatory power of the model is low ($R^2 \approx 3.7\%$)**, indicating that expenditure alone explains only a small fraction of the variation in test scores. This suggests that while spending matters, many other factors also influence student performance.

To gain a more comprehensive understanding of the factors affecting student achievement, future analyses should employ **multiple regression models** that incorporate additional explanatory variables such as student demographics, teacher qualifications, and school characteristics. We will cover multiple regression in subsequent sessions.

7.5 Report by Stargazer

`stargazer` produces well-formatted regression tables that can be easily included in reports or publications.

```
stargazer(model, type = "text", digits = 2)
```

```
===============================================
                        Dependent variable:
                    ---------------------------
                             TestScore
-----------------------------------------------
expenditure                   0.01***
                              (0.001)

Constant                     623.62***
                              (7.72)

-----------------------------------------------
Observations                    420
R2                             0.04
Adjusted R2                    0.03
Residual Std. Error      18.72 (df = 418)
F Statistic           15.87*** (df = 1; 418)
===============================================
Note:               *p<0.1; **p<0.05; ***p<0.01
```

References

- Introduction to Econometrics with R

8 Theoretical Framework of Linear Regression

🎯 **Study Objectives**

- Understand the key assumptions underlying the linear regression model.
- Understand the theoretical framework of linear regression, including the OLS estimator, its properties, and its distribution.
- Central Limit Theorem (CLT)
  - Apply the CLT to approximate the distribution of sample mean and calculating probabilities using distribution table.
  - Understand the implications of CLT for the sampling distribution of the OLS estimator.
- Distinguish the difference between exact distribution and sampling distribution of the OLS estimator.
- Understand R squared and adjusted R squared as measures of fit.
- Be able to identify violations of the linear regression assumptions using diagnostic plots.
- Understand how to address heteroskedasticity.

In this section, we will discuss the theoretical framework of linear regression, including the assumptions underlying the linear regression model and essential hypothesis tests.

Q: Why the assumptions matter?

A: In short, violation of the assumptions may lead to biased or inefficient estimates, incorrect inferences, and unreliable predictions.

8.1 Assumptions of Linear Regression

Q: What are the assumptions of linear regression?

A: The **Gauss-Markov Assumptions** ensure that the Ordinary Least Squares (OLS) estimator is the Best Linear Unbiased Estimator (BLUE).

1. **Linearity**: The relationship between the independent variable(s) and the dependent variable is linear in parameters.
2. **No perfect multicollinearity**: The independent variables are NOT perfectly correlated.
3. **Error has zero conditional mean**
4. **Error is homoskedastic**

   - Homoskedasticity: The variance of the error term is constant across all levels of the independent variable(s).

   In time series settings (we won't cover in this course), we also assume:

   - No autocorrelation: The error terms are not correlated with each other.

5. **Residuals are normally distributed**

   This assumption is useful for conducting hypothesis tests and constructing confidence intervals.

6. **Large outliers are unlikely**

   This assumption requires that observations with values of X_i, Y_i, or both that are far outside the usual range of the data are unlikely.

   Large outliers can make OLS regression results misleading as OLS are sensitive to outliers.

   Mathematically, this assumption can be stated as: X and Y have **finite kurtosis**.

8.2 Regression Framework and OLS Estimator

Multiple linear regression models is defined as

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + ... + \beta_k X_{ik} + \varepsilon_i$$

where

- $i = 1, \dots, n$ indexes the observations,
- Y_i is the dependent variable, and
- X_{ij} represents the j-th independent variable for observation i, with $j = 1, \dots, k$ and k denoting the total number of independent variables.
- The coefficients β_j are to be estimated, and
- ε_i is the error term.

The OLS estimator $\hat{\beta}^{OLS}$ minimizes the sum of squared residuals (SSR):

$$\hat{\beta}^{OLS} = \arg\min_{\beta} \sum_{i=1}^{n} e_i^2$$

where $e_i = Y_i - \hat{Y}_i$ is the residual for observation i.

$\hat{Y}_i$ is the predicted value of Y_i given the estimated coefficients.

$$\hat{\beta}^{OLS} = \arg\min_{\beta} \sum_{i=1}^{n} (Y_i - \beta_0 - \sum_{j=1}^{k} \beta_j X_{ij})^2$$

$\hat{\beta}^{OLS}$ can be obtained by the first order condition (setting the derivative of SSR with respect to each β_j to zero and solving the resulting system of equations).

Read OLS in Matrix Form for more details. We won't cover the OLS derivation in this course.

Here we give the direct formula for the OLS estimator:

$$\hat{\beta} = \left(\sum_{i=1}^{n} X_i X_i' \right)^{-1} \left(\sum_{i=1}^{n} X_i Y_i \right)$$

where

$$X_i = \begin{pmatrix} 1 \\ X_{i1} \\ X_{i2} \\ \vdots \\ X_{ik} \end{pmatrix}$$

is a $(k+1) \times 1$ vector of independent variables including the intercept for observation i.

Properties of OLS estimator:

- **Unbiasedness**: $E[\hat{\beta}^{OLS}] = \beta$ (if Gauss-Markov assumptions hold)
- **Efficiency**: Among all linear unbiased estimators, OLS has the smallest variance
- **Consistency**: As sample size $n \to \infty$, $\hat{\beta}^{OLS} \to \beta$ in probability

8.2.1 Exact Distribution of OLS Estimator

Under the assumption of normally distributed errors, $\varepsilon | X \sim N(0, \sigma^2 I)$, we have in matrix form

$$\hat{\beta} \sim N(\beta, \sigma^2 (X'X)^{-1}). \tag{8.1}$$

where

- $\sigma^2 = \text{Var}(\varepsilon_i \mid X)$ is the variance of the error term.
- X is the matrix of independent variables.

$$X = \begin{pmatrix} X_1' \\ X_2' \\ \vdots \\ X_n' \end{pmatrix}_{n \times (k+1)}$$

Equation 8.1 is called the **exact (finite-sample) distribution** of the OLS estimator.

$\text{Var}(\hat{\beta}) = \sigma^2 (X'X)^{-1}$ is called the **variance-covariance matrix** of the OLS estimator, often denoted as $V_{\hat{\beta}}$.

We estimate σ^2 with $\hat{\sigma}^2$:

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^n e_i^2}{n-k-1}.$$

The standard errors of $\hat{\beta}_j$, $j = 1, \dots, k$, are given by the square root of the (j, j) element of the variance-covariance matrix, $V_{\hat{\beta}}$.

The individual coefficient estimates $\hat{\beta}_j$ are normally distributed:

$$\hat{\beta}_j \sim N(\beta_j, \sigma^2 v_{jj})$$

8.3 Sampling Distribution of OLS Estimator

Becausethe OLS estimator $\hat{\beta}_0, \hat{\beta}_1, \ldots$ are computed from a randomly drawn sample of data, the estimators themselves are random variables with a probability distribution.

The sampling distribution describes the values they could take over different possible random samples. In small samples, theses sampling distributions are complicated, but in large samples, they are normal because of the Central Limit Theorem (CLT).

> Sampling Distribution vs. Exact Distribution
>
> - The distribution depends on the sample data (X_i, Y_i).
> - Based on Central Limit Theorem (CLT), we can establish the sampling distribution of $\hat{\beta}$ when sample size is large, i.e., $n \to \infty$.

8.3.1 Central Limit Theorem (CLT)

Theorem 8.1. *Suppose that $X_1, \ldots, X_n$ is an independent and identically distributed (iid) sequence with a finite mean $\mathbb{E}(X_i) = \mu$ and variance $Var(X_i) = \sigma^2$, where $0 < \sigma^2 < \infty$.*

Define a sample mean: $\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$.

*The **Lindeberg-Lévy CLT** states that*

$$\frac{\overline{X} - \mathbb{E}[\overline{X}]}{\sqrt{Var[\overline{X}]}} = \frac{\overline{X} - \mu}{\sqrt{\sigma^2/n}} = \sqrt{n} \cdot \frac{\overline{X} - \mu}{\sigma} \xrightarrow{d} N(0, 1)$$

where $\xrightarrow{d}$ denotes convergence in distribution.

Alternative expressions of CLT:

$$\sqrt{n}(\overline{X} - \mu) \xrightarrow{d} N(0, \sigma^2)$$

$$\overline{X} \xrightarrow{d} N(\mu, \frac{\sigma^2}{n})$$

Q: Why does CLT matter?

A:

- The central limit theorem implies that if N is large **enough**, we can **approximate** the distribution of $\overline{X}$ by the **standard normal distribution** with mean μ and variance σ^2/N **regardless of the underlying distribution of** X.
- The scaling by n is crucial.

$$\sigma^2_{\overline{X}} = \mathrm{Var}[\overline{X}] = \frac{\sigma^2}{n}$$

  This means as sample size increases, the sample variance shrinks at the rate of $1/n$.

> Tip
>
> **Interpretation:** As the sample size n increases,the mean becomes more concentrated around the true population mean μ.

- How large must n be for the distribution of $\overline{X}$ to be approximately normal?

  There is no definitive answer, but a common rule of thumb is that $n \geq 30$ is often sufficient for the CLT to provide a good approximation.

  Because the distribution of $\overline{X}$ approaches the normal distribution as n increases, $\overline{X}$ is said to have an **asymptotic normal distribution**.

8.3.2 CLT Example

The amount of time customers spend in a grocery store is a random variable with mean $\mu = 40$ minutes and standard deviation $\sigma = 15$ minutes. Assume each customer's time in the store is independent of others.

$$\mathbb{E}[X] = \mu = 40, \quad \text{and} \quad \mathrm{SD}(X) = \sigma = 15$$

Consider the following probabilities:

Example 8.1. Assuming X is normally distributed, what is the probability that a randomly selected customer spends more than 42 minutes in the store, i.e., compute $P(X > 42)$?

Use the normal distribution table to look up the probabilities.

Solution

Individual Customer

We define the standard normal variable Z as:

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$$

where $\mu = 40$ and $\sigma = 15$.

To compute $P(X > 42)$, we convert to the standard normal:

$$P(X > 42) = P(\frac{X - 40}{15} > \frac{42 - 40}{15}) = P(Z > 0.1333)$$

By symmetry of the normal distribution:

$$P(Z > 0.1333) = P(Z < -0.1333) = \Phi(-0.1333) = 0.4483$$

where $\Phi(\cdot)$ denotes the cumulative distribution function (CDF) of the standard normal distribution.

Interpretation: There is approximately a **44.83%** probability that a randomly selected customer will spend more than 42 minutes in the store.

Note that the **specific distribution** for X, i.e., normality here, is **required** to compute the probability for a single customer.

Example 8.2. Given a random sample of $n = 64$ customers, what is the probability of the average time spent by the 64 customers exceeds 42 minutes, i.e., compute $P(\overline{X}_{64} > 42)$?

Hint: apply CLT and justify why CLT can be applied here.

Solution

Sample Mean for $n = 64$

By the Central Limit Theorem, for **large** n **with finite mean and variance**, the sampling distribution of the sample mean $\overline{X}_{64}$ is approximately normal (regardless the distribution of X):

$$\overline{X}_{64} \sim N(\mu, \frac{\sigma^2}{n})$$

Plug in $\mu = 40$ and $\sigma = 15$:

$$\sigma_{\overline{X}_{64}} = \frac{\sigma}{\sqrt{n}} = \frac{15}{\sqrt{64}} = 1.875$$

According to CLT $\overline{X}_{64}$ is normally distributed:

$$\overline{X}_{64} \sim N\left(40, 1.875^2\right)$$

To calculate $P(\overline{X} > 42)$, we standardize the sample average to the standard normal variable Z, defined as:

$$Z = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} = \frac{\overline{X} - 40}{1.875}$$

We can rewrite the probability in terms of Z:

$$P(\overline{X} > 42) = P(\frac{\overline{X} - \mu}{\sigma/\sqrt{n}} > \frac{42 - 40}{1.875}) = P(Z > 1.067)$$

Using the standard normal distribution table:

$$P(Z > 1.067) = P(Z < -1.067)$$

Since the standard normal distribution table is rounded at two decimal places, we look up $P(Z < -1.07) = 0.1423$:

Interpretation: There is a **14.23%** probability that the average time spent by a random sample of 64 customers exceeds 42 minutes.

Example 8.3. Given a random sample of $n = 81$ customers, what is the probability of the average time spent by the 81 customers exceeds 42 minutes, i.e., compute $P(\overline{X}_{81} > 42)$?

Solution

Sample Mean for $n = 81$

$$\sigma_{\overline{X}_{81}} = \frac{\sigma}{\sqrt{n}} = \frac{15}{\sqrt{81}} = \frac{15}{9} = 1.667$$

According to CLT $\overline{X}_{81}$ is normally distributed:

$$\overline{X}_{81} \sim N(\mu = 40, \sigma_{\overline{X}_{81}} = 1.667^2)$$

Standardize the sample average

$$P(\overline{X} > 42) = P(\frac{\overline{X} - \mu}{\sigma_{\overline{X}_{81}}} > \frac{42 - 40}{1.667})$$

Rewrite the probability in terms of Z:

$$z = \frac{42 - 40}{1.667} = 1.20$$

$$P(\overline{X} > 42) = P(Z > 1.20)$$

Look up the normal distribution table

$$P(Z > 1.20) = P(Z < -1.2) \approx 0.1151$$

Interpretation: A random sample of 16 customers has a **11.51%** chance of yielding an average time above 42 minutes.

Example 8.4. Discussion: Compare your results in the three probabilities. Provide a brief interpretation of how the probabilities differ and why.

Solution

- For **a single customer**, there is a **44.83%** chance of spending more than 42 minutes.
- For the **average of 64 customers**, the probability drops to **14.23%**.
- For the **average of 81 customers**, it drops further to **11.51%**.
- **Why?** As sample size increases, the variability (standard deviation) of the sample mean decreases, making it less likely to observe extreme values like an average over 42 minutes. This demonstrates the Central Limit Theorem and the stabilizing effect of larger samples. (See Figure 8.1 for a graphical illustration.)
  - The individual distribution is wider (larger standard deviation), so the chance of extreme values is higher.
  - As sample size increases, the standard error decreases, so the distribution of the sample mean becomes narrower.
  - This means it becomes less likely to observe a sample mean far from the population mean.

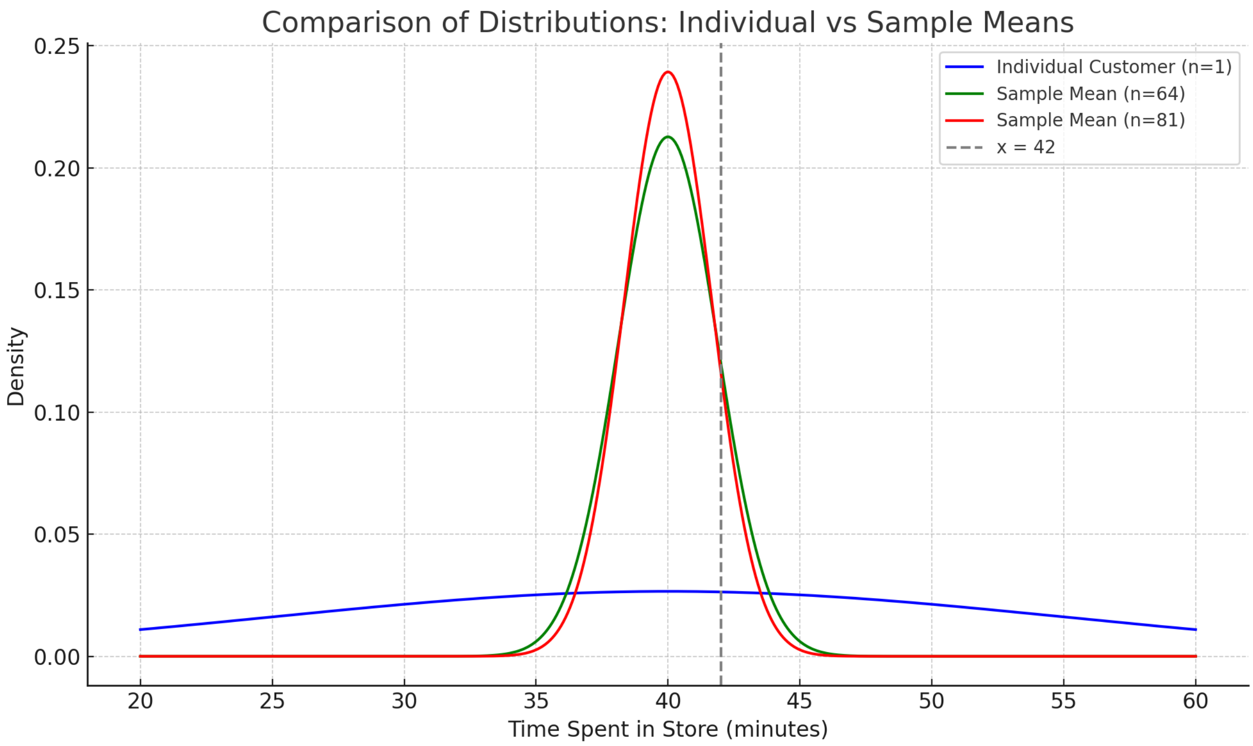


Figure 8.1

Key Takeaways from CLT

Generally, as the sample size increases, the **sampling distribution of the sample mean** becomes more concentrated around the population mean.
This reflects a reduction in variability: **larger samples yield more precise estimates of the population mean**.
Conversely, smaller sample sizes result in a wider sampling distribution, indicating greater variability in the sample mean.

8.3.3 Large Sample Distribution of OLS Estimator

Now we can establish the large sample distribution of the OLS estimator $\hat{\beta}$ based on CLT.

Under the Gauss-Markov assumptions, as $n \to \infty$,

$$\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} N(0, \sigma^2 E(X_i X_i')^{-1})$$

- Again, note the scaling by $\sqrt{n}$. It shows how fast the estimator $\hat{\beta}$ converges to the true parameter β as sample size n increases.

- Note the difference of the variance between the exact distribution and the large sample distribution.
  - The exact distribution depends on the **sample data** $X'X$
  - The large sample distribution depends on the **population moment** $E(X_i X_i')$.
- It suggests that $\hat{\beta}$ is **asymptotically normal** and **consistent**. "Consistency" means that as the sample size n increases, the estimator $\hat{\beta}$ converges in probability to the true parameter β.

8.4 Measure of Fit

We often use R^2 to measure the goodness of fit of a regression model.

- R^2 is called "the coefficient of determination."
- It represents the proportion of variance in the dependent variable that is predictable by the regression model.
- $0 \leq R^2 \leq 1$

R^2 is defined as:

$$R^2 = 1 - \frac{SSR}{SST} = \frac{SST - SSR}{SST} = \frac{SSE}{SST}$$

where

- $SSR = \sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2 = \sum_{i=1}^{n} e_i^2$ is the sum of squared residuals (unexplained variation)
- $SST = \sum_{i=1}^{n}(Y_i - \bar{Y})^2$ is the total sum of squares (total variation)
- $SSE = \sum_{i=1}^{n}(\hat{Y}_i - \bar{Y})^2$ is the explained sum of squares (explained variation)
- $\bar{Y} = \frac{1}{n}\sum_{i=1}^{n} Y_i$ is the mean of the dependent variable

Note that adding variables always increases R^2, even if the new variables are not statistically significant. → To address this, we use the **adjusted** R^2.

In a regression model with multiple explanatory variables, we often use **adjusted** R^2 that adjusts the number of explanatory variables.

"adjusted R^2" or "R-bar-squared", defined by

$$\overline{R}^2 = 1 - \frac{n-1}{n-k}(1 - R^2),$$

which imposes a penalty as k increases in a given sample size n.

8.5 Homoskedasticity

Homoskedasticity assumption requires constant variance of the error term across all levels of the independent variable(s).

$$\mathrm{Var}(\varepsilon_i | X) = \sigma^2$$

When the assumption is violated, we have heteroskedasticity:

$$\mathrm{Var}(\varepsilon_i | X) = \sigma_i^2$$

that is, the variance of the error term varies with the level of the independent variable(s).

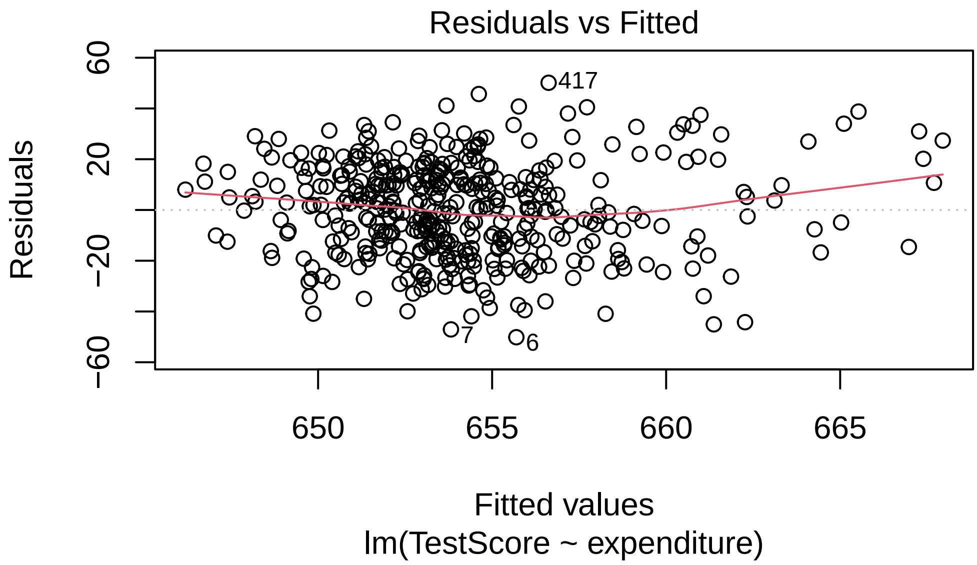
If the error has heteroskedasticity, the standard error under homoskedasticity assumption will be underestimated, leading to invalid t statistics, affecting hypothesis tests and confidence intervals.

8.5.1 Detection of Heteroskedasticity

8.5.2 Graphical Method

A common graphical method to detect heteroskedasticity is to plot the residuals against the fitted values or one of the independent variables.

Recall the CASchools example, we regress TestScore on expenditure.



We recap the figure summary below:

- The residuals appear reasonably balanced around zero, though the spread suggests substantial unexplained variation.
- The spread of residuals seems to **increase slightly with fitted values**, indicating potential heteroskedasticity (non-constant variance of errors). This suggests that the assumption of homoskedasticity may be violated, which could affect the reliability of our coefficient estimates standard errors.

8.5.3 Fix Heteroskedasticity

A quick fix is to use heteroskedasticity-robust standard errors, which are valid even when the homoskedasticity assumption is violated.

```
library(tidyverse)
library(sandwich) # for robust standard errors
library(lmtest) # for coeftest()
library(stargazer)

f_name <-
 ↪  "https://raw.githubusercontent.com/my1396/FIN5005-Fall2025/refs/heads/mai
cas <- read_csv(f_name,
    col_types = cols(
        county = col_factor(), # read as factor
        grades = col_factor()
    )
)
cas <- cas %>%
```

```
    mutate(TestScore = (read + math) / 2)
model <- lm(TestScore ~ expenditure, data = cas)
```

```
# OLS standard errors
se_ols <- sqrt(diag(vcov(model)))

# HC1 heteroskedasticity-robust standard errors
se_hc1 <- sqrt(diag(vcovHC(model, type = "HC1")))

# Stargazer output: same model, different SEs
stargazer(model, model,
          se = list(se_ols, se_hc1),
          digits = 4,
          column.labels = c("OLS SE", "Robust SE"),
          type = "text"
          )
```

```
===========================================================
                                  Dependent variable:
                              ----------------------------
                                       TestScore
                                   OLS SE       Robust SE
                                    (1)            (2)
-----------------------------------------------------------
expenditure                      0.0057***      0.0057***
                                  (0.0014)      (0.0016)


Constant                        623.6165***    623.6165***
                                  (7.7197)      (8.4664)


-----------------------------------------------------------
Observations                        420            420
R2                                 0.0366        0.0366
Adjusted R2                        0.0343        0.0343
Residual Std. Error (df = 418)    18.7239        18.7239
F Statistic (df = 1; 418)        15.8734***    15.8734***
===========================================================
Note:                           *p<0.1; **p<0.05; ***p<0.01
```

Under the heteroskedasticity-robust standard errors:

- No effects on the coefficient estimates.
- The Std. Error is slightly larger (0.001620 vs. 0.001443). The coefficient remains statistically significant at the 5% level.

Interpretation

The heteroskedasticity-robust results suggest that the OLS standard errors may have been **too optimistic (underestimated variability)**. While the statistical significance of the expenditure effect is slightly reduced, it remains highly significant, indicating a robust positive relationship between expenditure and test scores.

9 Hypothesis Testing and Confidence Intervals

We have been using t-statistics and p-values to for testing hypotheses such as whether a correlation coefficient is significantly different from zero or if a regression coefficient is significantly different from zero.

In the following section, we will introduce the procedure of hypothesis testing formally.

Note that we will mainly focus on two-sided hypothesis tests when testing for the significance of a single coefficient as they are most commonly used in practice.

We will use one-sided tests when introducing F-tests for the overall significance of a regression model and for model comparison.

Study objectives for hypothesis testing

The goal is be able to conduct hypothesis test manually given necessary information.
For example, in case of testing if a regression coefficient is significantly different from zero, you should be able to

1. Establish null and alternative hypotheses.
2. Calculate the test statistic given the coefficient estimate, standard error.
3. Find the critical value of your test statistic given the significance level, 5% is commonly used.

   You need to be familiar with the distribution table to find the critical value. The tables can be found HERE.
4. Decision rule.

   Compare the test statistic with the critical value to make decision on rejecting or not rejecting the null hypothesis.
5. Interpret the result in context. What it means in reality?

9.1 t-test for Single Coefficient

Using the simple linear regression model of California School Test Scores in the previous session as an example, test whether expenditure has a significant effect on test scores at the 5% significance level.

```
Call:
lm(formula = TestScore ~ expenditure, data = cas)

Residuals:
    Min      1Q  Median      3Q     Max 
-50.146 -14.206   0.689  13.513  50.127 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 6.236e+02  7.720e+00  80.783  < 2e-16 ***
expenditure 5.749e-03  1.443e-03   3.984 7.99e-05 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 18.72 on 418 degrees of freedom
Multiple R-squared:  0.03659,	Adjusted R-squared:  0.03428 
F-statistic: 15.87 on 1 and 418 DF,  p-value: 7.989e-05
```

Here is the hypothesis test step by step for the expenditure coefficient in the simple linear regression of California school test scores.

1. State Hypotheses
   - $H_0: \beta_{\text{exp}} = 0$ (expenditure has no effect on test scores)
   - $H_1: \beta_{\text{exp}} \neq 0$ (two-sided)
2. Calculate test statistic

$$t = \frac{\hat{\beta} - \beta}{\text{SE}(\hat{\beta})} \sim t_{df}$$

   where the test statistic follows a t-distribution with **degrees of freedom (df)** $= n - k - 1$. k is the number of predictors (excluding the intercept).

   Coming back to the regression output, we have estimate = 0.005749, value to test against with = 0, standard error = 0.001443.

   That is, $\hat{\beta} = 0.005749$, $\beta = 0$, and $\text{SE}(\hat{\beta}) = 0.001443$,

$$t = \frac{\hat{\beta} - \beta}{\mathrm{SE}(\hat{\beta})} = \frac{0.005749 - 0}{0.001443} = 3.984$$

where $n = 420$ (number of observations) and $k = 1$ (number of predictors), so $df = 420 - 1 - 1 = 418$.

3. Find critical value, $C_{\alpha/2}$.

   For significance level at 5%, i.e., $\alpha = 0.05$, the critical value is given by

$$C_{\alpha/2} = t_{0.975,df}$$

   Referring to the t-distribution table, $t_{0.975,418} \approx 1.96$.

   > **Finding critical value**
   >
   > Notice that when df is large, the critical value approaches that of the standard normal distribution, $z_{0.975} = 1.96$.
   > The rule-of-thumb is that when $df > 30$, you can use the standard normal critical values.

4. Decision rule

   - Reject H_0 if $|t| > 1.96$.
   - Fail to reject H_0 if $|t| \leq 1.96$.

   Since $3.984 > 1.96$, we reject H_0 and conclude that β_{exp} is statistically significantly different from zero.

5. Interpretation in context

   Expenditure has a statistically significant positive association with test scores at the 5% level. The point estimate implies that a \$1,000 increase in per-student expenditure is associated with about $0.005749 \times 1000 = 5.75$ points higher test scores.

9.1.1 P-value Approach

Statistical software often reports the p-value, which is the smallest significance level at which you would reject the null hypothesis.

- If the p-value $< \alpha$, reject H_0.
- If the p-value $\geq \alpha$, fail to reject H_0.
- Common significance levels are 1%, 5%, and 10%.
- **Interpretation:** The probability of rejecting the null hypothesis when the null hypothesis is true.

Q: How to find the p-value given test statistic?
A: For two-sided test, the p-value is given by

$$p\text{-value} = 2\,\mathbb{P}(T > |t|)$$

Example 9.1. A marketing manager wants to understand whether the number of social media posts influences monthly customer engagement for an online store. She runs a regression analysis and finds a t-statistic of $t = 2.457$ for the coefficient on the number of posts, with $df = 30$.

- Compute the two-sided p-value.
- Based on a 5% significance level, would you reject the null hypothesis that the number of posts has no effect on customer engagement?

Solution

The p-value is obtained by:

$$p\text{-value} = 2\,\mathbb{P}(T > |t|) = 2 \times 0.01 = 0.02$$

The p-value is 0.02, which is less than the common significance level of 0.05. Therefore, we reject the null hypothesis and conclude that the regression coefficient is statistically significant at the 5% level.

Interpretation: There is statistically significant evidence that the number of social media posts affects customer engagement for the online store.

Example 9.2. A financial analyst is studying the relationship between advertising expenditure and sales revenue for a chain of retail stores. She runs a regression and obtains an estimated coefficient for advertising spend. To test whether advertising has a statistically significant effect on sales, she computes a t-statistic of $t = 2.15$ with $df = 25$.

At the 5% significance level, should she reject the null hypothesis that advertising has no effect on sales?

Solution

The critical value at 5% significance level is $t_{0.975,25} = 2.060$. Since $2.15 > 2.060$, we reject the null hypothesis at the 5% significance level.

Interpretation: There is statistically significant evidence that advertising expenditure affects sales revenue for the retail chain.

Example 9.3. A store manager wants to investigate whether the amount spent on in-store promotions affects weekly sales. She runs a regression and obtains a t-statistic of $t = 0.82$ for the coefficient on promotion spending, with $df = 28$.

At the 5% significance level, would you reject the null hypothesis that promotion spending has no effect on weekly sales?

Solution

The critical value at 5% significance level is $t_{0.975,28} = 2.048$. Since $0.82 < 2.048$, we **fail to reject** the null hypothesis at the 5% significance level.

Interpretation:

There is no statistically significant evidence that the amount spent on in-store promotions affects weekly sales.

9.2 Confidence Interval for Single Coefficient

A 95 percent confidence interval for the slope is given by

$$\left(\hat{\beta} - C_{\alpha/2} \times SE(\hat{\beta}),\ \hat{\beta} + C_{\alpha/2} \times SE(\hat{\beta})\right)$$

where $C_{\alpha/2}$ is the critical value for a two-sided test at significance level α.

For the expenditure coefficient in the simple linear regression of California school test scores, $\hat{\beta} = 0.005749$, $SE(\hat{\beta}) = 0.001443$, and $C_{0.025} = 1.96$.

A 95% confidence interval is therefore

$$0.005749 \pm 1.96 \times 0.001443 \approx [0.0029,\ 0.0086],$$

which corresponds to roughly 2.9 to 8.6 points per $1,000.

The effect is statistically significant but modest in size, and the low R^2 indicates that many other factors also influence test scores.

10 Multiple Linear Regression

Study Objectives

- Understand the concept of multiple linear regression (MLR) and how it extends simple linear regression (SLR).
- Perform multiple regression analysis using R and interpret the results.
- Learn how to interpret the coefficients in a multiple regression model.
- Understand the importance of controlling for confounding variables in regression analysis.
- Conduct and interpret F-tests for the overall significance of a regression model.
- Explore alternative model specifications and their implications on regression results.

The multiple regression model extends the basic concept of the simple regression model to include multiple explanatory variables. This allows us to analyze the relationship between a dependent variable and several independent variables simultaneously.

A multiple regression model enables us to estimate the effect on Y_i of changing a regressor X_{1i} if the remaining regressors $X_{2i}, X_{3i}, \ldots, X_{ki}$ are held constant.

Just like in the simple regression model, we assume the true relationship between Y_i and $X_{1i}, X_{2i}, X_{3i}, \ldots, X_{ki}$ to be linear. The relation is given by the population regression function:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + \ldots + \beta_k X_{ki} + \varepsilon_i.$$

Now we go back to the example of test scores of students in California schools. Recall in the simple regression model, we regression test score on expenditure per student and found statistically significant positive effect. However, the R-squared (3.7%) was rather low and residuals plot shows substantial noise, indicating that there are other factors affecting test scores.

💡 We can improve the model by including more explanatory variables.

We expect that smaller class sizes lead to better test scores, as teachers can give more individual attention to each student. Therefore, we introduce class size (number of students per teacher) as an additional regressor.

The multiple regression model is then given by:

$$\text{TestScore}_i = \beta_0 + \beta_1 \times \text{expenditure}_i + \beta_2 \times \text{STR} + \varepsilon_i$$

where STR is the student-teacher ratio (representing class size).

10.1 Descriptive Analysis

10.1.1 Summary Statistics

```r
# Data Preparation
library(tidyverse)

f_name <-
 ↪  "https://raw.githubusercontent.com/my1396/FIN5005-Fall2025/refs/heads/m
cas <- read_csv(f_name,
    col_types = cols(
        county = col_factor(), # read as factor
        grades = col_factor()
    )
)
cas <- cas %>%
    mutate(
        TestScore = (read + math) / 2,
        STR = students / teachers
    )
# summary statistics
cas %>%
    select(TestScore, expenditure, STR) %>%
    summary()
```

```
   TestScore       expenditure         STR
 Min.   :605.6   Min.   :3926   Min.   :14.00
 1st Qu.:640.0   1st Qu.:4906   1st Qu.:18.58
```

```
 Median :654.5   Median :5215   Median :19.72
 Mean   :654.2   Mean   :5312   Mean   :19.64
 3rd Qu.:666.7   3rd Qu.:5601   3rd Qu.:20.87
 Max.   :706.8   Max.   :7712   Max.   :25.80
```

10.1.2 Scatter Plot

Scatter plot of TestScore against expenditure and STR:

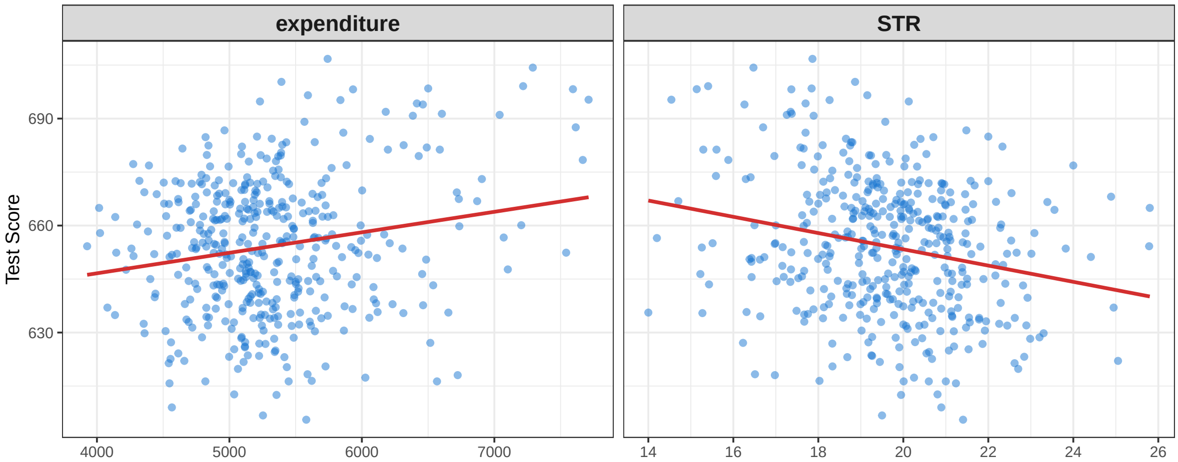


Figure 10.1 Scatter plots of TestScore vs expenditure and STR. STR stands for students-teach ratio, proxying class size.

There seems to be a negative correlation between class size (STR) and test scores, as expected.

Use `cor()` to compute the correlation matrix.

```
cor(cas %>% select(TestScore, expenditure, STR)) %>% round(2)
```

```
            TestScore expenditure   STR
TestScore        1.00        0.19 -0.23
expenditure      0.19        1.00 -0.62
STR             -0.23       -0.62  1.00
```

10.1.3 Run Multiple Regression

We estimate the model using OLS (`lm()` function) and obtain the following results:

```r
model_multiple <- lm(TestScore ~ expenditure + STR, data = cas)
summary(model_multiple)
```

```
Call:
lm(formula = TestScore ~ expenditure + STR, data = cas)

Residuals:
    Min      1Q  Median      3Q     Max 
-47.507 -14.403   0.407  13.195  48.392 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept) 675.577176  19.562221  34.535   <2e-16 ***
expenditure   0.002487   0.001823   1.364   0.1733    
STR          -1.763216   0.610914  -2.886   0.0041 ** 
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 18.56 on 417 degrees of freedom
Multiple R-squared:  0.05545,	Adjusted R-squared:  0.05092 
F-statistic: 12.24 on 2 and 417 DF,  p-value: 6.824e-06
```

10.1.4 Interpretation of the Coefficients

The estimated model is

$$\widehat{\text{TestScore}} = 675.58 + 0.0025 \cdot \text{expenditure} - 1.76 \cdot \text{STR}.$$

- **STR (–1.76): Holding expenditure constant**, an additional student per teacher lowers the test score on average by 1.76 points. This effect is statistically significant at the 1% level.
- **Expenditure (0.0025): Holding STR constant**, a one-unit increase in expenditure is associated with an increase of 0.0025 points in the test score. This effect is very small and statistically insignificant (p-value = 0.17).

10.1.5 Compare with Simple Regression

```r
library(stargazer)
model_simple <- lm(TestScore ~ expenditure, data = cas)
stargazer(
    model_simple, model_multiple,
    digits = 4,
    type = ifelse(knitr::is_html_output(), "html", "latex"),
    dep.var.labels = "Test Score",
    column.labels = c("SLR", "MLR"),
    covariate.labels = c("Expenditure", "STR"),
    header = FALSE,
    notes = "Standard errors in parentheses.",
    notes.append = TRUE
)
```

Table 10.1

	Dependent variable:	
	Test Score	
	SLR	MLR
	(1)	(2)
Expenditure	0.0057^{***}	0.0025
	(0.0014)	(0.0018)
STR		-1.7632^{***}
		(0.6109)
Constant	623.6165^{***}	675.5772^{***}
	(7.7197)	(19.5622)
Observations	420	420
R^2	0.0366	0.0555
Adjusted R^2	0.0343	0.0509
Residual Std. Error	18.7239 (df = 418)	18.5619 (df = 417)
F Statistic	15.8734^{***} (df = 1; 418)	12.2409^{***} (df = 2; 417)
Note:		*p<0.1; **p<0.05; ***p<0.01
		Standard errors in parentheses.

In the simple regression of TestScore on expenditure:

$$\widehat{\text{TestScore}} = 623.6 + 0.0057 \times \text{expenditure},$$

- Expenditure had a positive and statistically significant effect ($p < 0.001$).
- The effect size (0.0057 vs. 0.0025) was larger than in the multiple regression.
- The overall explanatory power was improved (Adjusted $R^2 \approx 0.0343$ in the SLR vs. $R^2 \approx 0.0509$ in the MLR).

Thus, once STR is included, the apparent effect of expenditure becomes much smaller and loses statistical significance.

Q: Why does the effect of expenditure change when STR is included?

A: Expenditure is no longer significant in the multiple regression because expenditure and STR are **correlated**. Districts that spend more on education often also have smaller class sizes ($\rho = -0.62$).

In the simple regression, the positive effect of expenditure partly reflected the fact that higher spending was associated with lower STR, which itself improves test scores. Once STR is explicitly controlled for, the "true" partial effect of expenditure (holding class size constant) is close to zero.

This illustrates the importance of multiple regression: it separates the effect of each regressor while holding others fixed, and it reveals that the real driver of test scores here is class size, not expenditure.

10.1.6 Regression diagnostics

- **Residual Standard Error (RSE):**
  - SLR: 18.72 (df = 418) → MLR: 18.56 (df = 417)
  - The MLR has a slightly smaller RSE, meaning it predicts test scores a bit more accurately.
- **Multiple R^2:**
  - SLR: 0.037 → MLR: 0.055
  - Adding STR increases the proportion of variation explained, though the overall fit remains modest.

- **Adjusted R^2:**
  - SLR: 0.034 → MLR: 0.051
  - The adjusted R^2 also improves, showing STR provides genuine explanatory power beyond expenditure.
- **F-statistic (overall significance):**
  - SLR: 15.87, p < 0.001 → MLR: 12.24, p < 0.001
  - Both models are statistically significant overall. The MLR has a smaller F-statistic because it includes more regressors, but still clearly improves explanatory power.

10.2 F-test for Significance of the Overall Regression

F-test is also known as the one-way analysis of variance (ANOVA) test. It tests whether the regression model as a whole is statistically significant.

F-test for that all k of the slope coefficients in a linear model are equal to zero, i.e., to test the exclusion of all explanatory variables except the intercept, β_0. Formally speaking.

$$\begin{aligned}
&\mathrm{H}_0 : \beta_1 = \beta_2 = \cdots = \beta_K = 0 \\
&\mathrm{H}_1 : \text{At least one of the } \beta_1, \beta_2, \ldots, \beta_K, \text{is not zero.}
\end{aligned}$$

This is referred to as the F-test for one-way ANOVA.

The test statistic is given by:

$$F = \left(\frac{n-k-1}{k}\right)\left(\frac{R^2}{1-R^2}\right) \sim F(k, n-k-1).$$

where F is F-distributed with k and $n-k-1$ degrees of freedom.

The F-statistic can be rewritten as:

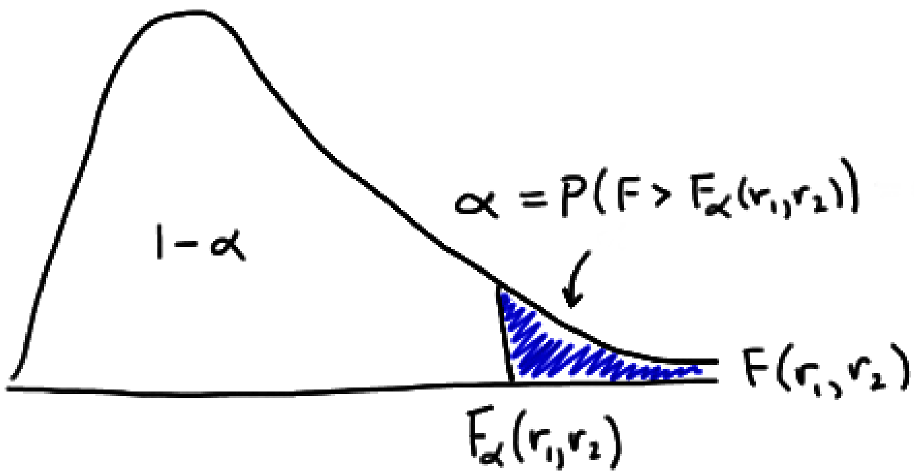
$$\begin{aligned}
F &= \frac{R^2/k}{(1-R^2)/(n-k-1)} = \frac{SSE/k}{SSR/(n-k-1)} \\
&= \frac{\text{MSE}}{\text{MSR}} \\
&= \frac{\text{Explained Variation per degree of freedom}}{\text{Residual Variation per degree of freedom}},
\end{aligned}$$

where

- SSE is the explained sum of squares (variation **explained** by the regression model).
    - $MSE = SSE/k$ is the mean square due to explained variation.
- SSR is the residual sum of squares (variation **not explained** by the regression model).
    - $MSR = SSR/(n-k-1)$ is the mean square due to residual variation.

We reject H_0 if $F > F_\alpha(k, n-k-1)$.

- $F_\alpha(k, n-k-1)$ is the $(1-\alpha)$ percentile in the $F(k, n-k-1)$ distribution, corresponding to the level of significance, α.



The p-value is found by:

$$\text{P-value} = \mathbb{P}(F > F_{\text{obs}}),$$

where F_{obs} is your calculated/observed test statistic based on your data sample.

- Large values of F give evidence against the validity of the null hypothesis. Note that a large F is induced by a large value of R^2.
- The logic of the test is that the F statistic is a measure of the loss of fit (namely, all of R^2) that results when we impose the restriction that all the slopes are zero. If F is large, then the hypothesis is rejected.

10.2.1 Example: F-test for the Multiple Regression Model

1. **Null and alternative hypotheses**

$$\begin{aligned}&H_0 : \beta_1 = \beta_2 = 0\\ &H_1 : \text{At least one of the } \beta_1, \beta_2, \text{is not zero.}\end{aligned}$$

   In plain language,

   - H_0: Expenditure and STR have no effect on test scores.
   - H_1: At least one of expenditure and STR has an effect on test scores.

2. **Calculate the test statistic**

$$F = \frac{(R^2/k)}{(1-R^2)/(n-k-1)}$$

   where k is the number of regressors (not counting the intercept) and n is the sample size.

   From the regression output, we have

   - $R^2/k = 0.05545/2 = 0.027725$.
   - $(1-R^2)/(n-k-1) = (1-0.05545)/417 \approx 0.00226511$

   Hence, the F statistic is

$$F = \frac{0.027725}{0.00226511} \approx 12.24.$$

3. **Critical value at** $\alpha = 5\%$

   Degrees of freedom for the F distribution are $(df_1, df_2) = (k, n-k-1) = (2, 417)$.

   Referring to the F-distribution table to find the critical value.

   According to the F table, there is no exact value for $df_2 = 417$, we use the value for $df_2 = \infty$ as an approximation.

   We use the The critical value $F_{0.95}(2, \infty) = 3.00$. The exact value is $F_{0.95}(2, 417)$ is expected to be slightly larger than 3.00. But since our calculated F statistic is much larger than 3.00, we can safely reject the null hypothesis. It does not matter if the exact critical value is 3.01 or 3.06.

4. **Decision rule and conclusion**

   Decision rule at 5 percent significance level: reject H_0 if $F > F_{\text{crit}}$.

   Since $F_{\text{obs}} = 12.24 > F_{\text{crit}} = 3.00$, we reject H_0 and conclude that at least one of expenditure and STR has a statistically significant effect on test scores.

5. **Interpretation in context**

   Statistically the regression is significant overall. That means at least one of the regressors, expenditure or STR, is linearly related to test scores.

   From the coefficient t tests in your summary we know STR is the significant predictor while expenditure is not significant once STR is included. In plain language this suggests that class size is driving the overall significance of the model rather than expenditure.

10.3 More Alternative Model Specifications

We now consider student demographics as additional regressors. We include the following variables:

- `eng`: percentage of students who are English learners.
- `lunch`: percentage of students who are eligible for free lunch.
- `calworks`: percentage of students whose families receive aid from the California Work Opportunity and Responsibility to Kids (CalWORKs) program.

Note

- Students eligible for CalWorks live in families with a total income below the threshold for the subsidized lunch program so both variables are indicators for the share of **economically disadvantaged children**.

We consider the following models:

$$
\begin{aligned}
(I)\ TestScore &= \beta_0 + \beta_1 \times STR + u, \\
(II)\ TestScore &= \beta_0 + \beta_1 \times STR + \beta_2 \times english + u, \\
(III)\ TestScore &= \beta_0 + \beta_1 \times STR + \beta_2 \times english + \beta_3 \times lunch + u, \\
(IV)\ TestScore &= \beta_0 + \beta_1 \times STR + \beta_2 \times english + \beta_4 \times calworks + u, \\
(V)\ TestScore &= \beta_0 + \beta_1 \times STR + \beta_2 \times english + \beta_3 \times lunch + \beta_4 \times calworks + u.
\end{aligned}
$$

```
library(sandwich) # for robust standard errors

# estimate different model specifications
spec1 <- lm(TestScore ~ STR, data = cas)
spec2 <- lm(TestScore ~ STR + english, data = cas)
spec3 <- lm(TestScore ~ STR + english + lunch, data = cas)
spec4 <- lm(TestScore ~ STR + english + calworks, data = cas)
spec5 <- lm(TestScore ~ STR + english + lunch + calworks, data =
 ↪  cas)

# gather robust standard errors in a list
rob_se <- list(
    sqrt(diag(vcovHC(spec1, type = "HC1"))),
    sqrt(diag(vcovHC(spec2, type = "HC1"))),
    sqrt(diag(vcovHC(spec3, type = "HC1"))),
    sqrt(diag(vcovHC(spec4, type = "HC1"))),
    sqrt(diag(vcovHC(spec5, type = "HC1")))
)

# generate a summarizing table using stargazer
is_html <- knitr::is_html_output()
stargazer(
    spec1, spec2, spec3, spec4, spec5,
    type = if (is_html) "html" else "latex",
    se = rob_se,
    digits = 3,
    header = FALSE,
    column.labels = c("(I)", "(II)", "(III)", "(IV)", "(V)"),
    model.numbers = FALSE,
    notes = "Standard errors in parentheses.",
    notes.append = TRUE,
    font.size = if (is_html) NULL else "small",
    column.sep.width = if (is_html) "" else "1pt",
    float = is_html   # PDF: emit a bare tabular so it fits
     ↪  inside adjustbox
)
```

	Dependent variable:				
	TestScore				
	(I)	(II)	(III)	(IV)	(V)
STR	−2.280***	−1.101**	−0.998***	−1.308***	−1.014***
	(0.519)	(0.433)	(0.270)	(0.339)	(0.269)
english		−0.650***	−0.122***	−0.488***	−0.130***
		(0.031)	(0.033)	(0.030)	(0.036)
lunch			−0.547***		−0.529***
			(0.024)		(0.038)
calworks				−0.790***	−0.048
				(0.068)	(0.059)
Constant	698.933***	686.032***	700.150***	697.999***	700.392***
	(10.364)	(8.728)	(5.568)	(6.920)	(5.537)
Observations	420	420	420	420	420
R^2	0.051	0.426	0.775	0.629	0.775
Adjusted R^2	0.049	0.424	0.773	0.626	0.773
Residual Std. Error	18.581 (df = 418)	14.464 (df = 417)	9.080 (df = 416)	11.654 (df = 416)	9.084 (df = 415)
F Statistic	22.575*** (df = 1; 418)	155.014*** (df = 2; 417)	476.306*** (df = 3; 416)	234.638*** (df = 3; 416)	357.054*** (df = 4; 415)

Note: *p<0.1; **p<0.05; ***p<0.01
Standard errors in parentheses.

10.3.1 Interpretations

- Adding control variables roughly halves the coefficient on **STR**.

- The estimate is sensitive to the set of control variables used.

- Conclusion: decreasing the student–teacher ratio *ceteris paribus* by one unit leads to an estimated average increase in test scores of about **1 point**.

  "Ceteris Paribus" means "all other things being equal", i.e., holding all other regressors constant.

- Adding student characteristics as controls increases
  - R^2 and $\bar{R}^2$ from **0.049 (spec (1))** up to **0.773 (spec (3) and spec (5))**.

  - These variables can therefore be considered suitable predictors of test scores.

- `calworks` is not statistically significant in all models.
  - Example: in **spec (5)**, the coefficient on `calworks` is not significantly different from zero at the 5% level since

    $$\left|\frac{-0.048}{0.059}\right| = 0.81 < 1.96.$$

    For $df = 415$, we can safely use the normal distribution table to find the critical value for a two-sided t-test at the 5% significance level as the sample size is large ($df > 30$).

    Commonly used critical values under normal distribution:
    * 1.96 for 5% significance level;

    * 1.64 for 10% significance level; and

    * 2.58 for 1% significance level.

- The effect of adding `calworks` to the base specification (spec (3)) on the coefficient of **size** and its standard error is negligible.

- We can therefore consider `calworks` as a **superfluous control variable**, given the inclusion of `lunch` in this model.

  The two variables are highly correlated ($\rho = 0.74$). See Figure 10.2 for a scatter plot.

```
cat("Correlation between lunch and calworks: \n\n")
cor(cas %>% select(lunch, calworks)) %>% round(2)
```

```
Correlation between lunch and calworks:

         lunch calworks
lunch     1.00     0.74
calworks  0.74     1.00
```

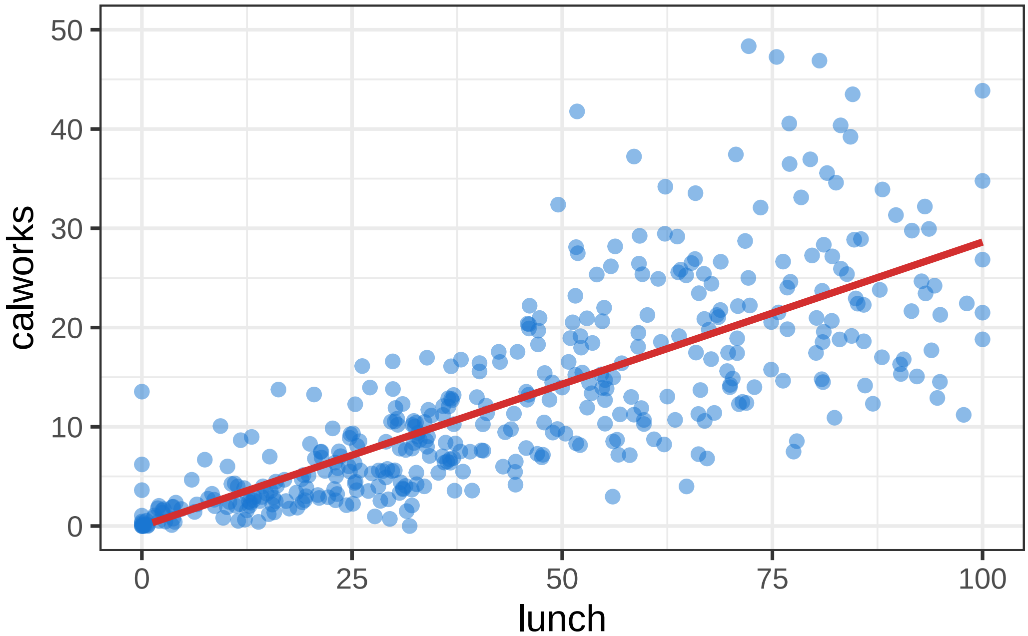


Figure 10.2 Scatter plots of `lunch` vs. `calworks`. Positive linear relationship between the two variables.

Ref:

- Section 7.6 Analysis of the Test Score Data Set, *Introduction to Econometrics with R*, https://www.econometrics-with-r.org/7-6-analysis-of-the-test-score-data-set.html

11 Lab 2: Regression with Dummy Variables

In this lab, we will run regressions with dummy variables (also known as binary variables or indicator variables).

🎯 In this script, you will learn how to:

- Load data from a local CSV file
- Create dummy variables from a numeric variable
- Run a regression with dummy variables
- Interpret the regression results
- Conduct an ANOVA test
  - to compare means across groups
  - to compare models

11.1 Instructions

Copy and paste **ALL code** in this page into a new R script file in RStudio. Save the file as `lab2_binary_variables.R` (we call it a script or a source file). Then, run the code step by step, answering the questions.

Alternatively, use Google Colab.

```
# load packages and dataset
pkgs <- c("tidyverse", "moments", "data.table", "stargazer")
missing <- setdiff(pkgs, rownames(installed.packages()))
if (length(missing) > 0) install.packages(missing)
invisible(lapply(pkgs, function(pkg)
  suppressPackageStartupMessages(library(pkg, character.only =
  TRUE))))
```

11.2 Load data into working directory

We continue to use the same dataset California school dataset (CASchools). Refer to Data Visualization: Dataset Overview for detailed description of the dataset.

Previously, we loaded the data directly from an online data repository. Now, we will load the data from a local CSV file.

Download the `CASchools_test_score.csv` file from Canvas and placed it in the same folder as where you put `lab2_binary_variables.R`. This is to make sure that the script can find the data file.

- Depending on your computer setting, you may or may not see the file extension `.csv`.
- You can rename the data file, note that the data file name needs to match the `f_name` used in the code below.

Troubleshooting: If you get an error message like

> Error: 'CASchools_test_score.csv' does not exist in current working directory ('path-to-file').

It means that your working directory is not set to the folder where you put the data file. You need to set your working directory to that folder. You can do this in RStudio by clicking on `Session` -> `Set Working Directory` -> `Set to Source File Location`. Then, run the code again.

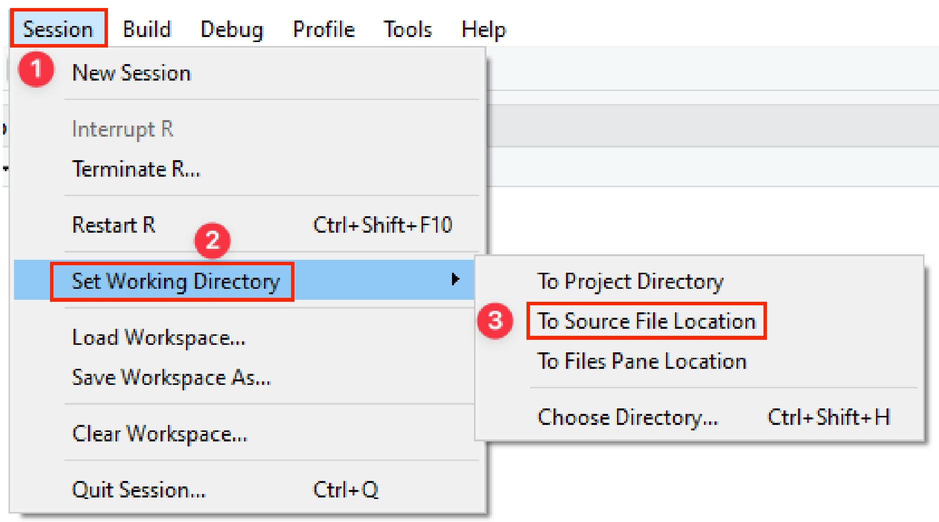


Figure 11.1 change-working-directory-r

```r
# load dataset
f_name <- "CASchools_test_score.csv"
cas <- read_csv(f_name)
# data preview, first and last 5 rows
cas %>% as.data.table() %>% print(topn = 5)
```

```
    district                            school      county grades students
       <num>                            <char>      <char> <char>    <num>
  1:   75119               Sunol Glen Unified     Alameda  KK-08      195
  2:   61499              Manzanita Elementary      Butte  KK-08      240
  3:   61549      Thermalito Union Elementary      Butte  KK-08     1550
  4:   61457 Golden Feather Union Elementary      Butte  KK-08      243
  5:   61523        Palermo Union Elementary      Butte  KK-08     1335
 ---
416:   68957         Las Lomitas Elementary   San Mateo  KK-08      984
417:   69518          Los Altos Elementary Santa Clara  KK-08     3724
418:   72611          Somis Union Elementary     Ventura  KK-08      441
419:   72744               Plumas Elementary        Yuba  KK-08      101
420:   72751           Wheatland Elementary        Yuba  KK-08     1778
    teachers calworks   lunch computer expenditure    income   english  read
       <num>    <num>   <num>    <num>       <num>     <num>     <num> <num>
  1:   10.90   0.5102  2.0408       67    6384.911 22.690001  0.000000 691.6
  2:   11.15  15.4167 47.9167      101    5099.381  9.824000  4.583333 660.5
  3:   82.90  55.0323 76.3226      169    5501.955  8.978000 30.000002 636.3
  4:   14.00  36.4754 77.0492       85    7101.831  8.978000  0.000000 651.9
  5:   71.50  33.1086 78.4270      171    5235.988  9.080333 13.857677 641.8
 ---
416:   59.73   0.1016  3.5569      195    7290.339 28.716999  5.995935 700.9
417:  208.48   1.0741  1.5038      721    5741.463 41.734108  4.726101 704.0
418:   20.15   3.5635 37.1938       45    4402.832 23.733000 24.263039 648.3
419:    5.00  11.8812 59.4059       14    4776.336  9.952000  2.970297 667.9
420:   93.40   6.9235 47.5712      313    5993.393 12.502000  5.005624 660.5
      math
     <num>
  1: 690.0
  2: 661.9
  3: 650.9
  4: 643.5
  5: 639.9
 ---
416: 707.7
417: 709.5
418: 641.7
419: 676.5
420: 651.0
```

Prepare dataset for analysis.

As done previously, we create the following variables:

- `TestScore`: average test score of students in a school (average of math and reading scores)
- `str`: student-teacher ratio

```
cas <- cas %>%
    mutate(
        TestScore = (read + math) / 2,
        STR = students / teachers
    )
```

11.3 Regression When X is a Binary Variable

Instead of using a continuous regressor X , we might be interested in running the regression

$$Y_i = \beta_0 + \beta_1 D_i + u_i, \tag{11.1}$$

where D_i is a binary variable, a so-called dummy variable. For example, we may defined D_i as follows:

$$D_i = \begin{cases} 1 & \text{if } STR \text{ in } i^{th} \text{ school district} < 20 \\ 0 & \text{if } STR \text{ in } i^{th} \text{ school district} \geq 20. \end{cases} \tag{11.2}$$

The regression model now is

$$TestScore_i = \beta_0 + \beta_1 D_i + u_i.$$

11.3.1 Data Visualization

```
# Create the dummy variable as defined above
# D=1 (low STR); D=0 (high STR)
cas$D <- cas$STR < 20
```

```
# Compute group means
means <- aggregate(TestScore ~ D, data = cas, mean)
means
```

A data.frame: 2 × 2

D <lgl>	TestScore <dbl>
FALSE	650.0768
TRUE	657.2462

```
ggplot(cas, aes(x = factor(D), y = TestScore)) +
    # scatter points, spreads the points horizontally so they
     ↪  don't overlap
    geom_jitter(width = 0.1, height = 0, size = 0.5, color =
     ↪  "steelblue") +
    geom_point(
        data = means, aes(x = factor(D), y = TestScore),
        color = "red", size = 2
    ) + # group means
    labs(
        x = expression(D[i]),
        y = "Test Score",
    ) +
    theme_minimal(base_size = 14)
```

```
Unable to display output for mime type(s): text/html
```

We see that small size class has higher average test score.

11.3.2 Regression with a Binary Variable

With D as the regressor, it is not useful to think of β_1 as a slope parameter since since $D_i \in \{0, 1\}$, i.e., we only observe two discrete values instead of a continuum of regressor values. There is no continuous line depicting the conditional expectation function $E(TestScore_i | D_i)$ since this function is solely defined for x-positions $D_i = 0$ and $D_i = 1$.

Therefore, **the interpretation** of the coefficients in this regression model is as follows:

- $E(Y_i | D_i = 0) = \beta_0$, so β_0 is the expected test score in districts where $D_i = 0$ and STR is larger or equal to 20.
- $E(Y_i | D_i = 1) = \beta_0 + \beta_1$, or $\beta_1 = E(Y_i | D_i = 1) - E(Y_i | D_i = 0)$. Thus, β_1 is the difference in group-specific expectations, i.e., the difference in expected test score between districts with $STR < 20$ and those with $STR \geq 20$.

We now estimate the dummy regression model as defined by Equations (11.1) and (11.2).

```r
dummy_model_slr <- lm(TestScore ~ D, data = cas)
stargazer(dummy_model_slr, type = "text", title = "Dummy
↪  Variable Regression Results", digits = 3)
```

```
Dummy Variable Regression Results
===============================================
                        Dependent variable:    
                    ---------------------------
                             TestScore         
-----------------------------------------------
D                            7.169***          
                              (1.847)          
                                               
Constant                    650.077***         
                              (1.393)          
                                               
-----------------------------------------------
Observations                    420            
R2                             0.035           
Adjusted R2                    0.032           
Residual Std. Error      18.741 (df = 418)     
F Statistic           15.073*** (df = 1; 418)  
===============================================
Note:               *p<0.1; **p<0.05; ***p<0.01
```

💡 Q: **Based on the regression output, answer the following questions:**

- What is the average test score in school districts with $STR \geq 20$?
- What is the average test score in school districts with $STR < 20$?
- What is the difference in average test score between school districts with $STR < 20$ and those with $STR \geq 20$?
- Is the difference statistically significant at the 5% significance level?

11.4 MLR with a Binary Variable

```
dummy_model_mlr <- lm(TestScore ~ computer + english + lunch +
 ↪  D, data = cas)
stargazer(dummy_model_slr, dummy_model_mlr,
    type = "text",
    title = "Regression Results",
    column.labels = c("SLR", "MLR"),
    digits = 3
)
```

```
Regression Results
====================================================================
                                  Dependent variable:
                 ------------------------------------------------
                                       TestScore
                              SLR                        MLR
                              (1)                        (2)
--------------------------------------------------------------------
computer                                                0.001
                                                       (0.001)

english                                               -0.139***
                                                       (0.034)

lunch                                                 -0.544***
                                                       (0.022)

D                          7.169***                    2.876***
                           (1.847)                     (0.952)

Constant                  650.077***                  678.590***
                           (1.393)                     (1.164)

--------------------------------------------------------------------
Observations                 420                         420
R2                          0.035                       0.770
Adjusted R2                 0.032                       0.768
Residual Std. Error    18.741 (df = 418)          9.176 (df = 415)
F Statistic        15.073*** (df = 1; 418) 347.915*** (df = 4; 415)
====================================================================
Note:                                    *p<0.1; **p<0.05; ***p<0.01
```

💡 Q: **Comparing the regression results of the simple regression with a binary variable and the multiple regression with additional control variables, answer the following questions:**

- How does the coefficient of the binary variable change when we add more control variables?
- Why does the estimated effect of small classes shrink when other control variables are included?
- Does the smaller coefficient in the multiple linear regression model mean small classes are unimportant, or does it mean we now have a less biased estimate of their effect? Explain.

11.5 ANOVA test

11.5.1 One-way ANOVA test

The purpose of a one-way ANOVA (analysis of variance) test is to determine the existence of a statistically significant difference among the means of three or more populations. The test actually uses variances to help determine if the population means are equal or not.

In what follows, we will use the one-way ANONA test to examine whether the average test scores differ across schools with different sizes of student-teacher ratios (STR).

We first **create a new categorical variable** `str_cat` based on the variable `str` (student-teacher ratio):

- Small size class: STR < 18
- Medium size class: 18 ≤ STR < 21
- Large size class: STR ≥ 21

Then, we run the one-way ANOVA test to see if the average test scores differ across schools with different sizes of student-teacher ratios (STR).

Null and alternative hypotheses:

- H_0: The average test scores are the same across schools with different sizes of student-teacher ratios (STR).
- H_1: At least one pair of average test scores are different.

```
cas <- cas %>%
    mutate(
        str_cat = case_when(
            STR < 18 ~ "Small",
            STR >= 18 & STR < 21 ~ "Medium",
            STR >= 21 ~ "Large"
```

```
        )
    )
cas$str_cat %>% table()
```

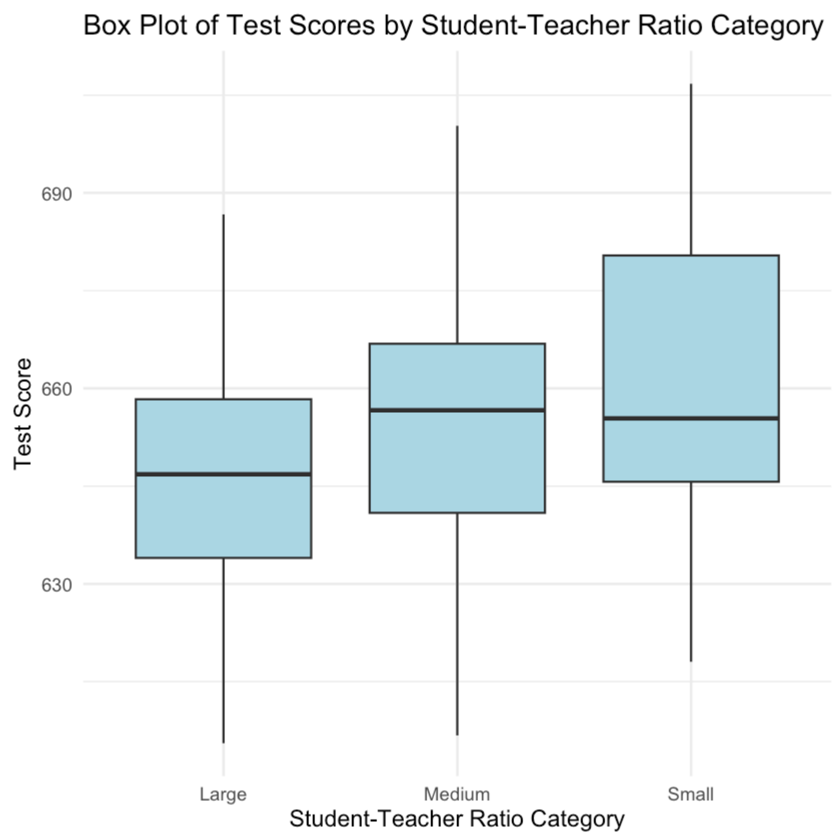
```
.
 Large Medium  Small
    92    253     75
```

```
# ANOVA test
anova_result <- aov(TestScore ~ str_cat, data = cas)
summary(anova_result)
```

```
             Df Sum Sq Mean Sq F value   Pr(>F)
str_cat       2   9101    4551   13.27 2.59e-06 ***
Residuals   417 143008     343
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Given the significance level $\alpha = 0.05$, we reject the null hypothesis (p-value < 0.05) and conclude that at least one pair of average test scores are different across schools with different sizes of student-teacher ratios (STR).

```
# Box plot of TestScore by str_cat
p_anova <- ggplot(cas, aes(x = str_cat, y = TestScore)) +
    geom_boxplot(fill = "lightblue") +
    labs(
        x = "Student-Teacher Ratio Category",
        y = "Test Score",
        title = "Box Plot of Test Scores by Student-Teacher Ratio
         ↪  Category"
    ) +
    theme_minimal(base_size = 14)
p_anova
```



11.5.2 Two-way ANOVA test

The two-way ANOVA test is used to compare the fits of two regression models.

- Test for Improvement: It tests whether adding more predictors (variables) to a model significantly improves the model's ability to explain the variability in the response variable.
- Model Selection: Helps in deciding between a simpler model with fewer variables and a more complex one with more variables.

For instance, consider we have two models:

- Model 1: $y = \beta_0 + \beta_1 x_1 + \varepsilon$, and
- Model 2: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \varepsilon$.

We can form a test by comparing the R^2 from the two models.

$$\begin{aligned} \mathrm{H}_0 &: \beta_2 = \beta_3 = 0 \\ \mathrm{H}_1 &: \beta_2 \neq 0 \text{ or } \beta_3 \neq 0, \end{aligned}$$

which imposes two exclusion restrictions.

Coming back to our California school TestScore example, we can use the two-way ANOVA test to see if the multiple regression model with additional control variables (Model 2) significantly improves the model's ability to explain the variability in TestScore compared to the simple regression model (Model 1).

```
anova_result2 <- anova(dummy_model_slr, dummy_model_mlr)
anova_result2
```

A anova: 2 × 6

	Res.Df <dbl>	RSS <dbl>	Df <dbl>	Sum of Sq <dbl>	F <dbl>	Pr(>F) <dbl>
1	418	146815.42	NA	NA	NA	NA
2	415	34940.45	3	111875	442.9261	6.146085e-129

The p-value is way less than 0.05, indicating that we reject the null hypothesis and conclude that adding the control variables (`computer`, `english`, and `lunch`) significantly improves the model's ability to explain the variability in TestScore.

Part III

Advanced

12 Interactions of Binary Variables

🎯 **Study Objectives**

- Interpret interaction terms of two binary variables in regression models.
- Understand the importance of controlling for changes in interacting variables when interpreting coefficients.
- Conduct and interpret a two-way ANOVA to compare nested regression models.
- Interpret interaction terms between a binary variable and a continuous variable in regression models.
- Visualize interaction effects using regression lines.

For this section of the course, we will be working with the `Boston` dataset, which contains 506 observations on housing values in various suburbs of Boston. This dataset provides detailed information on factors such as crime rates, property age, locations (proximity to the Charles River), and other socioeconomic variables.

We will use this data to **explore and analyze the determinants of housing prices**, applying regression techniques to understand how different factors influence the value of homes.

```r
# load packages and dataset
pkgs <- c("tidyverse", "MASS", "data.table", "stargazer",
 ↪  "cowplot")
missing <- setdiff(pkgs, rownames(installed.packages()))
if (length(missing) > 0) install.packages(missing)
invisible(lapply(pkgs, function(pkg)
 ↪  suppressPackageStartupMessages(library(pkg, character.only =
 ↪  TRUE))))
data(Boston) # Boston housing data
```

Use `?Boston` to see the description of the dataset.

Here are some variables in the dataset which we are going to use in this session.

Variables	Definitions
`medv`	median value of owner-occupied homes in $1000s.
`lstat`	the percentage of individuals with low socioeconomic status
`indus`	proportion of non-retail business acres per town; industrial, office, or commercial land not directly serving consumers.
`age`	proportion of owner-occupied units built prior to 1940.
`crim`	per capita crime rate by town.
`chas`	A dummy variable. Takes the value 1 if the Charles River (a short river in the proximity of Boston) passes through the suburb and is 0 otherwise.

12.1 Interactions of Two Binary Variables

Consider the following regression model

$$medv_i = \beta_0 + \beta_1\, chas_i + \beta_2\, old_i + \beta_3\, (chas_i \times old_i) + u_i \tag{12.1}$$

where $chas_i$ and old_i are dummy variables defined as follows:

$$chas_i = \begin{cases} 1 & \text{if the suburb is next to the Charles River} \\ 0 & \text{otherwise} \end{cases}$$

$$old_i = \begin{cases} 1 & \text{if } age_i \geq 95 \\ 0 & \text{otherwise} \end{cases}$$

where agi_i being the proportion of owner-occupied units built prior to 1940.

Instructions:

- Generate and append the binary variable *old* to the dataset Boston.
- Conduct the regression stated above and assign the result to `mod_bb`.
- Obtain a robust coefficient summary of the model. How do you interpret the results?

```
Boston <- Boston %>%
    mutate(old = ifelse(age >= 95, 1, 0))
mod_bb <- lm(medv ~ chas*old, data = Boston)
summary(mod_bb)
```

```
Call:
lm(formula = medv ~ chas * old, data = Boston)

Residuals:
    Min      1Q  Median      3Q     Max 
-17.771  -4.765  -1.683   2.776  33.433 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  23.6989     0.4503  52.624  < 2e-16 ***
chas          4.0582     1.6872   2.405   0.0165 *  
old          -7.1319     0.9493  -7.513 2.67e-13 ***
chas:old     10.5462     3.7577   2.807   0.0052 ** 
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.604 on 502 degrees of freedom
Multiple R-squared:  0.1301,    Adjusted R-squared:  0.1249 
F-statistic: 25.02 on 3 and 502 DF,  p-value: 4.244e-15
```

The estimated regression model is given by

$$\widehat{medv}_i = 23.70 + 4.06\,chas_i - 7.13\,old_i + 10.55\,(chas_i \times old_i)$$

Interpretation of the coefficients

We start with listing all possible combinations of the binary variables $chas_i$ and old_i.

Scenario	$chas_i$	old_i	Interpretation
1	0	0	Suburb not next to Charles River and new
2	0	1	Suburb not next to Charles River and old
3	1	0	Suburb next to Charles River and new
4	1	1	Suburb next to Charles River and old

Based on the estimated regression model, we can compute the expected value of `medv` for each of the four groups.

Scenario	$chas_i$	old_i	$chas_i \times old_i$	Expected value of `medv`
1	0	0	0	$\hat{\beta}_0 = 23.70$
2	0	1	0	$\hat{\beta}_0 + \hat{\beta}_2 = 23.70 + (-7.13) = 16.57$
3	1	0	0	$\hat{\beta}_0 + \hat{\beta}_1 = 23.70 + 4.06 = 27.76$
4	1	1	1	$\hat{\beta}_0 + \hat{\beta}_1 + \hat{\beta}_2 + \hat{\beta}_3 =$ $23.70 + 4.06 + (-7.13) + (10.55) = 31.18$

From the table above, we can interpret the coefficients as follows:

- $\hat{\beta}_0 = 23.70$ is the expected value of `medv` for suburbs that are not next to the Charles River and are new (i.e., `old = 0`).
- Comparing scenario 2 with 1: $\hat{\beta}_2 = -7.13$ is the difference in expected value of `medv` between new and old suburbs that are **NOT next to the Charles River**.

  In other words, old suburbs that are not next to the Charles River have an expected `medv` that is 7.13 lower than new suburbs that are **NOT next to the Charles River**.
- Comparing scenario 3 with 1: $\hat{\beta}_1 = 4.06$ is the difference in expected value of `medv` between suburbs that are next to the Charles River and those that are not, among **new** suburbs (i.e., `old = 0`).

  In other words, **new** suburbs that are next to the Charles River have an expected `medv` that is 4.06 higher than new suburbs that are not next to the Charles River.
- $\hat{\beta}_3 = 10.55$ has two interpretations:
  - If we compare scenario 4 with scenario 2, the difference in expected value of `medv` is given by $\hat{\beta}_1 + \hat{\beta}_3 = 4.06 + 10.55 = 14.61$.

    This means that **old** suburbs that are next to the Charles River have an expected `medv` that is 14.61 higher than **old** suburbs that are not next to the Charles River.

    → $\hat{\beta}_3$ can be interpreted as the difference in the effect of being next to the Charles River between old and new suburbs.
  - If we compare scenario 4 with scenario 3, the difference in expected value of `medv` is given by $\hat{\beta}_2 + \hat{\beta}_3 = -7.13 + 10.55 = 3.42$.

    This means that old suburbs that are **next to the Charles River** have an expected `medv` that is 3.42 higher than new suburbs that are **next to the Charles River**.

    → $\hat{\beta}_3$ can be interpreted as the difference in the effect of being old between suburbs that are next to the Charles River and those that are not.

> **Tip**
>
> **Controlling for changes in the interacting variables** is essential when interpreting the coefficient of interaction terms.
> A common approach is to substitute specific values for the interacting variables and compute the expected value of the dependent variable under different scenarios. This allows for a clearer interpretation of the coefficients by comparing how the expected value of the dependent variable varies across these scenarios.

12.1.1 Add More Controls Variables

We can also add more control variables to Equation 12.1 to improve the model fit (Adjusted $R^2 = 12\%$ as of now). For example, we can add `lstat` and `crim` as additional control variables.

$$medv_i = \beta_0 + \beta_1\, chas_i + \beta_2\, old_i + \beta_3\, (chas_i \times old_i) + \beta_4\, lstat_i + \beta_5\, crim_i + u_i \tag{12.2}$$

We estimate the above regression model as follows.

```r
mod_bb2 <- lm(medv ~ chas * old + lstat + crim, data = Boston)
# reorder the variables for stargazer output
vars.order <- c("chas", "old", "chas:old", "lstat", "crim")
stargazer(mod_bb, mod_bb2,
    type = "text",
    column.labels = c("Restricted", "Unrestricted"),
    order = paste0("^", vars.order, "$")
)
```

```
===================================================================
                                 Dependent variable:
                ---------------------------------------------------
                                         medv
                        Restricted              Unrestricted
                           (1)                       (2)
-------------------------------------------------------------------
chas                     4.058**                  4.000***
                         (1.687)                  (1.183)

old                     -7.132***                 1.748**
                         (0.949)                  (0.776)
```

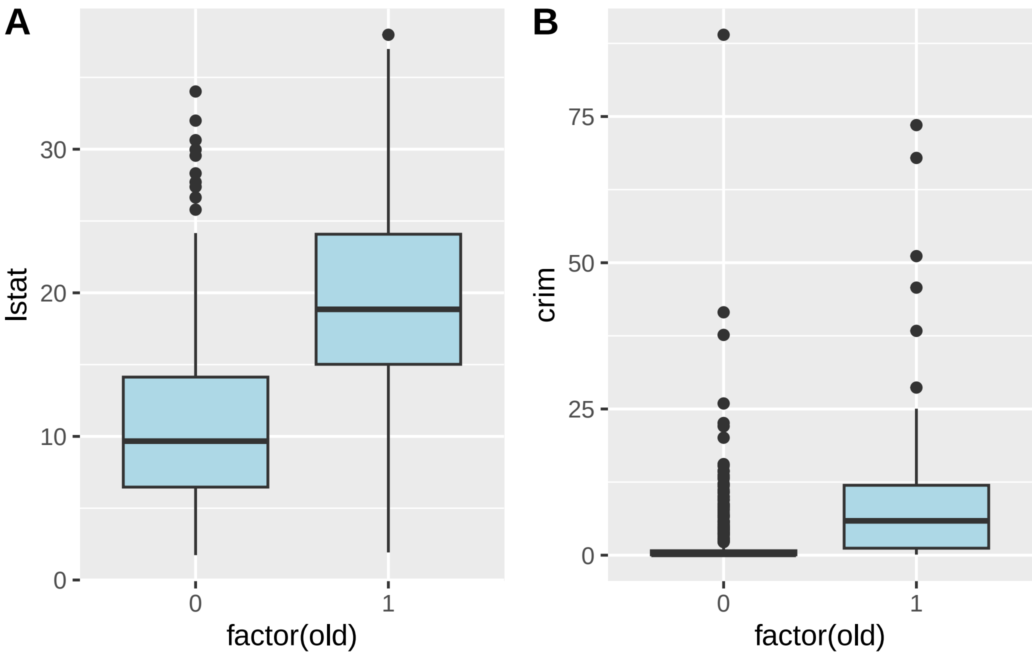
```
chas:old                      10.546***                 4.008
                               (3.758)                 (2.651)

lstat                                                 -0.949***
                                                       (0.046)

crim                                                   -0.079**
                                                       (0.036)

Constant                      23.699***               34.106***
                               (0.450)                 (0.573)

--------------------------------------------------------------------
Observations                     506                     506
R2                              0.130                   0.574
Adjusted R2                     0.125                   0.570
Residual Std. Error    8.604 (df = 502)         6.033 (df = 500)
F Statistic       25.017*** (df = 3; 502) 134.705*** (df = 5; 500)
====================================================================
Note:                                   *p<0.1; **p<0.05; ***p<0.01
```

Comparing the two models:

- The coefficient of `old` reversed signs, from negative (-7.13) to positive (1.75), suggesting that after accounting for crime and low socioeconomic status, older suburbs have slightly higher values.
- Negative coefficients for `lstat` (-0.95) and `crim` (-0.079) indicate that higher poverty and crime reduce house values.

Plot `lstat` and `crim` group by `old` to see how the distribution of these two variables differ between new and old suburbs.

```r
p1 <- ggplot(Boston, aes(x = factor(old), y = lstat)) +
    geom_boxplot(fill = "lightblue")
p2 <- ggplot(Boston, aes(x = factor(old), y = crim)) +
    geom_boxplot(fill = "lightblue")
plot_grid(p1, p2, labels = c("A", "B"), ncol = 2)
```



We see that old suburbs tend to have higher `lstat` and `crim` values compared to new suburbs. This suggests that controlling for these variables is important when examining the relationship between `medv`, `chas`, and `old`.

12.1.2 Two-Way ANOVA Mathematical Formulation

We now conduct a two-way ANOVA to formally compare the two models `mod_bb` and `mod_bb2`.

```
anova(mod_bb, mod_bb2)
```

```
Analysis of Variance Table

Model 1: medv ~ chas * old
Model 2: medv ~ chas * old + lstat + crim
  Res.Df   RSS Df Sum of Sq      F    Pr(>F)
1    502 37161
2    500 18200  2     18961 260.45 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

1. State the null and alternative hypotheses.

$$H_0 : \beta_4 = 0 \text{ and } \beta_5 = 0$$
$$H_1 : \beta_4 \neq 0 \text{ or } \beta_5 \neq 0$$

   In plain language:

- H_0: The additional variables `lstat` and `crim` do not improve the model fit.
- H_1: At least one of the additional variables `lstat` or `crim` improves the model fit.

2. Calculate the F-statistic.

   The F-statistic is based on the R^2 of the two regressions:

$$F = \left(\frac{n - k_U - 1}{p}\right)\left(\frac{R_U^2 - R_R^2}{1 - R_U^2}\right) \sim F(p, n - k_U - 1).$$

   where

   - $n = 506$ is the number of observations;
   - $k_U = 5$ is the number of predictors in the unrestricted model (where H_1 is allowed to be true), excluding the intercept;
   - $p = 2$ is the number of restrictions (i.e., the number of additional variables in the unrestricted model);
   - $R_U^2 = 0.574$ is the R^2 of the unrestricted model;
   - $R_R^2 = 0.130$ is the R^2 of the restricted model.

   Plug in the numbers, we get

$$F = \left(\frac{506 - 5 - 1}{2}\right)\left(\frac{0.574 - 0.130}{1 - 0.574}\right) = 260.56$$

3. Find the critical value

   The F-statistic follows an $F(2, 502)$ distribution under the null hypothesis. At 5% significance level, we use the the critical value $F_{2,\infty} = 3.00$.

4. Decision rule.

   Since the calculated F-statistic (260.56) is much larger than the critical value (3.00), we reject the null hypothesis.

   This suggests that at least one of `lstat` and `crim` significantly improves the model fit.

5. Conclusion.

   We conclude that model (2) with additional control variables `lstat` and `crim` provides a significantly better fit to the data compared to model (1) without these controls.

12.2 Interactions of a Binary Variable and a Continuous Variable

An interaction between a binary variable and a continuous variable shows how the effect of the continuous variable changes depending on the group. This helps us see differences that a simple line wouldn't capture. For example, the effect of business area on house prices might be different in new versus old suburbs.

Now consider the regression model

$$medv_i = \beta_0 + \beta_1 \times indus_i + \beta_2 \times old_i + \beta_3 \times (indus_i \times old_i) + u_i$$

where $indus_i$ being the proportion of non-retail business acres in suburb i; old_i is the same binary variable as defined above, indicating if the suburb is new.

Instructions:

- Estimate the above regression model and assign the result to `mod_bc`.
- What is the relationship between `medv` and `indus` for new suburbs? What about old suburbs?
- Is the effect of `indus` on `medv` statistically different between new and old suburbs?
- Plot `medv` against `indus` and add the regression lines for both states of the binary variable `old`.

```
mod_bc <- lm(medv ~ indus * old, data = Boston)
summary(mod_bc)
```

```
Call:
lm(formula = medv ~ indus * old, data = Boston)

Residuals:
    Min      1Q  Median      3Q     Max
-12.379  -5.066  -1.588   3.015  33.046

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 30.06350    0.72882  41.250   <2e-16 ***
indus       -0.65844    0.06569 -10.024   <2e-16 ***
old         -7.48438    3.07918  -2.431   0.0154 *
```

```
indus:old    0.37115    0.17558   2.114   0.0350 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.024 on 502 degrees of freedom
Multiple R-squared:  0.2434,    Adjusted R-squared:  0.2389
F-statistic: 53.83 on 3 and 502 DF,  p-value: < 2.2e-16
```

The estimated regression model is given by

$$\widehat{medv}_i = 30.06 - 0.66 \times indus_i - 7.48 \times old_i + 0.37 \times (indus_i \cdot old_i)$$

Interpretation of the coefficients

- For new suburbs (i.e., `old = 0`), the relationship between `medv` and `indus` is given by

  $$E(medv_i | old_i = 0) = \hat{\beta}_0 + \hat{\beta}_1 \times indus_i = 30.06 - 0.66 \times indus_i$$

  Thus, for new suburbs, a one-unit increase in `indus` is associated with a decrease in the expected value of `medv` by 0.66.

- For old suburbs (i.e., `old = 1`), the relationship between `medv` and `indus` is given by

  $$\begin{aligned} E(medv_i | old_i = 1) &= (\hat{\beta}_0 + \hat{\beta}_2) + (\hat{\beta}_1 + \hat{\beta}_3) \times indus_i \\ &= (30.06 - 7.48) + (-0.66 + 0.37) \times indus_i \\ &= 22.58 - 0.29 \times indus_i \end{aligned}$$

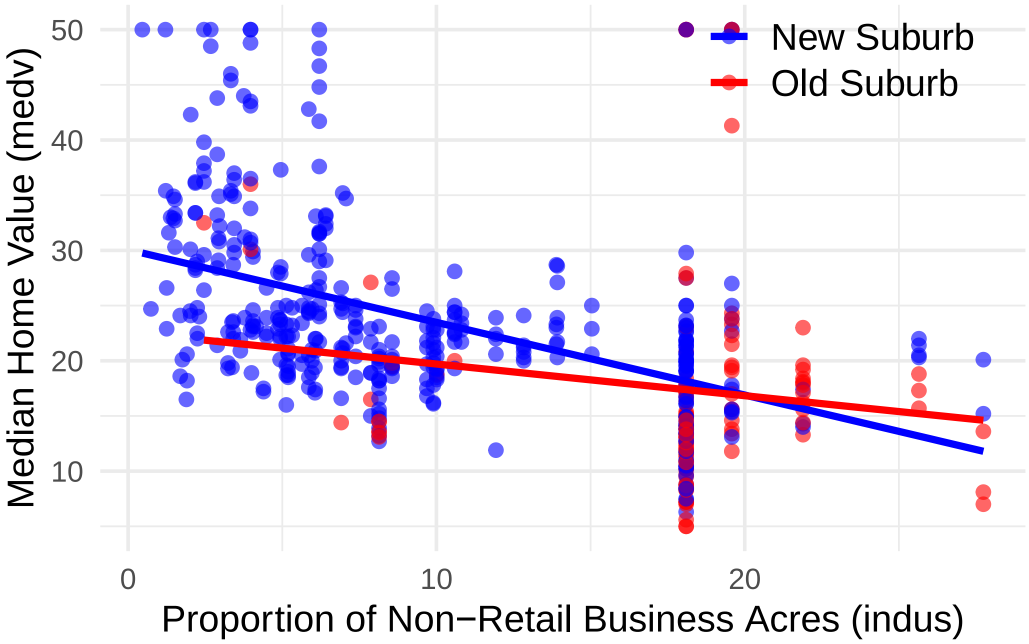
  Thus, for old suburbs, a one-unit increase in `indus` is associated with a decrease in the expected value of `medv` by 0.29. The negative effect is smaller in old suburbs due to the positive interaction term ($\hat{\beta}_3 = 0.37$).

- The effect of `indus` on `medv` is statistically different between new and old suburbs at 5% significance level given that the coefficient of the interaction term ($\hat{\beta}_3$) is statistically significant.

Visualization

Now we plot `medv` against `indus` with regression lines for both states of the binary variable `old`.

```r
ggplot(Boston, aes(x = indus, y = medv, color = factor(old))) +
    geom_point(alpha = 0.6) +
    geom_smooth(method = "lm", formula = y ~ x, se = FALSE) +
    labs(
        x = "Proportion of Non-Retail Business Acres (indus)",
        y = "Median Home Value (medv)",
        color = "Old Suburb (old)"
    ) +
    scale_color_manual(values = c("0" = "blue", "1" = "red"),
     ↪  labels = c("New Suburb", "Old Suburb")) +
    theme_minimal(base_size = 14) +
    theme(
        legend.title = element_blank(),
        legend.text = element_text(size = 14),
        legend.position = c(0.8, 0.9)
    )
```



We see that the slope of the regression line for new suburbs (blue) is steeper than that for old suburbs (red), indicating a stronger negative relationship between `medv` and `indus` in new suburbs compared to old suburbs. This visual representation aligns with our earlier interpretation of the regression coefficients.

13 Regression with a Binary Dependent Variable

🎯 **Study Objectives**

- Understand why linear regression is inappropriate for binary dependent variables.
- Learn the logit and probit models for analyzing binary outcomes.
- Interpret coefficients in logistic regressions.
- Apply logit/probit models using real-world data (HMDA mortgage application dataset).
- Understand the concept of predicted probabilities.
- Compare linear probability model (LPM), logit, and probit approaches.

13.1 Why Binary Dependent Variables Require Special Treatment

In many economic and financial applications, the dependent variable is **binary** (takes only two values: 0 or 1). Examples include:

- **Mortgage approval**: Did the bank approve the mortgage application? (Yes = 1, No = 0)
- **Labor force participation**: Is the individual employed? (Yes = 1, No = 0)
- **Default risk**: Did the borrower default on the loan? (Yes = 1, No = 0)
- **Market entry**: Does the firm enter the market? (Yes = 1, No = 0)
- **Product choice**: Does the consumer purchase the product? (Yes = 1, No = 0)

13.1.1 Why Linear Regression Fails

If we try to use ordinary linear regression (OLS) with a binary dependent variable $Y_i \in \{0, 1\}$:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \cdots + \beta_k X_{ki} + u_i$$

This approach is called the **Linear Probability Model (LPM)** because it estimates the probability that $Y_i = 1$ as a linear function of the predictors.

Problems with LPM:

1. **Predicted probabilities outside [0,1]**: OLS can produce fitted values $\hat{Y}_i < 0$ or $\hat{Y}_i > 1$, which makes no sense for probabilities.
2. **Heteroskedasticity**: The error term variance depends on X, violating the homoskedasticity assumption. This means standard errors are incorrect.
3. **Nonlinear relationship**: The effect of X on the probability of $Y = 1$ is unlikely to be constant across all values of X. For example, increasing income from \$20,000 to \$30,000 likely has a different effect on mortgage approval than increasing from \$200,000 to \$210,000.

13.1.2 The Solution: Logit and Probit Models

To address these problems, we use **nonlinear models** that ensure predicted probabilities stay within [0,1]:

- **Logit model**: Uses the logistic cumulative distribution function (CDF)
- **Probit model**: Uses the standard normal cumulative distribution function (CDF)

13.2 The HMDA Dataset: Mortgage Lending Discrimination

We will use data from the **Federal Reserve Bank of Boston** collected under the **Home Mortgage Disclosure Act (HMDA)**. The dataset contains information on mortgage applications in the Boston area in 1990. The dataset contains 2380 observations with 14 variables.

Research Question: Do banks discriminate against minority applicants when making mortgage lending decisions, after controlling for economic factors?

13.2.1 Load the Data

```
# Load required packages
pkgs <- c("AER", "tidyverse", "stargazer", "margins",
 ↪  "data.table")
missing <- setdiff(pkgs, rownames(installed.packages()))
if (length(missing) > 0) install.packages(missing)
invisible(lapply(pkgs, function(pkg)
 ↪  suppressPackageStartupMessages(library(pkg, character.only =
 ↪  TRUE))))

# Load HMDA data
data("HMDA")

# Preview the data
data.table(HMDA)
```

```
      deny pirat hirat     lvrat  chist  mhist  phist unemp selfemp insurance
    <fctr> <num> <num>     <num> <fctr> <fctr> <fctr> <num>  <fctr>    <fctr>
   1:    no 0.221 0.221 0.8000000     5     2     no   3.9     no       no
   2:    no 0.265 0.265 0.9218750     2     2     no   3.2     no       no
   3:    no 0.372 0.248 0.9203980     1     2     no   3.2     no       no
   4:    no 0.320 0.250 0.8604651     1     2     no   4.3     no       no
   5:    no 0.360 0.350 0.6000000     1     1     no   3.2     no       no
  ---
2376:    no 0.310 0.250 0.8000000     1     1     no   3.2    yes       no
2377:    no 0.300 0.300 0.7770492     1     2     no   3.2     no       no
2378:    no 0.260 0.200 0.5267606     2     1     no   3.1     no       no
2379:   yes 0.320 0.260 0.7538462     6     1    yes   3.1     no       no
2380:   yes 0.350 0.260 0.8135593     2     2     no   4.3     no       no
      condomin   afam single hschool
        <fctr> <fctr> <fctr>  <fctr>
   1:       no     no     no     yes
   2:       no     no    yes     yes
   3:       no     no     no     yes
   4:       no     no     no     yes
   5:       no     no     no     yes
  ---
2376:       no     no     no     yes
2377:      yes     no    yes     yes
```

```
2378:       no     no     no     yes
2379:      yes    yes    yes     yes
2380:      yes     no    yes     yes
```

13.2.2 Variable Definitions

The key variables in the HMDA dataset are:

- We will mainly use `deny`, `pirat`, and `afam` in our analysis.

Variable	Definition
`deny`	Was the mortgage application denied? (yes/no) — **dependent variable**
`pirat`	Payments-to-income ratio (monthly loan payments / monthly income)
`afam`	Is the applicant African American? (yes/no)
`hirat`	Housing expense-to-income ratio (monthly housing expenses / monthly income)
`lvrat`	Loan-to-value ratio (loan amount / assessed property value)
`chist`	Credit history: consumer credit score
`mhist`	Mortgage history: mortgage credit score
`phist`	Public record history: coded as “no” or “yes” (any bankruptcies, tax liens, etc.)
`insurance`	Was mortgage insurance denied? (yes/no)
`selfemp`	Is the applicant self-employed? (yes/no)
`single`	Is the applicant single? (yes/no)
`hschool`	Does the applicant have high school education? (yes/no)
`unemp`	1989 Massachusetts unemployment rate in applicant’s industry (%).
`condominium`	Is the property a condominium? (yes/no)

13.2.3 Data Exploration

13.2.3.1 Frequency Tables and Contingency Tables

Let's examine the denial rates overall and by race.

We first look at the overall denial counts and rates.

```
# Overall denial counts
cat("Overall Denial Counts:\n")
with(HMDA, table(deny))

# Overall denial rate
cat("===========================\n")
cat("Overall Denial Rate:\n")
with(HMDA, prop.table(table(deny))) %>% round(3)
```

```
Overall Denial Counts:
deny
  no  yes
2095  285
===========================
Overall Denial Rate:
deny
  no  yes
0.88 0.12
```

The table shows that about 12% of mortgage applications were denied overall. The majority of applications (88%) were approved.

Next, we examine denial counts and rates by race.

A **contingency table** is an effective method to see the association between two categorical variables. It counts the number of observations in each of the four possible scenarios. When dealing with just one categorical variable, this is referred to as a **frequency table**, which count the number of observations for each category.

The following gives a 2x2 contingency table for mortgage denial by African-American or not.

```
# Denial rate by race
cat("Denail counts by race:\n")
with(HMDA, table(deny, afam))
cat("===========================\n")
cat("Denial rate by race:\n")
with(HMDA, prop.table(table(deny, afam), margin = 2)) %>%
 ↪  round(3)
```

```
Denail counts by race:
     afam
deny    no  yes
  no  1852  243
  yes  189   96
===========================
Denial rate by race:
     afam
deny     no   yes
  no  0.907 0.717
  yes 0.093 0.283
```

Some observations from the table:

- The majority of applicants are non-African American.
- African American applicants have a higher denial rate (about 28%) compared to non-African American applicants (about 9%).

> This descriptive evidence suggests that *the likelihood of denial may be systematically higher for African American applicants*. However, these simple proportions do not control for other relevant factors such as income, credit history, or loan-to-value ratio. Logistic regression will allow us to model this relationship more rigorously while accounting for these additional variables.

13.2.3.2 Summary Statistics

```r
summary(HMDA)
```

```
  deny          pirat             hirat             lvrat          chist
 no :2095   Min.   :0.0000   Min.   :0.0000   Min.   :0.0200   1:1353
 yes: 285   1st Qu.:0.2800   1st Qu.:0.2140   1st Qu.:0.6527   2: 441
            Median :0.3300   Median :0.2600   Median :0.7795   3: 126
            Mean   :0.3308   Mean   :0.2553   Mean   :0.7378   4:  77
            3rd Qu.:0.3700   3rd Qu.:0.2988   3rd Qu.:0.8685   5: 182
            Max.   :3.0000   Max.   :3.0000   Max.   :1.9500   6: 201
 mhist    phist          unemp         selfemp    insurance  condomin
 1: 747   no :2205   Min.   : 1.800   no :2103   no :2332   no :1694
 2:1571   yes: 175   1st Qu.: 3.100   yes: 277   yes:  48   yes: 686
 3:  41              Median : 3.200
 4:  21              Mean   : 3.774
                     3rd Qu.: 3.900
```

```
                      Max.   :10.600
  afam      single     hschool
 no :2041   no :1444   no :  39
 yes: 339   yes: 936   yes:2341
```

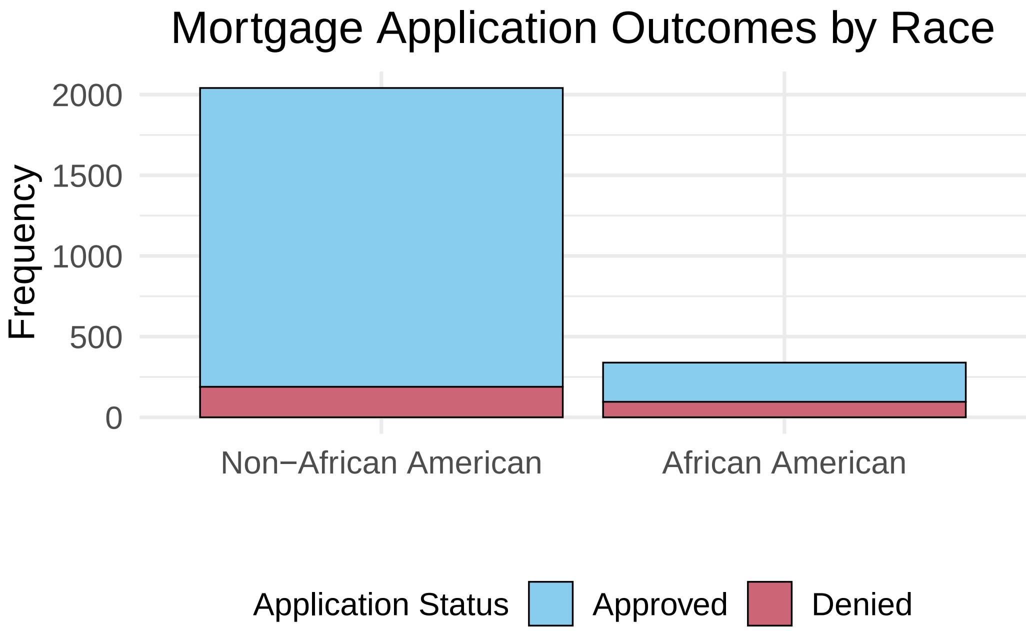
Summary statistics by race.

```
# Summary statistics by race
HMDA %>%
    group_by(afam) %>%
    summarise(
        n = n(),
        denial_rate = mean(deny == "yes"),
        mean_pirat = mean(pirat),
        mean_lvrat = mean(lvrat)
    )
```

```
# A tibble: 2 x 5
  afam      n denial_rate mean_pirat mean_lvrat
  <fct> <int>       <dbl>      <dbl>      <dbl>
1 no     2041      0.0926      0.327      0.726
2 yes     339      0.283       0.351      0.809
```

13.2.4 Visualizing the Contingency Table

We can visualize the contingency table using a **Stacked Bar Plot**.



The stacked bar graph shows:

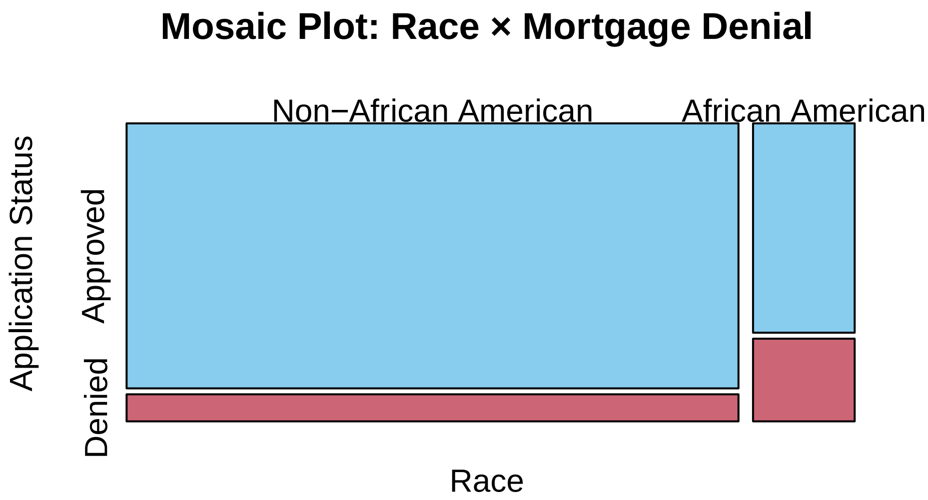
- The sample sizes of African American and non-African American applicants
- The distribution of approved vs. denied applications within each racial group
- African American applicants appear to have a higher proportion of denials

Issue: When the groups have very different sizes, it can be hard to compare proportions using absolute frequencies.

Remedy: Use a **mosaic plot** to visualize **relative frequencies**.

13.2.4.1 Mosaic Plot

A mosaic plot replaces absolute frequencies with relative frequencies, making it easier to compare proportions across groups.



Interpretation:

- The widths of the boxes are proportional to the percentage of each racial group in the sample.
- The heights represent the denial rates within each group.
- We can see that the "Denied" box for African American applicants is taller than for non-African American applicants, indicating a higher denial rate.

13.2.5 Measures of Risk and Association for Binary Outcomes

To quantify the difference in mortgage denial rates between racial groups, we can calculate several measures of risk and association.

```
# Calculate risk (denial rate) for each group
risk_table <- contingency_table %>%
    prop.table(margin = 2) %>%
    as.data.frame.matrix()
risk_table %>% round(3)
# Extract probabilities
p_non_afam <- risk_table["Denied", "Non-African American"] #
↪  P(deny=1 | afam=0)
p_afam <- risk_table["Denied", "African American"] # P(deny=1 |
↪  afam=1)

cat("\nRisk (Denial Rate) by Race:\n")
cat("===========================\n")
cat(sprintf("Non-African American: %.4f (%.2f%%)\n", p_non_afam,
↪  p_non_afam * 100))
cat(sprintf("African American:     %.4f (%.2f%%)\n", p_afam,
↪  p_afam * 100))
```

```
        Non-African American African American
Approved                0.907          0.717
Denied                  0.093          0.283

Risk (Denial Rate) by Race:
===========================
Non-African American: 0.0926 (9.26%)
African American:     0.2832 (28.32%)
```

1. **Risk Difference (Excess Risk)**

   The **risk difference** or **excess risk** (ER) is the difference in denial rates between the two groups:

   $$ER = P(\text{deny} = 1|\text{afam} = 1) - P(\text{deny} = 1|\text{afam} = 0)$$

   ```r
   ER <- p_afam - p_non_afam
   cat("\nRisk Difference (Excess Risk):\n")
   cat("==============================\n")
   cat(sprintf("ER = %.4f - %.4f = %.4f (%.2f percentage
    ↪  points)\n", p_afam, p_non_afam, ER, ER * 100))
   ```

   ```
   Risk Difference (Excess Risk):
   ==============================
   ER = 0.2832 - 0.0926 = 0.1906 (19.06 percentage points)
   ```

   Interpretation: African American applicants have a denial rate that is 19.06 percentage points higher than non-African American applicants.

   - If $ER = 0$: no difference in risk between groups
   - If $ER > 0$: higher risk for the treatment group (African Americans, `afam=1`)

     By contrast, the control group is non-African Americans (`afam=0`).
   - If $ER < 0$: lower risk for the treatment group

2. **Risk Ratio (Relative Risk)**

   The **risk ratio** or **relative risk** (RR) is the ratio of denial rates:

   $$RR = \frac{P(\text{deny} = 1|\text{afam} = 1)}{P(\text{deny} = 1|\text{afam} = 0)}$$

```r
RR <- p_afam / p_non_afam
cat("\nRisk Ratio (Relative Risk):\n")
cat("===========================\n")
cat(sprintf("RR = %.4f / %.4f = %.4f\n", p_afam, p_non_afam,
    RR))
```

```
Risk Ratio (Relative Risk):
===========================
RR = 0.2832 / 0.0926 = 3.0581
```

Interpretation: African American applicants have a denial rate that is 3.06 times higher than non-African American applicants.

- If $RR = 1$: no difference in risk
- If $RR > 1$: higher risk for the treatment group (African Americans, `afam=1`)
- If $RR < 1$: lower risk for the treatment group

3. **Odds Ratio**

The **odds ratio** (OR) compares the odds of denial between the two groups.

First, calculate the odds for each group:

$$\text{odds} = \frac{P(\text{deny} = 1)}{P(\text{deny} = 0)} = \frac{P(\text{deny} = 1)}{1 - P(\text{deny} = 1)}$$

```r
# Calculate odds for each group
odds_non_afam <- p_non_afam / (1 - p_non_afam)
odds_afam <- p_afam / (1 - p_afam)

cat("\nOdds by Race:\n")
cat("=============\n")
cat(sprintf("Non-African American: %.4f\n", odds_non_afam))
cat(sprintf("African American:     %.4f\n", odds_afam))
```

```
Odds by Race:
=============
Non-African American: 0.1021
African American:     0.3951
```

This means that the odds of being denied a mortgage are approximately **0.10 for non-African American applicants** and **0.40 for African American applicants**.

The **odds ratio** is:

$$OR = \frac{\text{odds}(\text{afam} = 1)}{\text{odds}(\text{afam} = 0)} = \frac{P(\text{deny} = 1|\text{afam} = 1)/[1 - P(\text{deny} = 1|\text{afam} = 1)]}{P(\text{deny} = 1|\text{afam} = 0)/[1 - P(\text{deny} = 1|\text{afam} = 0)]}$$

```
OR <- odds_afam / odds_non_afam
cat("\nOdds Ratio:\n")
cat("===========\n")
cat(sprintf("OR = %.4f/%.4f = %.4f\n", odds_afam,
  odds_non_afam, OR))
```

```
Odds Ratio:
===========
OR = 0.3951/0.1021 = 3.8712
```

Interpretation: The odds of mortgage denial for African American applicants are 3.87 times higher than the odds for non-African American applicants.

- If $OR = 1$: no difference in odds
- If $OR > 1$: higher odds for the treatment group (African Americans, `afam=1`)
- If $OR < 1$: lower odds for the treatment group

Important Distinction

- **Risk difference** measures the absolute difference in probabilities (additive scale)
- **Risk ratio** and **odds ratio** measure relative differences (multiplicative scale)
- The odds ratio is particularly useful because it directly relates to the coefficient in logistic regression:

  When we estimate a logistic model with `afam` as the predictor, $e^{\beta_{\text{afam}}}$ will approximately equal the odds ratio we calculated here.

13.3 Model Specification: The Boston Mortgage Example

We want to model the probability that a mortgage application is denied as a function of applicant characteristics.

13.3.1 Preparing the Data

First, we create a binary numeric variable for denial (1 = denied, 0 = approved) and recode some factors:

```
# Create binary dependent variable
HMDA$deny_binary <- ifelse(HMDA$deny == "yes", 1, 0)

# Create binary variable for African American
HMDA$afam_binary <- ifelse(HMDA$afam == "yes", 1, 0)

# Check the variables
head(HMDA[, c("deny", "deny_binary", "pirat", "afam",
 ↪  "afam_binary")])
```

```
  deny deny_binary pirat afam afam_binary
1   no           0 0.221   no           0
2   no           0 0.265   no           0
3   no           0 0.372   no           0
4   no           0 0.320   no           0
5   no           0 0.360   no           0
6   no           0 0.240   no           0
```

13.3.2 Model 1: Linear Probability Model

Let's start with a simple linear regression to see its limitations.

The OLS regression of the binary dependent variable, $deny$ against the payment-to-income ratio (pirat), is estimated as:

$$deny = \beta_0 + \beta_1 \cdot pirat + \varepsilon$$

```
lpm_simple <- lm(deny_binary ~ pirat, data = HMDA)
coeftest(lpm_simple, vcov. = vcovHC, type = "HC1")
```

```
t test of coefficients:

             Estimate Std. Error t value  Pr(>|t|)
(Intercept) -0.079910   0.031967 -2.4998   0.01249 *
pirat        0.603535   0.098483  6.1283 1.036e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

$$\widehat{deny} = -0.080 + 0.604 \cdot pirat$$

The estimated coefficient on P/I ratio is positive, and the population coefficient is statistically significantly different from 0 at the 1% level (the t-statistic is 6.12). If P/I ratio increases by 0.1, the probability of denial increases by approximately $0.604 \times 0.1 \approx 0.060$, that is, by 6 percentage points. Thus applicants with higher debt payments as a fraction of income are more likely to have their application denied.

Now we plot the data and the regression line to visualize the model.

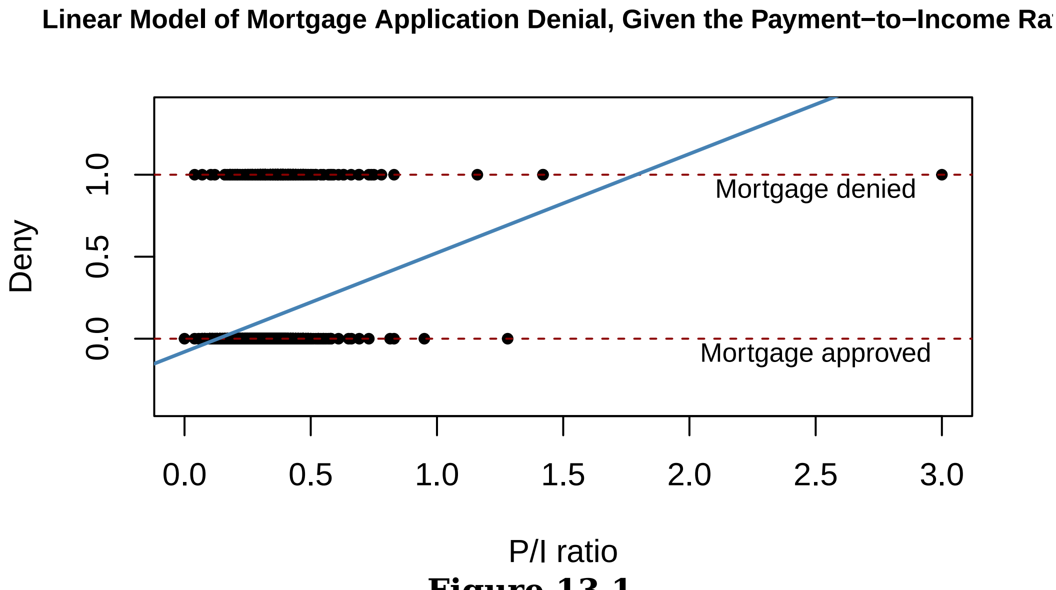


Figure 13.1

> Shortcomings with the linear probability model
>
> 1. The linear probability model can predict probabilities outside the [0, 1] range. For example, for high values of P/I ratio (`pirat`), the predicted probability exceeds 1. However, probabilities must lie between 0 and 1.
> 2. The relationship between P/I ratio and the probability of denial may NOT be linear in reality.

It is reasonable to expect the marginal effects of P/I ratio on denial probability to diminish as P/I ratio increases.

Although a change in P/I ratio from 0.3 to 0.4 might have a large effect on the probability of denial, once the P/I ratio is already very high (e.g., 0.9 to 1.0), increasing P/I ratio further will have litte effect.

3. The error term in the linear probability model is *heteroskedastic*, violating OLS assumptions.

   This means that standard errors and hypothesis tests based on OLS are invalid. → This issue can be addressed using heteroskedasticity robust standard errors.

13.3.3 Model 2: Logit Regression with a single predictor

The estimated model is

$$\mathrm{P}(deny_i = 1 | pirat_i) = F(\beta_0 + \beta_1 \cdot pirat_i) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 \cdot pirat_i)}},$$

where F is the logistic distribution function.

Note that the left-hand side is the probability that the i-th mortgage application is denied ($deny_i = 1$), given the payment-to-income ratio ($pirat_i$).

The strength of the logit model is that it ensures predicted probabilities are always between 0 and 1.

```
# Estimate logit model
logit_simple <- glm(deny_binary ~ pirat,
    family = binomial(link = "logit"),
    data = HMDA
)
coeftest(logit_simple, vcov. = vcovHC, type = "HC1")
```

```
z test of coefficients:

            Estimate Std. Error  z value  Pr(>|z|)
(Intercept) -4.02843    0.35898 -11.2218 < 2.2e-16 ***
pirat        5.88450    1.00015   5.8836 4.014e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

$$\widehat{\mathrm{P}}(deny_i = 1|pirat_i) = F(-4.02 + 5.88 \cdot pirat_i) = \frac{1}{1 + e^{-(-4.02+5.88 \cdot pirat_i)}}$$

Visualizing the logit model:

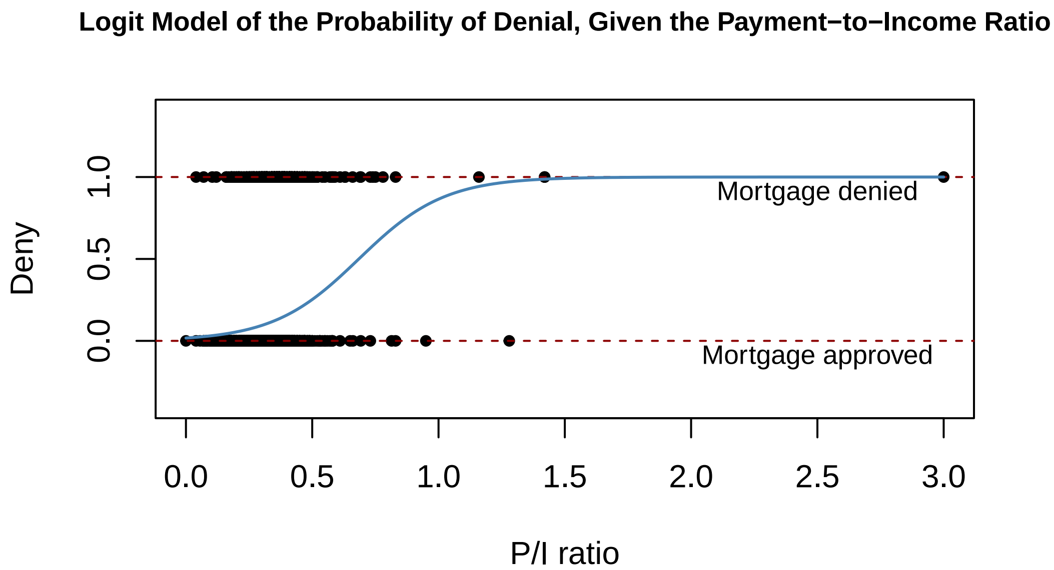


Figure 13.2 The logit model uses the logistic distribution function to model the probability of denial as a function of the payment-to-income ratio. Unlike the linear probability model, the logit model ensures predicted probabilities remain within the [0,1] range.

The estimated logit regression function has a stretched "S" shape: It is nearly 0 and flat for small values of P/I ratio, it turns and increases for intermediate values, and it flattens out again and is nearly 1 for large values.

Example 13.1. Predicted probability.

What is the probability of denial given a P/I ratio of 0.3? What about for P/I ratio being 0.4 and 0.5?

Solution

Solution 13.1. For P/I ratio of 0.3, the estimated probability of denial based on the estimated logit model is:

$$\widehat{P}(deny = 1|pirat = 0.3) = \frac{1}{1 + e^{-(-4.02+5.88 \cdot 0.3)}} \approx 9.4\%$$

That is, the probability of denial is approximately 9.4%.

For P/I ratio of 0.4:

$$\widehat{P}(deny = 1|pirat = 0.4) = \frac{1}{1 + e^{-(-4.02+5.88\cdot 0.4)}} \approx 15.7\%$$

That is, the probability of denial is approximately 15.7%.

For P/I ratio of 0.5:

$$\widehat{P}(deny = 1|pirat = 0.5) = \frac{1}{1 + e^{-(-4.02+5.88\cdot 0.5)}} \approx 25.2\%$$

Main takeaway: As P/I ratio increases, the probability of denial increases, but not linearly.

Interpretation on the logit coefficients:

The estimated model is

$$\widehat{P}(deny_i = 1 \mid pirat_i) = \frac{1}{1 + e^{-(-4.02+5.88\cdot pirat_i)}}$$

which can be written in **log-odds form** as

$$\log\left(\frac{P(deny_i = 1)}{1 - P(deny_i = 1)}\right) = -4.02 + 5.88 \cdot pirat_i$$

1. **Interpreting coefficients (log-odds scale):**

   The slope coefficient, $\hat{\beta}_1 = 5.88$, means that a **one-unit increase** in `pirat` increases the **log-odds** of mortgage denial by **5.88**, holding all else constant.

2. **Interpreting odds ratios (exponentiating coefficients):**

   To obtain a more intuitive interpretation, we exponentiate the coefficient:

   $$e^{\hat{\beta}_1} = e^{5.88} \approx 357.7$$

   This means that for a **one-unit increase** in the payment-to-income ratio, the **odds** of mortgage denial are about **357.7 times larger**.

   - Because a one-unit increase in `pirat` is very large in practice, it is often more meaningful to interpret smaller changes.

   For example, for a **0.1 increase** in `pirat`:

   $$e^{5.88\times 0.1} \approx 1.80$$

   This means that a 0.1 increase in the payment-to-income ratio increases the **odds of denial by about 80%**.

Hypothesis Testing:

Using the normal distribution of parameter estimates, we can use the standard normal table rather than the t table for critical points to test hypotheses about the coefficients.

The z-statistic for testing $H_0 : \beta_1 = 0$ is:

$$z = \frac{\hat{\beta}_1 - 0}{SE(\hat{\beta}_1)} = \frac{5.88 - 0}{1} \approx 5.88$$

Since $z > 1.96$, we reject the null hypothesis at the 5% significance level and conclude that there is a statistically significant positive relationship between payment-to-income ratio and the probability of mortgage denial.

13.3.4 Model 3: Linear Probability Model with Multiple Predictors

$$deny_i = \beta_0 + \beta_1 \cdot pirat_i + \beta_2 \cdot afam_i$$

```r
# Estimate LPM
lpm_multi <- lm(deny_binary ~ pirat + afam_binary, data = HMDA)
summary(lpm_multi)
```

```

Call:
lm(formula = deny_binary ~ pirat + afam_binary, data = HMDA)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.62526 -0.11772 -0.09293 -0.05488  1.06815 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) -0.09051    0.02079  -4.354 1.39e-05 ***
pirat        0.55919    0.05987   9.340  < 2e-16 ***
afam_binary  0.17743    0.01837   9.659  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3123 on 2377 degrees of freedom
Multiple R-squared:  0.076, Adjusted R-squared:  0.07523 
F-statistic: 97.76 on 2 and 2377 DF,  p-value: < 2.2e-16
```

Interpretation:

- $\hat{\beta}_1$: A one-unit increase in the payment-to-income ratio increases the probability of denial by approximately $\hat{\beta}_1$.
- $\hat{\beta}_2$: African American applicants have a probability of denial that is $\hat{\beta}_2$ percentage points higher than non-African American applicants, holding `pirat` constant.

13.3.5 Model 4: Logit Model with Multiple Predictors

Now we estimate a logit model including both P/I ratio (`pirat`) and African-American binary (`afam_binary`) as predictors:

$$P(deny_i = 1|pirat_i, afam_i) = F(\beta_0+\beta_1 \cdot pirat_i+\beta_2 \cdot afam_i) = \frac{1}{1 + e^{-(\beta_0+\beta_1 \cdot pirat_i+\beta_2 \cdot afam_i)}}.$$

```r
# Estimate logit model
logit_multi <- glm(deny_binary ~ pirat + afam_binary,
    family = binomial(link = "logit"),
    data = HMDA
)
coeftest(logit_multi, vcov. = vcovHC, type = "HC1")
```

```
z test of coefficients:

             Estimate Std. Error  z value  Pr(>|z|)    
(Intercept) -4.12556    0.34597 -11.9245 < 2.2e-16 ***
pirat        5.37036    0.96376   5.5723 2.514e-08 ***
afam_binary  1.27278    0.14616   8.7081 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The estimated logit model is:

$$\begin{aligned}\widehat{P}(deny_i = 1|pirat_i, afam_i) &= F(-4.13 + 5.37 \cdot pirat_i + 1.27 \cdot afam_i) \\ &= \frac{1}{1 + e^{-(-4.13+5.37 \cdot pirat_i+1.27 \cdot afam_i)}}\end{aligned}$$

Interpretation of Logit Coefficients

The coefficients in a logit model represent the change in the **log-odds** of denial:

$$\log\left(\frac{P(deny=1)}{1-P(deny=1)}\right) = -4.13 + 5.37 \cdot pirat_i + 1.27 \cdot afam_i$$

1. Interpreting coefficients (log-odds scale):

- $\hat{\beta}_1 > 0$: A one-unit increase in `pirat` increases the log-odds of denial by $\hat{\beta}_1$, holding other variables constant.
- $\hat{\beta}_2 > 0$: Being African American increases the log-odds of denial by $\hat{\beta}_2$ compared to non-African Americans, holding other variables constant.

2. Interpreting odds ratios (exponentiating coefficients):

Alternatively, we can interpret the **odds ratio** by taking the exponential of the coefficients:

- **For `pirat`**: $e^{\hat{\beta}_1}$ represents the **multiplicative change in odds** for a one-unit increase in the payment-to-income ratio.
  For example, if the P/I ratio increases by 0.2, then $e^{5.37\times 0.2} = 2.93$, that is, the odds of mortgage denial are approximately 2.94 times higher, holding `afam` constant.
- **For `afam`**: $e^{\hat{\beta}_2}$ is the **odds ratio comparing African American applicants to non-African American applicants**.
  Based on the estimates, $e^{1.2} = 3.56$, that is, African American applicants have odds of denial that are 3.56 times higher than non-African American applicants, holding `pirat` constant.

Example 13.2. Suppose we have:

- Odds of denial for a non-African American applicant with `pirat = 0.3` is 0.15
- The odds ratio for African Americans compared to non-African American is $e^{\hat{\beta}_2} = 3.56$

Calculate the **expected odds** for an African American applicant with the same `pirat = 0.3`.

Solution

Solution 13.2.

$$\text{odds}(\text{afam} = 1) = \text{odds}(\text{afam} = 0) \times e^{\hat{\beta}_2} = 0.15 \times 3.56 = 0.534$$

The **expected odds** of denial for an African American applicant with `pirat = 0.3` is 0.534.

13.3.6 Model 5: Probit Model

In contrast to the logit model, which uses the logistic CDF, the probit model uses the standard normal distribution function to model the probability of denial.

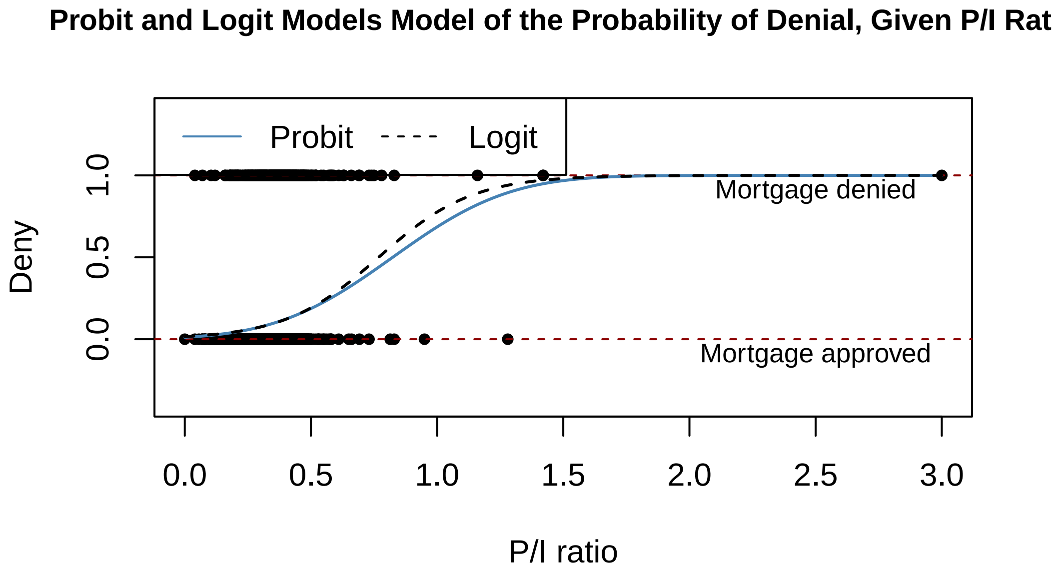
$$P(deny_i = 1|X_i) = \Phi(\beta_0 + \beta_1 \cdot pirat_i + \beta_2 \cdot afam_i)$$

where $\Phi(\cdot)$ is the standard normal CDF.

```
# Estimate probit model
probit_multi <- glm(deny_binary ~ pirat + afam_binary,
    family = binomial(link = "probit"),
    data = HMDA
)
coeftest(probit_multi, vcov. = vcovHC, type = "HC1")
```

```
z test of coefficients:

             Estimate Std. Error  z value  Pr(>|z|)
(Intercept) -2.258787   0.176608 -12.7898 < 2.2e-16 ***
pirat        2.741779   0.497673   5.5092 3.605e-08 ***
afam_binary  0.708155   0.083091   8.5227 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

13.3.7 Comparison: Logit, and Probit



Practical Recommendation:

- Logit and probit typically give very similar results.
- Logit is more common in economics and finance due to the convenient odds ratio interpretation.

13.3.8 Predicted Probabilities

To compare effects across models, we calculate **predicted probabilities** at specific values of the predictors.

Example 13.3. For the three multivariate models (models 3–5), what is the predicted probability of denial for:

- An African American applicant with `pirat = 0.3`?
- A non-African American applicant with `pirat = 0.3`?

Solution

Solution 13.3.

For linear probability model (LPM):

- African American applicant (`afam_binary = 1`):

$$\widehat{deny} = -0.09 + 0.56 \times 0.3 + 0.18 \times 1 = 0.258$$

- Non-African American applicant (`afam_binary = 0`):

$$\widehat{deny} = -0.09 + 0.56 \times 0.3 + 0.18 \times 0 = 0.078$$

For logit model:

- African American applicant:

$$\hat{P}(deny = 1 | pirat = 0.3, afam = 1) = \frac{1}{1 + e^{-(-4.13+5.37\times0.3+1.27\times1)}} \approx 0.22$$

- Non-African American applicant:

$$\hat{P}(deny = 1 | pirat = 0.3, afam = 1) = \frac{1}{1 + e^{-(-4.13+5.37\times0.3+1.27\times0)}} \approx 0.07$$

For probit model:

- African American applicant:

$$\hat{P}(deny = 1 | pirat = 0.3, afam = 1) = \Phi(-2.26+2.74\times0.3+0.71\times1) = \Phi(-0.728)$$

Refer to the standard normal table for $\Phi(-0.728) \approx 0.23$.

- Non-African American applicant:

$$\hat{P}(deny = 1 | pirat = 0.3, afam = 0) = \Phi(-2.26+2.74\times0.3+0.71\times0) \approx 0.075$$

Refer to the standard normal table for $\Phi(-1.438) \approx 0.075$.

13.4 Summary: Logit vs. Probit vs. LPM

	Linear	Logit	Probit
Function	Linear	Logistic CDF	Normal CDF
Predicted probabilities	Can be outside [0,1]	Always in [0,1]	Always in [0,1]
Interpretation	Direct (probability)	Log-odds / Odds ratio	Z-score
Heteroskedasticity	Always present	Accounted for	Accounted for

	Linear	Logit	Probit
When to use	Quick approximation	Preferred for most applications	Similar to logit; standard in some fields